

The Environment (Protection) Rules, 1986

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[The Environment \(Protection\) Rules, 1986](#)

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[In exercise of the powers conferred by sections 6 and 25 of the Environment \(Protection\) Act, 1986 \(29 of 1986](#)

[\), the Central Government hereby makes the following rules, namely:-](#)

[1.](#)

[Short title and commencement. - \(i\) These rules may be called The Environment \(Protection\) Rules, 1986.](#)

[\(ii\)](#)

[They shall come into force on the date of their publication in the Official Gazette.](#)

2.

Definitions. - In these rules, unless the context otherwise requires,-

(a)

"Act" means the Environment (Protection) Act, 1986 (29 of 1986

);

(aa)

["areas" means all areas where the hazardous substances are handled;]

(b)

"Central Board" means the Central Board for the Prevention and Control of Water Pollution constituted under section 3 of the Water (Prevention and Control of Pollution) Act, 1974 (6 of 1974

);

(c)

"Form" means a Form set forth in Appendix A to these rules;

(d)

"Government Analyst" means a person appointed or recognised as such under section 13;

(e)

"person" in relation to any factory or premises means a person or occupier or his agent who has control over the affairs of the factory or premises and includes in relation to any substance, the person in possession of the substance;

(ee)

["prohibited substance" means the substance prohibited for handling;]

[Inserted by G.S.R. 931(E), dated 27.10.1989 (w.e.f. 27.10.1989).]

(f)

"recipient system" means the part of the environment, such as soil, water, air or other which receives the pollutants;

(ff)

["restricted substance" means the substance restricted for handling;]

[Inserted by G.S.R. 931(E), dated 27.10.1989 (w.e.f. 27.10.1989).]

(g)

"section" means a section of the Act;

(h)

"Schedule" means a Schedule appended to these rules;

(i)

"standards" means standards prescribed under these rules;

(j)

"State Board" means a State Board for the Prevention and Control of Water Pollution constituted under section 4 of the Water (Prevention and Control of Pollution) Act, 1974 (6 of 1974

), or a State Board for the Prevention and Control of Air Pollution constituted under section 5 of the Air (Prevention and Control of Pollution) Act, 1981 (14 of 1981

).

3.

Standards for emission or discharge of environmental pollutants. - (1) For the purposes of protecting and improving the quality of the environment and preventing and abating environmental pollution, the standards for emission or discharge of environmental pollutants from the industries, operations or processes shall be as specified in [Schedules I to IV].

[* * *]

[Proviso omitted by S.O. 23(E), dated 16.1.1991 (w.e.f. 16.1.1991).]

(2)

Notwithstanding anything contained in sub-rule (1), the Central Board or a State Board may specify more stringent standards from those provided in [Schedules I to IV]

[Substituted by G.S.R. 422(E), dated 19.5.1993 (w.e.f. 19.5.1993).]

in respect of any specific industry, operation or process depending upon the quality of the recipient system and after recording reasons, therefor, in writing.

(3)

[The standards for emission or discharge of environmental pollutants specified under sub-rule (1) or sub-rule (2) shall be complied with by an industry, operation or process within a period of one year of being so specified.

[(3-A)(i) Notwithstanding anything contained in sub-rules (1) and (2), on and from the 1st day of January, 1994, emission or discharge of environmental pollutants from the [industries, operations or processes other than those industries, operations or processes for which standards have been specified in Schedule I shall]

[Inserted omitted by S.O. 23(E), dated 16.1.1991 (w.e.f. 16.1.1991).]

[not exceed the relevant parameters and standards specified in Schedule VI:

Provided that the State Boards may specify more stringent standards for the relevant parameters with respect to specific industry or locations after recording reasons therefor in writing;

(ii)

The State Board shall while enforcing the standards specified in Schedule VI follow the guidelines specified in Annexures I and II in that Schedule.]

[(3-B) The combined effect of emission or discharge of environmental pollutants in an area, from industries, operations, processes, automobiles and domestic sources, shall not be permitted to exceed the relevant concentration in ambient air as specified against each pollutant in columns (3) to (5) of Schedule VII.]

[Substituted by G.S.R. 7, dated 22.12.1998 (w.e.f. 2.1.1999). Earlier it was inserted by G.S.R. 176(E), dated 2.4.1996 (w.e.f. 3.4.1996).]

(4)

[Notwithstanding anything contained in sub-rule (3),-

(a)

the Central Board or a State Board, depending on the local conditions or nature of discharge of environment pollutants, may, by order, specify a lesser period than a period specified under sub-rule (3) within which the compliance of standards shall be made by an industry, operation or process;

(b)

the Central Government in respect of any specific industry, operation or process, by order, may specify any period other than a period specified under sub-rule (3) within which the compliance of standards shall be made by such industry, operation or process.

(5)

Notwithstanding anything contained in sub-rule (3), the standards for emission or discharge of environmental pollutants specified under sub-rule (1) or sub-rule (2) in respect of an industry, operation or process before the commencement of the Environment (Protection) Amendment Rules, 1991, shall be complied with by such industry, operation or process by the 31st day of December, 1991.]

[Inserted omitted by S.O. 23(E), dated 16.1.1991 (w.e.f. 16.1.1991).]

[Inserted by G.S.R. 422(E), dated 19.5.1993 (w.e.f. 19.5.1993).]

(6)

[Notwithstanding anything contained in sub-rule (3), an industry, operation or process which has commenced production on or before 16th May, 1981 and has shown adequate proof of at least commencement of physical work for establishment of facilities to meet the specified standards within a time-bound programme, to the satisfaction of the concerned State Pollution Control Board, shall comply with such standards latest by the 31st day of December, 1993.

(7)

Notwithstanding anything contained in sub-rule (3) or sub-rule (6) an industry, operation or process which has commenced production after the 16th day of May, 1991 but before the 31st day of December, 1991 and has shown adequate proof of at least commencement of physical work for establishment of facilities to meet the specified standards within a time-bound programme, to the satisfaction of the concerned State Pollution Control Board, shall comply with such standards latest by the 31st day of December, 1992.]

[Added by G.S.R. 95(E), dated 12.2.1992 (w.e.f. 12.2.1992).]

(8)

[With effect from the date specified hereunder, the following coal based thermal power plants shall be supplied with, and shall use, raw or blended or beneficiated coal with ash content not exceeding thirty-four per cent, on quarterly average basis, namely :-]

[Substituted by Notification No. G.S.R. 02 (E) dated 2.1.2014]

(a)

a stand-alone thermal power plant (of any capacity), or a captive thermal power plant of installed capacity of 100 MW or above, located beyond 1000 kilometres from the pit-head or, in an urban area or an ecologically sensitive area or a critically polluted industrial area, irrespective of its distance from the pit-head, except a pit-head power plant, with immediate effect]

[Substituted by Notification No. G.S.R. 02 (E) dated 2.1.2014]

(b)

a stand-alone thermal power plant (of any capacity), or a captive thermal power plant of installed capacity of 100 MW or above, located between 750 - 1000 kilometres from the, 2015;, pit-head, with effect from the 1st day of January]

[Substituted by Notification No. G.S.R. 02 (E) dated 2.1.2014]

(c)

a stand-alone thermal power plant (of any capacity), or a captive thermal power plant of installed capacity of 100 MW or above, located between 500-749 kilometres from the pit-head, with effect from the 5th day of June, 2016:

Provided that in respect of a thermal power plant using Circulating Fluidised Bed Combustion or Atmosphere Fluidised Bed Combustion or Pressurized Fluidised Bed Combustion or Integrated Gasification Combined Cycle technologies or any other clean technologies as may be notified by the Central Government in the Official Gazette, the provisions of clauses (a), (b) and (c) shall not be applicable]

[Substituted by Notification No. G.S.R. 02 (E) dated 2.1.2014]

Explanation : For the purpose of this rule]

[Substituted by Notification No. G.S.R. 02 (E) dated 2.1.2014]

(i)

'beneficiated coal' means coal containing higher calorific value but lower ash than the original ash content in the raw coal obtained through physical separation or washing process]

[Substituted by Notification No. G.S.R. 02 (E) dated 2.1.2014]

(ii)

'captive thermal power plant' means a power plant which is set up by an industry to generate electricity for its exclusive use]

[Substituted by Notification No. G.S.R. 02 (E) dated 2.1.2014]

(iii)

'critically polluted industrial area' means an industrial cluster or area where pollution levels have reached or likely to reach critical level, and has been identified as such by the Central Government or the State Government or the Central Pollution Control Board or a State Pollution Control Board]

[Substituted by Notification No. G.S.R. 02 (E) dated 2.1.2014]

(iv)

'ecologically sensitive area' means an area whose ecological balance is prone to be easily disturbed and has been identified and notified by the Central Government]

[Substituted by Notification No. G.S.R. 02 (E) dated 2.1.2014]

(v)

'installed capacity' shall be calculated by adding, individual capacity of all units within a boundary]

[Substituted by Notification No. G.S.R. 02 (E) dated 2.1.2014]

(vi)

'pit-head power plant' means any captive or stand-alone power station having captive transportation system for its exclusive use for transportation of coal from the loading point at the mining end, up to the uploading point at the power station without using the normal public transportation system]

[Substituted by Notification No. G.S.R. 02 (E) dated 2.1.2014]

(vii)

'stand-alone thermal power plant' means a power plant which is set up to generate electricity for feeding to electricity

grid or for locations that are not fitted with an electricity distribution system; and]

[Substituted by Notification No. G.S.R. 02 (E) dated 2.1.2014]

(viii)

'urban area' means an area limit of a city having a population of more than one million according to the last census.]

[Substituted by Notification No. G.S.R. 02 (E) dated 2.1.2014]

4.

Directions. - (1) Any direction issued under section 5 shall be in writing.

(2)

The direction shall specify the nature of action to be taken and the time within which it shall be complied with by the person, officer or the authority to whom such direction is given.

[(3-a)] The person, officer or authority to whom any direction is sought to be issued shall be served with a copy of the proposed direction and shall be given an opportunity of not less than fifteen days from the date of service of a notice to file with an officer designated in this behalf the objections, if any, to the issue of the proposed direction.

[(3-b)] Where the proposed direction is for the stoppage or regulation of electricity or water or any other service affecting the carrying on any industry, operation or process and is sought to be issued to an officer or an authority, a copy of the proposed direction shall also be endorsed to the occupier of the industry, operation or process, as the case may be, and objections, if any, filed by the occupier with an officer designated in this behalf shall be dealt with in accordance with the procedures under sub-rules (3- a) and (4) of this rule:

Provided that no opportunity of being heard shall be given to the occupier if he had already been heard earlier and the proposed direction referred to in sub-rule (3- b) above for the stoppage or regulation of electricity or water or any other service was the resultant decision of the Central Government after such earlier hearing.]

[Inserted by S.O. 64(E), dated 18.1.1988 (w.e.f. 18.1.1988).]

(4)

The Central Government shall within a period of 45 days from the date of receipt of the objections, if any, or from the date up to which an opportunity is given to the person, officer or authority to file objections whichever is earlier, after considering the objections, if any, received from the person, officer or authority sought to be directed and for reasons to be recorded in writing, confirm, modify or decide not to issue the proposed direction.

(5)

In a case where the Central Government is of the opinion that in view of the likelihood of a grave injury to the environment it is not expedient to provide an opportunity to file objections against the proposed direction, it may, for reasons to be recorded in writing, issue directions without providing such an opportunity.

(6)

Every notice or direction required to be issued under this rule shall be deemed to be duly served-

(a)

where the person to be served is a company, if the document is addressed in the name of the company at its registered office or at its principal office or place of business and is either,-

(i)

sent by registered post, or

(ii)

delivered at its registered office or at the principal office or place of business;

(b)

where the person to be served is an officer serving Government, if the document is addressed to the person and a copy thereof is endorsed to the Head of the Department and also to the Secretary to the Government, as the case may be, incharge of the Department in which for the time being the business relating to the Department in which the officer is employed is transacted and is either,-

(i)

sent by registered post, or

(ii)

is given or tendered to him;

(c)

in any other case, if the document is addressed to the person to be served and-

(i)

is given or tendered to him; or

(ii)

if such person cannot be found, is affixed on some conspicuous part of his last known place of residence or business or is given or tendered to some adult member of his family or is affixed on some conspicuous part of the land or building, if any, to which it relates; or

(iii)

is sent by registered post to that person.

Explanation. - For the purposes of this sub-rule,-

(a)

"company" means any body corporate and includes a firm or other association of individuals;

(b)

"a servant" is not a member of the family.

5.

Prohibition and restriction on the location of industries and the carrying on processes and operations in different areas. -

(1) The Central Government may take into consideration the following factors while prohibiting or restricting the location of industries and carrying on of processes and operations in different areas:-

(i)

Standards for quality of environment in its various aspects laid down for an area.

(ii)

The maximum allowable limits of concentration of various environmental pollutants (including noise) for an area.

(iii)

The likely emission or discharge of environmental pollutants from an industry, process or operation proposed to be prohibited or restricted.

(iv)

The topographic and climatic features of an area.

(v)

The biological diversity of the area which, in the opinion of the Central Government, needs to be preserved.

(vi)

Environmentally compatible land use.

(vii)

Net adverse environmental impact likely to be caused by an industry, process or operation proposed to be prohibited or restricted.

(viii)

Proximity to a protected area under the Ancient Monuments and Archaeological Sites and Remains Act, 1958 or a sanctuary, National Park, game reserve or closed area notified, as such under the Wild Life (Protection) Act, 1972, or places protected under any treaty, agreement or convention with any other country or countries or in pursuance of any decision made in any international conference, association or other body.

(ix)

Proximity to human settlements.

(x)

Any other factor as may be considered by the Central Government to be relevant to the protection of the environment in an area.

(2)

While prohibiting or restricting the location of industries and carrying on of processes and operations in an area, the Central Government shall follow the procedure hereinafter laid down.

(3)

(a)

Whenever it appears to the Central Government that it is expedient to impose prohibition or restrictions on the location of an industry or the carrying on of processes and operations in an area, it may, by notification in the Official Gazette and in such other manner as the Central Government may deem necessary from time to time, give notice of its intention

to do so.

(b)

Every notification under clause (a) shall give a brief description of the area, the industries, operations, processes in that area about which such notification pertains and also specify the reasons for the imposition of prohibition or restrictions on the location of the industries and carrying on of processes or operations in that area.

(c)

Any person interested in filing an objection against the imposition of prohibition or restriction on carrying on of processes or operations as notified under clause (a) may do so in writing to the Central Government within sixty days from the date of publication of the notification in the Official Gazette.

(d)

The Central Government shall, within a period of one hundred and twenty days from the date of publication of the notification in the Official Gazette, consider all the objections received against such notification and may [within [five hundred and forty five days][, and in respect of the States of Assam, Meghalaya, Arunachal Pradesh, Mizoram, Manipur, Nagaland, Tripura, Sikkim and Jammu and Kashmir in exceptional circumstance and for sufficient reasons within a further period of one hundred and eighty days]

[Inserted by Notification No. S.O. 2537(E), dated 8.8.2017, (w.e.f. 19.11.1986).]

[from such date of publication,]

[Inserted by G.S.R. 562(E), dated 27.5.1992.]

impose prohibition or restrictions on location of such industries and the carrying on of any process or operation in an area.

(4)

[Notwithstanding anything contained in sub-rule (3), whenever it appears to the Central Government that it is in public interest to do so, it may dispense with the requirement of notice under clause (a) of sub-rule (3).]

[Inserted by G.S.R. 320(E), dated 16.3.1994 (w.e.f. 16.3.1994).]

6.

[Procedure for taking samples.

- The Central Government or the officer empowered to take samples under section 11 shall collect the sample in sufficient quantity to be divided into two uniform parts and effectively seal and suitably mark the same and permit the person from whom the sample is taken to add his own seal or mark to all or any of the portions so sealed and marked. In case where the sample is made up in containers or small volumes and is likely to deteriorate or be otherwise damaged if exposed, the Central Government or the officer empowered shall take two of the said samples without opening the containers and suitably seal and mark the same. The Central Government or the officer empowered shall dispose of the samples so collected as follows:-

(i)

one portion shall be handed over to the person from whom the sample is taken under acknowledgment; and

(ii)

the other portion shall be sent forthwith to the environmental laboratory for analysis.]

7.

Service of notice. - The Central Government or the officer empowered shall serve on the occupier or his agent or person in charge of the place a notice then and there in Form I of his intention to have the sample analysed.

8.

Procedure for submission of samples for analysis, and the form of laboratory report thereon. - (1) Sample taken for analysis shall be sent by the Central Government or the officer empowered to the environmental laboratory by registered post or through special messenger alongwith Form II.

(2)

Another copy of Form II, together with specimen impression of seals of the officer empowered to take samples alongwith the seals/marks, if any, of the person from whom the sample is taken shall be sent separately in a sealed cover by registered post or through a special messenger to the environmental laboratory.

(3)

The findings shall be recorded in Form III in triplicate and signed by the Government Analyst and sent to the officer from whom the sample is received for analysis.

(4)

On receipt of the report of the findings of the Government Analyst, the officer shall send one copy of the report to the person from whom the sample was taken for analysis, the second copy shall be retained by him for his records and the third copy shall be kept by him to be produced in the Court before which proceedings, if any, are instituted.

9.

Functions of environmental laboratories. - The following shall be the functions of environmental laboratories:-

(i)

to evolve standardised methods for sampling and analysis of various types of environmental pollutants;

(ii)

to analyse samples sent by the Central Government or the officers empowered under sub-section (1) of section 11;

(iii)

to carry out such investigations as may be directed by the Central Government to lay down standards for the quality of environment and discharge of environmental pollutants, to monitor and to enforce the standards laid down;

(iv)

to send periodical reports regarding its activities to the Central Government;

(v)

to carry out such other functions as may be entrusted to it by the Central Government from time to time.

10.

Qualifications of Government Analyst. - A person shall not be qualified for appointment or recognised as a Government Analyst unless he is a -

(a)

graduate in science from a recognised university with five years' experience in a laboratory engaged in environmental investigations, testing or analysis; or

(b)

post-graduate in science or a graduate in engineering or a graduate in medicine or equivalent with two years' experience in a laboratory engaged in environmental investigations, testing or analysis; or

(c)

post-graduate in environmental science from a recognised university with two years' experience in a laboratory engaged in environmental investigations, testing or analysis.

11.

Manner of giving notice. - The manner of giving notice under clause (b) of section 19 shall be as follows, namely:-

(1)

The notice shall be in writing in Form IV;

(2)

The person giving notice may send notice to,-

(a)

if the alleged offence has taken place in a Union territory:-

(A)

the Central Board; and

(B)

the Ministry of Environment and Forests (represented by the Secretary of the Government of India);

(b)

if the alleged offence has taken place in a State:-

(A)

the State Board; and

(B)

the Government of the State (represented by the Secretary to the State Government incharge of environment); and

(C)

the Ministry of Environment and Forests (represented by the Secretary to the Government of India);

(3)

The notice shall be sent by registered post-acknowledgment due; and

(4)

The period of sixty days mentioned in clause (b) of section 19 of the Environment (Protection) Act, 1986 (29 of 1986

) shall be reckoned from the date it is first received by one of the authorities mentioned above.

12.

[Furnishing of information to authorities and agencies in certain cases.

- Where the discharge of environmental pollutant in excess of the prescribed standard occurs or is apprehended to occur due to any accident or other unforeseen act or event, the person in charge of the place at which such discharge occurs or is apprehended to occur shall forthwith intimate the fact of such occurrence or apprehension of such occurrence to all the following authorities or agencies, namely:-

(i)

The officer-in-charge of emergency or disaster relief operations in a district or other region of a State or Union territory specified by whatever designation, by the Government of the said State or Union territory, and in whose jurisdiction the industry, process or operation is located.

(ii)

The Central Board or a State Board, as the case may be, and its regional officer having local jurisdiction who have been delegated powers under sections 20, 21, 23 of the Water (Prevention and Control of Pollution) Act, 1974 (6 of 1974

) and section 24 of the Air (Prevention and Control of Pollution) Act, 1981 (14 of 1981

).

(iii)

The statutory authorities or agencies specified in column 3 in relation to places mentioned in column 2 against thereof of [Schedule V].]

13.

[Prohibition and restriction on the handling of hazardous substances in different areas.

(1)

The Central Government may take into consideration the following factors while prohibiting or restricting the handling of hazardous substances in different areas:-

(i)

the hazardous nature of the substance (either in qualitative or quantitative terms) as far as may be in terms of its damage causing potential to the environment, human beings, other living creatures, plants and property;

(ii)

the substances that may be or likely to be readily available as substitutes for the substances proposed to be prohibited or restricted;

(iii)

the indigenous availability of the substitute, or the state of technology available in the country for developing a safe substitute;

(iv)

the gestation period that may be necessary for gradual introduction of a new substitute with a view to bringing about a total prohibition of the hazardous substance in question; and

(v)

any other factor as may be considered by the Central Government to be relevant to the protection of environment.

(2)

While prohibiting or restricting the handling of hazardous substances in an area including their imports and exports the Central Government shall follow the procedure hereinafter laid down:-

(i)

Whenever it appears to the Central Government that it is expedient to impose prohibition or restriction on the handling of hazardous substances in an area, it may, by notification in the Official Gazette and in such other manner as the Central Government may deem necessary from time to time, give notice of its intention to do so.

(ii)

Every notification under clause (i) shall give a brief description of the hazardous substances and the geographical region or the area to which such notification pertains and also specify the reasons for the imposition of prohibition or restriction on the handling of such hazardous substances in that region or area.

(iii)

Any person interested in filing an objection against the imposition of prohibition or restrictions on the handling of hazardous substances as notified under clause (i) may do so in writing to the Central Government within thirty days from the date of publication of the notification in the Official Gazette.

(iv)

The Central Government shall within a period of sixty days from the date of publication of the notification in the Official Gazette consider all the objections received against such notification and may impose prohibition or restrictions on the handling of hazardous substances in a region or an area.]

14.

[Submission of environment statement.

- Every person carrying on an industry, operation or process requiring consent under section 25 of the Water (Prevention and Control of Pollution) Act, 1974 (

6 of 1974

) or under section 21 of the Air (Prevention and Control of Pollution) Act, 1981 (

14 of 1981

) or both or authorisation under the Hazardous Wastes (Management and Handling) Rules, 1989 issued under the Environment (Protection) Act, 1986 (

29 of 1986

) shall submit an environmental [statement] [for the financial year ending on the 31st March in Form V to the concerned State Pollution Control Board on or before the [thirtieth day of September]

[Inserted by G.S.R. 329(E), dated 13.10.1992 (w.e.f. 13.3.1992).]

[every year, beginning 1993.]

[Inserted by G.S.R. 329(E), dated 13.10.1992 (w.e.f. 13.3.1992).]

[I]

[Schedule renumbered as Schedule I by S.O. 82(E), dated 16-2-1987 (w.e.f. 16-2-1987).]

(See rule 3)

Sl. No.

Industry

Parameter

Standards

1

2

3

4

1.

Caustic soda industry

Concentration not to exceed, milligramme per litre (except for pH and flow)

Total concentration of mercury in the final effluent*

0.01

Mercury bearing waste-water generation (flow)

10 kilolitres/tonne of caustic soda produced

pH

5.5

- 9.0

*Final effluent is the combined effluent from (a) cell house, (b) brine plant, (c) chlorine handling, (d) hydrogen handling, (e) hydrochloric acid plant

2.

Man-made fibres

(synthetic)

Concentration not to exceed, milligramme per litre (except for pH)

Suspended solids

100

[BOD (3 days at 27

[Substituted by G.S.R. 176(E), dated 2-4-1996 (w.e.f. 3-4-1996).]

[°C)]

30

Petroleum Oil refinery

pH

5.5-9.0

3. []

[Substituted by G.S.R. 186(E), dated 18-3-2008 (w.e.f. 18-3-2008).]

[A.

Effluent]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

[Limiting value for concentration (mg/l except for pH)]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

1. [pH]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

6. [0-8.5]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

2. [Oil & Grease]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

5. [0]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

3. [BOD3days, 27°C]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

15. [0]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

4. [COD]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

125. [0]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

5. [Suspended Solids]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

20. [0]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

6. [Phenols]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

0. [35]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

7. [Sulphide]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

0. [5]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

8. [CN]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

0. [20]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

9. [Ammonia as N]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
15. [O]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
10. [TKN]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
40. [O]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
11. [P]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
3. [O]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
12. [Cr (Hexavalent)]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
0. [1]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
13. [Cr (Total)]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
2. [O]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
14. [Pb]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
0. [1]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
15. [Hg]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
0. [01]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
16. [Zn]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
5. [O]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
17. [Ni]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
1. [O]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
18. [Cu]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
1. [O]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
19. [V]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
0. [2]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
20. [Benzene]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
0. [1]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
21. [Benzo
(a)-Pyrene]
[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]
0. [2]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

[Notes. -]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

(i) [

Concentration limits shall be complied with at the outlet, discharging effluent (excluding discharge from sea water cooling systems) to receiving environment (surface water bodies, marine systems or public sewers). In case of application of treated effluent directly for irrigation/horticulture purposes (within or outside the premises of refinery), make-up water for cooling systems, fire fighting, etc., the concentration limits shall also be complied with at the outlet before taking the effluent for such application.

However, any use in the process such as use of sour water in desolater is excluded for the purpose of compliance.]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

(ii) [In case of circulating seawater cooling, the blow-down from cooling systems shall be monitored for pH and oil & grease (also hexavalent & total chromium, if chromate treatment is given to cooling water) and shall conform to the concentration limits for these parameters. In case of reuse of treated effluent as cooling water make-up, all the parameters (as applicable for treated effluent) shall be monitored and conform to the prescribed standards.]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

(iii) [In case of once through cooling with seawater, the oil & grease content in the effluent from cooling water shall not exceed 1.0 mg/l]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

[B. Emissions

Limiting concentration in mg/Nm, unless stated

Fuel Type

Existing refineries

New Refinery/Furnace/Boiler

(Furnace, Boiler and captive Power Plant)

SulphurDioxide (SO₂)

Gas

50

50

Liquid

1700

850

Oxides of Nitrogen (NO_x)

Gas

350

250

Liquid

450

350

Particulate Matter (PM)

Gas

10

5

Liquid

100

50

Carbon Monoxide (CO)

Gas

150

100

Liquid

200

150

Nickel and Vanadium (Ni+V)

Liquid

5

5

Hydrogen Sulphide

(H₂S) in fuel gas

Liquid/gas

150

150

Sulphur content in liquid fuel, weight %

Liquid/gas

1.0

0.5

Notes. -

- (i) In case of mixed fuel (gas and liquid) use, the limit shall be computed based on heat supplied by gas and liquid fuels.
- (ii) All the furnaces/boilers with heat input of 10 million kilo calories/hour or more shall have continuous systems for monitoring of SO₂, and NO_x. Manual monitoring for all the emission parameters in such furnaces or boilers shall be carried out once in two months.
- (iii) All the emission parameters in furnaces/boilers having heat input less than 10 million kilo calories/hour will be monitored once in three months.
- (iv) In case of continuous monitoring, one hourly average concentration values shall be complied with 98% of the time in a month. Any concentration value obtained through manual monitoring, if exceeds the limiting concentration value, shall be considered as non-compliance.
- (v) Data on Nickel and Vanadium content in the liquid fuel (in ppm) shall be reported. Nickel and Vanadium in the liquid fuel shall be monitored at least once in six months, if liquid fuel source quality are not changed. In case of changes, measurement is necessary after every change.

(FCC Regenerators)

Limiting concentration in mg/Nm³, unless stated

Existing refineries hydro-processed FCC feed

Other than hydro-processed FCC feed

New Refinery/FCC Commissioned

Sulphur Dioxide (SO₂)

500

1700

500

(for hydro-processed feed) 850 (for other feed)

Oxides of Nitrogen (NO_x)

400

450

350

Particulate Matter (PM)

100

100

50

Carbon Monoxide (CO)

400

400

300

Nickel and Vanadium (Ni+V)

2

5

2

Opacity, %

30

30

30

Notes. -

(i) In case part feed is hydro-processed, the emission values shall be calculated proportional to the feed rates of untreated and treated feeds.

(ii) FCC regenerators shall have continuous systems for monitoring of SO₂ and NO_x. One hourly average concentration values shall be complied with 98% of the time in a month, in case of continuous monitoring. Manual monitoring for all the emission parameters shall be carried out once in two months.

(iii) Any concentration value obtained through manual monitoring, if exceeds the limiting concentration value, shall be considered as non-compliance.

(iv) Data on Sulphur (weight in %), Nickel (PPM) and Vanadium (PPM) content in the feed to FCC shall be reported regularly.

(v) Limit of Carbon Monoxide emissions shall be complied with except during annual shut down of CO boiler for statutory maintenance.

Plant capacity

Existing SRU

New SRU or Refinery Commissioned

(Tonnes/day)

[Sulphur recovery Units (SRU)]

Sulphur recovery, %

Above 20

98.7

99.5

H₂S mg/Nm³,

15

10

Sulphur recovery, %

5-20

96

98

Sulphur recovery, %

1-5

94

96

Oxides of Nitrogen NO_x, mg/Nm³

All capacity

350

250

Carbon Monoxide (CO), mg/Nm³

All capacity

150

100

Notes. -

(i) Sulphur recovery units having capacity above 20 tonnes per day shall have continuous systems for monitoring of SO₂. Manual monitoring for all the emission parameters shall be carried out once in a month.

(ii) Data on Sulphur Dioxide emissions (mg/Nm³) shall be reported regularly.

(iii) Sulphur recovery efficiency shall be calculated on monthly basis, using quantity of sulphur in the feed to SRU and quantity of sulphur recovered.

C. Fugitive Emission

Storage of Volatile Liquids: General Petroleum Products

(1)

Storage tanks with capacity between 4 to 75 m³ and Total Vapour Pressure (TVP) of more than 10 kpa should have Fixed Roof Tank (FRT) with pressure valve vent.

(2)

Storage tanks with the capacity between 75 to 500 m³ and Total Vapour Pressure (TVP) of 10 to 76 kpa should have Internal Floating Roof Tank (IFRT) or External Floating Roof Tank (EFRT) or Fixed Roof Tank with vapour control or vapour balancing system.

(3)

Storage tanks with the capacity of more than 500 m³ and Total Vapour Pressure (TVP) of 10 to 76 kpa should have Internal Floating Roof Tank or External Floating Roof Tank or Fixed Roof Tank with vapour control system.

(4)

The tanks with the capacity of more than 75 m³ and Total Vapour Pressure (TVP) of more than 76 kpa should have Fixed Roof Tank with vapour control system.

(5)

Requirement for seals in Floating Roof Tanks:

(i)

(a)

IFRT and EFRT shall be provided with double seals with minimum vapour recovery of 96%.

(b)

Primary seal shall be liquid or shoe mounted for EFRT and vapour mounted for IFRT. Maximum seal gap width will be 4 cm and maximum gap area will be 200 cm²/m of tank diameter.

(c)

Secondary seal shall be rim mounted. Maximum seal gap width will be 1.3 cm and maximum gap area will be 20 cm²/m of tank diameter.

(d)

Material of seal and construction shall ensure high performance and durability.

(ii)

Fixed Roof Tanks shall have vapour control efficiency of 95% and vapour balancing efficiency of 90%.

(iii)

Inspection and maintenance of storage tanks shall be carried out under strict control. For the inspection, API RP 575 may be adopted. In-service inspection with regard seal gap should be carried out once in every six months and repair to be implemented in short time. In future, possibility of on-stream repair of both seals shall be examined.

Storage of Volatile Liquids: Benzene Storage

(1)

FRT with vapour to incineration with 99.9% of removal efficiency for Volatile Organic Compounds (VOC) shall be provided.

(2)

IFRT/EFRT with double seals, emission-reducing roof fitting and fitted with fixed roof with vapour removal efficiency of at least 99% shall be provided.

Solvents for Lube-Base Oil production (Furfural, NMP, MEK, Toluene and MIBK)

IFRT with double seals and inert gas blanketing with vapour removal efficiency of at least 97% shall be provided.

1

2

3

4

Emission control for Road Tank Truck/

Rail Tank Wagon Loading

Loading of Volatile Products

Gasoline and Naphtha:

(i) VOC reduction, %

(i)

99.5

(ii) Emission, gm/m³

(ii)

5

Benzene:

(i) VOC reduction, %

(i)

99.99

(ii) Emission, mg/m³

(ii)

20

Toluene/Xylene:

(i) VOC reduction, %

(i)

99.98

(ii) Emission, mg/m³

(ii)

150

Note. -

(i) It shall be applicable for Gasoline, Naphtha, Benzene, Toluene and Xylene loading.

(ii) Road Tank Truck shall have Bottom loading and Rail Tank Wagon shall have Top submerged loading.

(iii) Annual leak testing for vapour collection shall be done.

Standards for Equipment Leaks

(1)

Approach: Approach for controlling fugitive emissions from equipment leaks shall have proper selection, installation and maintenance of non-leaking or leak-tight equipment. Following initial testing after commissioning, the monitoring for leak detection is to be carried out as a permanent on-going Leak Detection and Repair (LDAR) programme. Finally detected leaks are to be repaired within allowable time frame.

(2)

Components to be Covered: Components that shall be covered under LDAR programme include (i) Block valves; (ii) Control valves; (iii) Pump seals; (iv) Compressor seals; (v) Pressure relief valves; (vi) Flanges-Heat Exchangers; (vii) Flanges-Piping; (viii) Connectors-Piping; (ix) Open ended lines; and (x) Sampling connections. Equipment and line sizes more than 1.875 cm or ¾ inch are to be covered.

(3)

Applicability: LDAR programme would be applicable to components (given at 2 above) for following products/compounds: (i) hydrocarbon gases; (ii) Light liquid with vapour pressure @ 20°C 1.0 kPa; and (iii) Heavy liquid with vapour pressure @ 20°C between 0.3 to 1.0 kPa.

(4)

While LDAR will not be applicable for heavy liquids with vapour pressure 0.3 kPa, it will be desirable to check for liquid dripping as indication of leak.

(5)

Definition of leak: A leak is defined as the detection of VOC concentration more than the values (in ppm) specified below at the emission source using a hydrocarbon analyzer according to measurement protocol (US EPA-453/R-95-017, 1995). Protocol for equipment leak emission estimates may be referred to :

Component

General Hydrocarbon (ppm)

Benzene (ppm)

Till 31st Dec., 2008
w.e.f. January 01, 2009
Till 31st Dec., 2008
w.e.f. January 01, 2009

Pump/Compressor

10000

5000

3000

2000

Valves/Flanges

10000

3000

2000

1000

sOther components

10000

3000

2000

1000

(6)

In addition, any component observed to be leaking by sight, sound or smell, regardless of concentration (liquid dripping, visible vapor leak) or presence of bubbles using soap solution should be considered as leak.

(7) Monitoring Requirements and Repair Schedule: Following frequency of monitoring of leaks and schedule for repair of leaks shall be followed:

Component

Frequency of monitoring

Repair schedule

Quarterly

(semiannual after two consecutive periods with 2% leaks and annual after 5 periods with 2% leaks)

Repair will be started within 5 working days and shall be completed within 15 working days after detection of leak for general hydrocarbons. In case of benzene, the leak shall be attended immediately for repair.

Pump seals

Quarterly

Compressor seals

Quarterly

Pressure relief devices

Quarterly

Pressure relief devices (after venting)

Within 24 hours

Heat Exchangers

Quarterly

Process drains

Annually

Components that are difficult to monitor

Annually

Pump seals with visible liquid dripping

Immediately

Immediately

Any component with visible leaks

Immediately

Immediately

Any component after repair/
replacement

Within five days

(8)

The percentage leaking components should not be more than 2% for any group of components, monitored excluding pumps/compressors. In case of pumps/compressors, it should be less than 10% of the total number of pumps/compressors or three pumps and compressors, whichever is greater.

(9)

Emission Inventory: Refinery shall prepare an inventory of equipment components in the plant. After the instrumental measurement of leaks, emission from the components will be calculated using stratified emission factors (USEPA) or any other superior factors. The total fugitive emission will be established.

(10)

Monitoring: Following types of monitoring methods may be judiciously employed for detection of leaks: (i) Instrumental method of measurement of leaks; (ii) Audio, Visual and Olfactory (AVO) leak detection; and (iii) Soap bubble method.

(11)

Data on time of measurement and concentration value for leak detection; time of repair of leak; and time of measurement concentration value after repair of leak should be documented for all the components.

(12)

Pressure relief and blow down systems should discharge to a vapour collection and recovery system or to flare.

(13)

Open-ended lines should be closed by a blind flange or plugged.

(14)

Totally closed-loop should be used in all routine samples.

(15)

Low emission packing should be used for valves.

(16)

High integrity sealing materials should be used for flanges.

D. Emission Standards for VOC from Wastewater Collection and Treatment

(1)

All contaminated and odorous wastewater streams shall be handled in closed systems from the source to the primary treatment stages (oil-water separator and equalization tanks).

(2)

The collection system shall be covered with water seals (traps) on sewers and drains and gas tight covers on junction boxes.

(3)

Oil-water separators and equalization tanks shall be provided with floating/fixed covers. The off-gas generated shall be treated to remove at least 90% of VOC and eliminate odour. The system design shall ensure safety (prevention of formation of explosive mixture, possible detonation and reduce the impact) by dilution with air/inert gas, installing LEL detector including control devices, seal drums, detonation arrestors, etc. The system shall be designed and operated for safe maintenance of the collection and primary treatment systems.

(4)

Wastewater from aromatics plants (benzene and xylene plants) shall be treated to remove benzene total aromatics to a level of 10, 20 ppm respectively before discharge to effluent treatment system without dilution.]

[Substituted by Notification No. G.S.R. 186 (E) dated 18.3.2008 (w.e.f. 19.11.1986)]

4. [Sugar industry

[Substituted by Notification No. G.S.R. 35(E), dated 14.1.2016 (w.e.f. 19.11.1986).]

Effluents

All concentration values are in milligramme per litre except for pH

pH

5.5 - 8.5

Total Suspended Solids (TSS), milligramme per litre

100 (for disposal on land)30 (for disposal in surface waters)

Biological Oxygen Demand, BOD [3 days at 270C], milligramme per litre

30 (for disposal in surface waters)

Oil Grease milligramme per litre

10

Total Dissolved Solids (TDS), milligramme per litre

2100

Final waste-water discharge limit

200 litre per tonne of cane crushed

(Final treated effluent discharge restricted to 100 litre per tonne of cane crushed and Waste water from spray pond overflow or cooling tower blow down to be restricted to 100 litre per tonne of cane crushed and only single outlet point form unit is allowed.)

Emissions

The particulate matter emissions from the stack shall be less than 150 milligramme per normal cubic metre]

4.

(1)

Treated effluent Irrigation protocol and waste water conservation or waste water management in Sugar industries

(i)

Loading rates for different soil textures

S.N

Soil Texture

Loading rate in m³/Ha/Day

1

Sandy

225 to 280

2

Sandy loam

170 to 225

3

Loam

110 to 170

4

Clay loam

55 to 110

5

Clay

35 to 55

(ii)

Waste water conservation and pollution control management

1. Establishment of cooling arrangement and polishing tank for recycling the excess condensate water to process or utilities or allied units.

2. Effluent Treatment Plant to be stabilized one month prior to the start of the crushing season and continue to operate one month after the crushing season.

3. During no demand period for irrigation, the treated effluent to be stored in a seepage proof lined pond having 15 days holding capacity only.

4. Flow meter to be installed in all water abstraction points and usage of fresh water to be minimized.

5. Suitable Air pollution control devices to be installed to meet the particulate matter emission standard.

5.

Thermal power plants

Maximum,

limiting concentration, milligramme per litre (except for pH and

temperature)
Condenser
cooling waters
pH
6.5
- 8.5
(once
through cooling system)
Temperature
Not
more than 5°C higher than the intake water temperature
Free
available chlorine
0.5
Boiler
blowdowns
Suspended
solids
100
Oil
and grease
20
Copper
(total)
1.0
Iron
(total)
1.0
Cooling
tower blowdown
Free
available chlorine
0.5
Zinc
1.0
Chromium
(total)
0.2
Phosphate
5.0
Other
corrosion inhibiting material
Limit
to be established on case by case basis by Central Board in case
of Union territories and State Boards in case of States
Ash
pond effluent
pH
6.5
- 8.5
Suspended

solids

100

Oil

and grease

20

[Standards for Discharge of Effluents from Textile Industry

[Substituted by Notification No. G.S.R. 978 (E), dated 10.10.2016 (w.e.f. 19.11.1986).]

S.

No.

Industry

Parameter

Standard

(applicable

for all modes of disposal*)

1

2

3

4

6

All

Integrated textile units, units of Cotton/ Woollen/ Carpets/

Polyester, Units having Printing/ Dyeing/ Bleaching process or
manufacturing and Garment units.

TREATED

EFFLUENTS

Maximum

concentration values in mg/l except for pH, colour, and SAR

pH

6.5

to 8.5

Suspended

Solids

100

Colour,

P.C.U. (Platinum Cobalt Units)

150

Bio-Chemical

Oxygen Demand

(BOD₃)

30

Oil

and Grease

10

Chemical

Oxygen Demand (COD)

250

Total

Chromium as (Cr)

2.0

Sulphide

(as S)

2.0

Phenolic

Compounds (as C₆H₅OH)

1.0

Total

Dissolved Solids, Inorganic (TDS)

2100**

Sodium

Absorption Ratio (SAR)

26**

Ammonical

Nitrogen (as N)

50

Notes:

1. *In case of direct disposal into rivers and lakes, the Central Pollution Control Board (CPCB) or State Pollution Control Boards/ Pollution Control Committees (SPCBs/ PCCs) may specify more stringent standards depending upon the quality of the recipient system.

2. **Standards for TDS and SAR shall not be applicable in case of marine disposal through proper marine outfall.

3. The treated effluent shall be allowed to be discharged in the ambient environment only after exhausting options for reuse in industrial process/ irrigation in order to minimise freshwater usage.

4. Any textile unit attached with the Common Effluent Treatment Plant (CETP) shall achieve the inlet and treated effluent quality standards as specified in serial number 55 of Schedule-I to the Environment (Protection) Rules, 1986 and shall also be jointly and severally responsible for ensuring compliance.

5. The standalone Micro, Small and Medium Enterprises (MSMEs) as per the MSME Development Act, 2006 shall meet the values specified above.

6. The standalone large scale units shall meet the values specified above; however, CPCB or SPCBs/ PCCs with the approval of CPCB, may mandate Zero Liquid Discharge in Large scale units in environmentally sensitive/ critical areas.

7. The TDS value with respect to treated effluent shall be 2100 milligramme per litre; however, in case where TDS in intake water is above 1100 milligramme per litre, a maximum contribution up to 1000 milligramme per litre shall be permitted provided the maximum value of 3100 milligramme per litre is not exceeded in the treated effluent.]

The special parameters are to be stipulated by the Central Board in case of Union territories State Boards in case of States depending upon the dye used in the industry. Where the industry uses chrome dyes, sulphur dyes and or phenolic compounds in the dyeing/ printing process, the limits on chromium of 2 mg/litre, sulphides of 2 mg/litre, and phenolic compounds of 5 mg/litre, respectively shall be imposed.

Where the quality requirement of the recipient system so warrants, the limit of BOD should be lowered upto 30 according to the requirement by the State Boards for the States and the Central Board for the Union territories.

A limit on sodium absorption ratio of 26 should be imposed by the State Boards for the the Central Board for the Union territories if the disposal of the effluent is to be made on land.

7. Composite woolen mills

Concentration not to exceed, milligramme per litre

(except for pH and bio-assay)

Common:

Suspended solids

100

PH

5.5 - 9.0

[BOD (3 days at 27°C)]

[Substituted by G.S.R. 176(E), dated 2-4-1996 (w.e.f. 3-4-1996).]

100

Oil and grease

10

Bio-assay

90% survival of fish after 96 hours

Special:

Total chromium (as Cr)

2

Sulphide(as S)

2

PhenolicCompounds (as C₆H₅OH)

5

Sl.

No.

Industry

Parameter

Standards

1

2

3

4

The special parameters are to be stipulated by the Central Board in case of Union territories and State Boards in case of State depending upon the dye used in the industry. Where the industry uses chrome dyes, sulphur dyes and or/phenolic compounds in the dyeing/printing process, the limits on chromium of 2 mg/litre, sulphides of 2 mg/litre and phenolic compounds of 5 mg/litre respectively shall be imposed.

Where the quality requirement of the recipient system so warrants, the limit of BOD should be lowered upto 30 according to the requirement by the State Boards for the State and the Central Board for the Union territories.

A limit on sodium absorption ratio of 26 should be imposed by the State Boards for the States and the Central Board for the Union territories if the disposal of the effluent is to be made on land.

[TABLE]

[Substituted by Notification No. G.S.R. 266 (E) dated 30.3.2012 (w.e.f. 19.11.1986)]

Sl.No.

Industry

Parameter

Standard

(1)

(2)

(3)

(4)

8

Dye and Dye

Intermediate Industry

A.

Emission Standards (Process)

Limiting

concentration in milligramme/Normal cubic metre (mg/Nm³), unless otherwise stated

Oxides of Sulphur

(SO_x)

200

HCl (Acid Mist)

35

Ammonia (NH₃)

30

Chlorine (Cl₂)

15

Note:All

process vents shall have chimney height of atleast two metres above the shed or building where equipment is installed.

B

. Effluent Standards

Limiting

concentration not to exceed in milligramme/litre (mg/l), except for pH, Temperature, Colour and Bioassay.

disposal in

surface water

marine disposa

on land for

irrigation

pH

6.0 to 8.5

5.5 ? 9.0

5.5 ? 9.0

Suspended Solids

100

-

200

Biochemical

Oxygen Demand - BOD (3 days, 27°C)

30

100

100

Chemical Oxygen

Demand (COD)

250

250

-

Ammonical

Nitrogen as N

50

50

-

Temperature

shall not exceed

5°C above the receiving water

-

Colour (Hazen

unit)

400

-

-

Mercury (Hg)

0.01

0.01

-

Hexavalent

Chromium (Cr+6) 0.1

0.1

1.0

-

Total Chromium

(Cr)

2.0

2.0

-

Copper (Cu)

2.0

3.0

-

Zinc (Zn)

5.0

15.0

-

Nickel (Ni)

3.0

5.0

-

Lead (Pb)

0.1

2.0

-

Manganese (Mn)

2.0

2.0

-

Cadmium (Cd)

0.2

2.0

-

Chloride (Cl-)

1000

-

-

Sulphate(SO₄²⁻)

1000

-

-

Phenolic Compounds

as C₆H₅OH

1.0

5.0

-

Oil Grease

10.0

10.0

10.0

Bio-assay Test

(with 1:8 dilution of effluents)

90% survival of

Test animals after 96 hours* in 100% effluent

-

-

*

The Bioassay test shall be conducted as per IS: 6582: 1971.

Note:

(i)

In case of disposal of effluent on land by industry directly or through a CETP, the industry or, CETP as the case may be, shall be required to install piezometers for monitoring of groundwater.

Atleast, two piezometers for three hectares shall be installed for a plot size above 10 hectares with a minimum of 16 piezometers. It shall be one per hectare within a minimum of six piezometers for a plot of size smaller than 10 hectares, in consultation with the concerned State Pollution Control Board for siting of piezometers.(ii) The standards for Chloride and Sulphate shall be applicable only for discharge of treated effluent into inland surface water courses. However, when discharged on land for irrigation, the norms for Chloride shall not be more than 600 mg/l over and above the contents of raw water and the Sodium Absorption Ratio (SAR) shall not exceed 26.(iii) Treated /untreated effluent shall be stored in holding tank(s) in such a manner which would not cause pollution of groundwater.

C.

Emission Standards for Captive Incinerator

Limiting

concentration in mg/Nm³, unless otherwise stated

Sampling

Duration in minutes unless otherwise stated

Particulate Matter

50

30

or more (for sampling of 300 litres of emission)

HCl (Acid Mist)

50

30

SO₂

200

30

Carbon Monoxide

100

daily

average

Total Organic

Carbon

20

30

Total Dioxins And
Furans

0.1
ng TEQ/Nm³
8

hours

Sb+As+Pb+Cr+Co+Cu+

Mn+Ni+V+Cd+Th+Hg and their compounds

1.5

2

hours

Note:

i.

All monitored values shall be corrected to 11% oxygen on dry basis.ii. The CO₂ concentration in tail gas shall not be less than 7%.

iii.

In case, halogenated organic waste is less than 1% by weight in input waste, all the facilities in twin chamber incinerator shall be designed so as to achieve a minimum temperature of 850 ± 250°C in primary chamber and 950°C in secondary combustion chamber and with a gas residence time in secondary combustion chamber not less than two seconds.

or

All

the facilities in single chamber incinerator for gaseous hazardous waste shall be designed so as to achieve a minimum temperature of 950 °C in the combustion chamber with a gas residence time not less than two seconds.

iv.

In case halogenated organic waste is more than 1% by weight in input waste, waste shall be incinerated only in twin chamber incinerators and all the facilities shall be designed to achieve a minimum temperature of 850± 250°C in primary chamber and 1100 °C in secondary combustion chamber with a gas residence time in secondary combustion chamber not less than two seconds.

v.

Scrubber meant for scrubbing emissions from incinerator shall not be used as quencher.

vi.

Incineration plants shall be operated, (i.e., combustion chambers) with such temperature, retention time and turbulence, as to achieve Total Organic Carbon (TOC) content in the incineration ash and residue less than 3%, and the loss on ignition for ash and residue is less than 5% of the dry weight. In case of non-conformity, ash and residue, as the case may be shall be re-incinerated.

vii. The

incinerator shall have a chimney of at least thirty metres height.

D.

Effluent Standards for Incinerator

Note:

(i)

Effluent from scrubber (s) and floor washing shall flow through closed conduit or pipe network and be treated to comply with the effluent standards mention at ?B? above.

(ii) The built up

in Total Dissolved Soilds (TDS) in wastewater of floor washings shall not exceed 1000 mg/l over and above the TDS of raw water used.

E.

Stormwater

Note:

(i)

Stormwater shall not be allowed to mix with scrubber water and/or floor washings.

(ii)

Stormwater shall be channelized through separate drains passing through a HDPE lined pit having holding capacity of 10 minutes (hourly average) of rainfall.?.

The standards for chlorides and sulphates are applicable or discharge into inland and surface water courses. However. when discharged on land for irrigation, the limit for chloride shall not be more than 600 milligrammes per litre and the sodium absorption ratio shall not exceed 26.

[S.No.

[Substituted by Notification No. G.S.R. 266 (E) dated 30.3.2012 (w.e.f. 19.11.1986)]

Industry

Parameter

Standard

(1)

(2)

(3)

(4)

9

Electroplating,

Anodizing Industry

A-Effluent

Standards

Limiting

Concentration in mg/l, except for pH and Temperature

(i) Compulsory

Parameters

pH

6.0 to 9.0

Temperature

Shall not exceed

5°C above the ambient temperature of the receiving body

Oil and Grease

10

Suspended Solids

100

Total Metal*

10

Trichloroethane

0.1

Trichloroethane

0.1

(ii) Specific

Parameter as per process

a.

Nickel and Chrome plating

Ammonical

Nitrogen, as N

50

Nickle, as Ni

Hexavalent

Chrommium, as Cr

0.1

Total Chromium, as

Cr

2

Sulphides, as S

2

Sulphates, as SO₄

-2

400

Phosphates, as P

5

Copper as Cu

3

b.

Zinc plating

Cyandies, (as CN⁻)

0.2

Ammonical

Nitrogen, as N

50

Total Residual

Chlorine, as Cl

1

Hexavalent

Chromium, as Cr

0.1

Total Chromium, as

Cr

2

Zinc, as ZN

5

Lead, as Pb

0.1

Iron, as Fe

3

c.

Cadmium plating

Cyanides, (as CN-)

0.2

Ammonical

Nitrogen, as N

50

Total Residual

Chlorine, as Cl

1

Hexavalent

Chromium, as Cr

0.1

Total Chromium, as

Cd

2

Cadmium, as Cd

2

d.

Anodizing

Ammonical

Nitrogen, as N

50

Total Residual

Chlorine, as Cl

1

Aluminium

5

Fluorides, as F

15

Sulphates, as SO₄

2-

400

Phosphates, as P

5

e.

Copper, Tin plating

Cyanides, (as CN-)

0.2

Copper, as Cu

3

Tin

2

f.

Precious Metal plating

Cyanides. (as CN-)

0.2

Total Residual

Chlorine, as Cl

1

B.-Emission

Standards+

Limiting

concentration in mg/m³ unless stated

(i)

Compulsory parameters

Acid mist (HCl

H₂ SO₄)**

50

(ii)

Specific parameters as per process

a.

Nickle Chromium plating

Nickle**

5

Hexavalent

Chromium**

0.5

b.

Zinc, Cooper or Cadmium plating

Lead**

10

Cyanides,

(Total)**

5

* 'Total Metal'

shall account for combined concentration of

Zn+Cu+Ni+Al+Fe+Cr+Cd+Pd+Sn+Ag in the effluent.+ Emission

standards shall be applicable to electroplating units having

water consumption atleast 5 m³/day. These units shall channelize

their emission through a stack or chimney having height at least

10 metres above ground level or 3 metres above top of shed or

building of the unit, whichever is more.** The existing units

shall comply with the norms of asterisked pollutants by 1st

January 2013. However, new units shall comply with the norms with

effect from commissioning of plant.

C.

Stormwater

Note :(i)

Stormwater for a unit (having plot size atleast 200 square

metres) shall not be allowed to mix with scrubber water, effluent

and/or floor washings.(ii) Stormwater within the battery

limits of a unit shall be channelized through separate drain/pipe

passing through a High Density Polyethylene (HDPE) lined pit

having holding capacity of ten minutes (Hourly average) of

rainfall.]

[Substituted by Notification No. G.S.R. 496(E), dated 9.5.2016 (w.e.f. 19.11.1986).]

[S. No.

Industry

Parameter

Standards

(1)

(2)

(3)

(4)

10.

Cement Plant (without co processing), Standalone Clinker

Grinding Plant or, Blending Plant

A ? Emission Standards

(i) Rotary Kiln ?without co processing

Date of Commissioning

Location

Concentration not to exceed, in mg/Nm³

(a)

(b)

(c)

Sulphur Dioxide (SO₂) in mg/ Nm³

Irrespective of date of commissioning

Anywhere in the country

100, 700 and 1000 when pyritic sulphur in the limestone is less than 0.25%, 0.25 to 0.5% and more than 0.5% respectively.

Oxides of Nitrogen (NO_x) in mg/ Nm³

After the date of notification (25.8.2014)

Anywhere in the country

(1) 600

Before the date of notification (25.8.2014)

Anywhere in the country

(2) 800 for rotary kiln with In

Line Calciner (ILC) technology.

(3) 1000 for rotary kiln using mixed stream of ILC, Separate

Line Calciner (SLC) and suspension preheater technology or SLC

technology alone or without calciner.

(i)

The timeline for implementation of emission standards for all the parameters i.e. Sulphur Dioxide (SO₂), Oxides of Nitrogen (NO_x) and Particulate Matter (PM), with respect to Rotary Kiln without co-processing shall be up to the 31st March, 2017.

(ii)

The emission standards for Sulphur Dioxide (SO₂) shall be reviewed after a period of five years from the date of notification of these rules.

(iii)

The word 'NO₂' shall be substituted by 'NO_x' wherever it occurs in the notification vide G.S.R. 612(E) dated 25th August, 2014.]

[S. No.

Industry

Parameter

Standards

(1)

(2)

(3)

(4)

10A.

Cement Plant with coprocessing of wastes

A- Emission Standards

Rotary Kiln - with co-processing of Wastes

Date of Commissioning

Location

Concentration not to exceed, in mg/ Nm³

(a)

(b)

(c)

Particulate Matter (PM)*

on or after the date of notification (25.8.2014)

anywhere in the country

30

before the date of notification (25.8.2014)

critically polluted area or urban centres with population

above 1.0 lakh or within its periphery of 5.0 kilometer radius

30

other than critically polluted area or urban centres

30

SO₂*

irrespective of date of commissioning

anywhere in the country

100, 700 and 1000 when pyritic sulphur in the limestone is

less than 0.25%, 0.25 to 0.5% and more than 0.5% respectively.

NO_x*

After the date of notification (25.8.2014)

anywhere in the country

(1) 600

Before the date of notification

anywhere in the country (25.8.2014)

(2) 800 for rotary kiln with In

Line Calciner (ILC) technology.

(3) 1000 for rotary kiln using mixed stream of ILC, Separate

Line Calciner (SLC) and suspension pre-heater technology or SLC

technology alone or without calciner.

HCl

10 mg/ Nm³

HF

1 mg/Nm³

TOC

10 mg/ Nm³**

Hg and its compounds

0.05 mg/ Nm³

Cd + Tl and their compounds

0.05 mg/ Nm³

Sb + As + Pb + Co + Cr + Cu + Mn + Ni + V and

their compounds

0.5 mg/ Nm³

Dioxins and Furans

0.1 ngTEQ/ Nm³

Note: The abbreviations used

in the Table shall mean as under: SO₂- Sulphur

dioxide; NO_x- Oxides of Nitrogen; HCl - Hydrogen

Chloride; HF - Hydrogen Fluoride; TOC - Total Organic Carbon; Hg

- Mercury; Cd - Cadmium; Tl - Thallium; Sb - Antimony; As - Arsenic; Pb - Lead; Co - Cobalt; Cr - Chromium; Cu - Copper; Mn - Manganese; Ni - Nickel; and V - Vanadium.

* The concentration values and timeline for implementation in respect of PM, SO₂ and NO_x shall be governed in accordance with the provisions under notification published vide GSR No. 612 (E), dated the 25th August, 2014 and amended from time to time.

** Permitting authority may prescribe separate standards on case to case basis, if Total Organic Carbon (TOC) does not result from the co-processing of waste.

The height of each individual stack connected to Kiln, Clinker Cooler, Cement Mill, Coal Mill, Raw Mill, Packaging section, etc. shall be of a minimum of 30 metres or, as per the formula $H = 14 (Q_1)^{0.3}$ and $H = 74 (Q_2)^{0.27}$ whichever is more, where ?H?

is the height of stack in metres and ?Q₁? is the maximum quantity of SO₂ expected to be emitted in kg/hr and ?Q₂? is the maximum quantity of PM expected to be emitted in tonnes/hr through the stack at 100 percent rated capacity of the plant;

The monitored values of SO₂, NO_x, HCl, HF, TOC, Metals and Dioxins and Furans at main kiln stack shall be corrected to 10% Oxygen, on dry basis and the norms for SO₂, NO_x, HCl, HF, TOC, Metals and Dioxins and Furans shall be applicable to main kiln stack and the norms for Particulate Matter (PM) shall be applicable to all the stacks in the plant. PM, SO₂, NO_x shall be monitored continuously. HCl, HF, TOC, Metals and Dioxins and Furans shall be monitored once in a year; Scrubber meant for scrubbing emissions shall not be used as quencher and plants having separate stack for gaseous emission for the scrubbing unit, the height of this stack shall be at least equal to the main stack.

B - Service waste

water (with co-processing of wastes)

All efforts shall be made by the industry for 'zero discharge' of service wastewater and in case, the industry prefers to discharge service wastewater, the following norms shall be complied with:

Concentration not to exceed, milligram per litre (except pH and temperature)

pH

5.5 to 9.0

Suspended Solids

100

Oil and Grease

10

Temperature

not more than 5°C higher than the intake

water temperature

C - Storm water

Storm-water shall not be

allowed to mix with effluent, treated sewage, scrubber water and
or or floor washings.

Storm-water within battery limits of industry shall be
channelized through separate drain(s).]

[Inserted by Notification No. G.S.R. 497(E), dated 10.5.2016 (w.e.f. 19.11.1986).]

11. Stone crushing

unit]

Suspended

particulate matter

The

suspended particulate matter measured between 3 metres and 10
metres from any process equipment of a stone crushing unit shall
not exceed 600 micro-grammes per cubic metre.

[* * *]

13. [

[Substituted by Notification No. G.S.R. 221 (E) dated 18.3.2011 (w.e.f. 19.11.1986)]

Rubber

Processing and Rubber Product Industry

A.

Effluent Standards (i) Natural Rubber Processing : Centrifuging
and Creaming Units

Limiting

value for concentration in mg/l, except for pH

Inland Surface

Water

Land for

Irrigation Public Sewar

pH

6.0-8.5

6.0-8.5

Suspended Solids

100

200

BOD, 3 days at

27°C

30

100

COD

250

-

Oil and Grease

10

10

Total Kjeldahi

Nitrogen, as N

100

*

Free Ammonia

5

*

Ammonical

Nitrogen, as N

50

*

Sulphides, as S

2

*

Total Dissolved

Solids

2100

2100

(ii)

Natural Rubber Processing : Craps and Crumb Units

pH

6.0-8.5

6.0-8.5

Suspended Solids

100

*

Colour

Colourless

*

Odour

Absent

*

BOD, 3 days at

27°C

30

100

COD

250

*

Oil and Grease

10

10

Total Kjeldahi

Nitrogen, as N

50

*

Ammonical

Nitrogen, as N

25

*

Sulphides, as S

2

*

Total Dissolved

Solids

2100

2100

(iii)

Rubber Products (Moulded, Extruded or
Calendared/Fabricated/Rubber Reclamation Unit Latex based Unit)

pH

6.2-8.5

6.0-8.5

Suspended Solids

50

100

Oil and Grease

10

10

BOD, 3 days at

27°C

50

*

Lead*

0.1

*

Zinc as Zn*

5

*

Total Chromium

0-05

*

(iv)

Tyre and Tube Industry

pH

6.0-8.5

6.0-8.5

Suspended Solids

50

*

Oil and Grease

10

10

(v)

Synthetic Rubber Industry

pH

6.0-8.5

6.0-8.5

Colour

Absent

*

Odour

Absent

*

BOD, 3 days at

27°C

50

*

COD

250

*

Oil and Grease

10

10

*Norms for these

parameters shall be prescribed by the concerned State Pollution Control Board/Pollution Control Committee on case basis.

B. Emission

Standards* (Rubber Product Industry i.e. Moulded, Extruded or Calendered/Fabricated/Rubber Reclamation Unit/Latex based Unit)

Concentration not

to exceed in mg/Nm³

Particulate Matter

150

Volatile Organic

Compounds

50

*These emission

standards shall not be applicable to SSI units.

NOTE :

All

rubber units shall channelize their fugitive emission through a stack having height of 12 metres or 2 metres above roof top of shed/building whichever is more.]

14. Small Pulp and

Paper Industry

Concentration not

to exceed milligramme per litre (except for pH and sodium absorption ratio)

*Discharge into

inland surface water Disposal on land

pH

5.5-9.0

Suspended solids

100

BOD

30

pH

5.5-9.0

Suspended solids

100

BOD

100

Sodium Absorption

Ratio

26

[Absorbable

Organic Halogens (AOX) in effluent discharge

[Inserted by Notification No. G.S.R. 546(E), dated 30.8.2005 (w.e.f. 19.11.1986).]

3.00

kg/ton of paper produced with effect from the date of publication
of this notification 2.00 kg/ton of paper produced with effect
from the 1st day of March, 2006;]

Explanation. - These standards shall apply to all small scale Pulp and Paper Mills having capacity below 24,000 MT per annum.

15. Fermentation Industry (Distilleries, Maltries and Breweries)

Concentration in the effluent not to exceed milligramme per litre (except for pH and colour and odour)

pH

5.5 - 9.0

Colour and odour

[All efforts should be made to remove colour and unpleasant odour as far as practicable.

Suspended solids

100

[BOD (3 days at 27

[Substituted by G.S.R. 176(E), dated 2-4-1996 (w.e.f. 3-4-1996).]

°C)]

- Disposal into inland surface water/river/streams

30mg/l

- Disposal on land or for irrigation

100 mg/l.]

Note (1). - Waste-water generation shall not exceed 250 cubic metre per tonne of paper produced.

[***]

[Notes 2 to 7 and entries relating thereto omitted by G.S.R. 176(E), dated 2-4-1996 (w.e.f. 3-4-1996).]

16. Leather

Tanneries

Concentration in

the effluent not to exceed milligramme per litre (except for pH
and per cent sodium)

Inland

Surface Waters

Public

Sewers

Land

for Irrigation

Marine

Coastal Areas

(a)

(b)

(c)

(d)

Suspended Solids

100

600

200

100

[BOD (3 days at

27°C)]

30

350

100

100

pH

6.0-9.0

6.0-9.0

6.0-9.0

6.0-9.0

Chlorides (as Cl)

1000

1000

600

-

Hexavalent

0.1

0.2

0.1

1.0

Chromium (Cr+6)

2.0

2.0

2.0

2.0

Total Chromium (as

Cr)

Sulphides (as S)

2.0

5.0

-

5.0

Sodium per cent

-

60

60

-

Boron (as B)

2.0

2.0

2.0

-

Oil and grease

10

20

10

20

17. [

[Substituted by Notification No. G.S.R. 1607(E), dated 29.12.2017 (w.e.f. 19.11.1986).]

Fertilizer Industry

A .- Effluent Standards

(i) Straight Nitrogenous Fertilizer

Plant/Ammonia (Urea Plant), Calcium Ammonium Nitrate and Ammonium

Nitrate Fertilizers

Limiting concentration not to exceed in
milligram/litre (mg/l), except for pH
pH

6.5 to 8.5

Suspended Solids

100

Oil and Grease

10

Ammonical Nitrogen as N

50

Total Kjeldhal Nitrogen (TKN) as N

75

Free Ammonical Nitrogen as N

2.0

CN concentration

0.1

Nitrate Nitrogen as N

Urea Plant

10

Other than Urea Plant

20

(ii) Straight Phosphatic Fertilizer Plant

pH

6.5 to 8.5

Suspended Solids

100

Oil and Grease

10

Fluoride

10

Dissolved Phosphate as P

5.0

(iii) Complex Fertilizer Plant and / or

NP/NPK(N-Nitrogen, P-Phosphorus and K-Potassium)

pH

6.5 to 8.5

Suspended Solids

100

Oil and Grease

10

Ammonical Nitrogen as N

50

Total Kjeldhal Nitrogen (TKN) as N

75

Free Ammoniacal Nitrogen as N

4.0

Total Kjeldhal Nitrogen (TKN) as N

75

Free Ammoniacal Nitrogen as N

4.0

Nitrate Nitrogen as N

20

Dissolved Phosphate as P

5.0

Fluoride as F-

10

Note:(i)

Chromium salt shall not be used in cooling tower as algacide.

(ii) The effluent shall be analysed for Vanadium and Arsenic once in a year and analysis report shall be submitted to the concerned State Pollution Control Board / Pollution Control Committee.

B .- Emission Standards

(i) Straight Nitrogenous

(a) Ammonia Plant- Reformer

Oxides of Nitrogen (as NO₂)

400 mg/Nm³

(b) Urea Plant ? Prilling Tower

Particulate Matter

Pre 1982 units

150 mg/Nm³

Post 1982 units

50 mg/Nm³**

(ii) Ammonium Nitrate/ Calcium Ammonium

Nitrate/NPK plant

Particulate Matter

Existing Plant

150 mg/Nm³

New Plant

100 mg/ Nm³

Ammonium as NH₃

Existing Plant

300 mg/Nm³

New Plant

150 mg/Nm³

Total Fluoride as F-

10 mg/Nm³(only NPK Plant)

(iii) Phosphatic Fertilizer Plants ?

Phosphoric Acid Plants/ Rock grinding and Acidulation SSP Plants

Particulate Matter

125 mg/Nm³

Total Fluoride as F-

20 mg/Nm³

(iv) Nitric Acid Plant

Oxides of Nitrogen (as NO₂)

400 mg/Nm³

*Values to be

reported at 3% O₂

** Total emission of

0.5 kg/ tonne of product.

Note:(i)

Fluoride norms shall be applicable only for NPK plant.

(ii) Plant

commissioned on or after the date of notification, shall be treated as 'New Plant'.

(iii) The height of

the stack emitting Sulphur Dioxide, Oxides of Nitrogen or Oxides of Phosphorus or acid mist shall be a minimum of 30 metres or as per the formula $H=14 (Q)^{0.3}$, whichever is more, where H is the height of stack in metres and Q is the maximum quantity of SO_2 NO_x or P_2O_5 equivalent expected to be emitted in kg/hr through the stack at 100 per cent rated capacity of the tail gas plant(s) and calculated as per the norms of gaseous emission.

(iv) Tail Gas plants

having more than one stream or unit of Sulphuric Acid, Nitric Acid or Phosphoric Acid at one location, the combined capacity of all the streams or units for a particular acid shall be taken into consideration for determining the stack height and applicability of emission standards individually.

(iii) Tail gas plants having separate stack for gaseous emission for the scrubbing unit, the height of this stack shall be equal to main stack or 30 metres, whichever is higher.]

18. Iron Ore

Mining and Ore Processing

A.

Emission Standards for Stack for De-dusting Unit

Particulate
matter

100mg/Nm³

Stack
height**

15.0 m

**

Stack height for De-dusting unit shall be calculated as $H=74 Q^{0.27}$, where H and Q are stack height in metre and particulate matter (PM) emission in tonner/hr respectively, i.e.

Q

(kg/hr) Up to 2.712.72 ? 7.867.87 ?

17.9617.97 ? 35.29

H (metre) 15202530

Note:-Stack

attached to De-dusting unit shall have minimum height of 15.0 metres and would be atleast 2.50 metres above the top-most point of the nearby building/shed or plant in the mine.

B.

Fugitive Emission Standards

Particulate
matter

1200

µ100mg/Nm³

Note:-Fugitive emission shall be monitored in the predominant downwind direction at a distance 0 25.0+2.0 metres from the source of

fugitive emission as per following :

Area

Monitoring

location

Mine face/Benches

Drilling

excavation and loading applicable for operation benches above waterable

Haul Roads/Service

Roads

Haul roads to ore

processing plant, waste dumps and loading areas and service road

Crushing plant

Run-off mine

unloading at hopper, crushing areas, screens and transfer points.

Screening Plant

Screens, conveying

and transportation of ore discharge points

Ore Storage and

Loading

Intermediate stock

bin/pile areas ore stock bin/pile areas , wagon/truck loading areas

Waste dump

Active

waste/reject dumps

C.

Effluent Standards

pH

5.5-9.0

Suspended

solids (non-rainy day)

50

mg/l

Suspended

solids (rainy day)

100

mg/l

Oil

and grease

10

mg/l

Note:-(I)

All efforts shall be made to reuse and re-circulate the treated effluent.(ii) The aforesaid effluent standards shall be complied with for sewage, service water, beneficiation of ore washwater and surface run-off put together

19. Calcium

Carbide

Particulate Matter

Emission :

-Kiln

250 milligramme

per normal cubic metre

-Arc Furnace

150 milligramme

per normal cubic metre

20. Carbon Black

Particulate Matter

Emission :

150 milligramme

per normal cubic metre

21. [] Copper, Lead

or Zinc Smelting Plant

Emission standards

a. Concentrator

Existing Unit

New Unit

Particulate Matter

(mg/Nm³)

100

75

b. Sulphur-Dioxide

Recovery Unit Limiting Concentration in mg/Nm³ Plant capacity

for 100% convertible concentration of Sulphuric Acid (time/day)

Existing Unit

New Unit

Sulphur-Dioxide

(So₂)

Upto

300

1370

1250

Above

300

1250

950

Acid Mist/Sulphur

Trioxide

Upto

300

90

70

Above

300

70

50

Note

:-1. Capacity in above stipulation means the installed

capacity of Sulphuric Acid Plant.2. Scrubbing units shall

have on-line pH meters with auto recording facility.3. Plant

commissioned on or after the date of notification, shall be

termed as 'New Unit.' 4. The height of the Stack emitting Sulphur Dioxide or acid mist shall be a minimum of 30 meters or as per the formula $H = 14(Q)^{0.3}$ (whichever is more), where 'H' is the height of stack in metres; and 'Q' is the maximum quantity of SO₂ in kg/hr, expected to be emitted through the stack at 110 per cent rated capacity of the Tail Gas plant(s) and calculated as per the norms of gaseous emission. 5. Tail Gas plants having more than one stream or unit of sulphuric acid at one location, the combined capacity of all the streams or units shall be taken into consideration for determining the stack height and applicability of emission standards. 6. Tail Gas plants having separate stack for gaseous emission for the scrubbing unit, the height of this stack shall be equal to main stack or 30 metres, whichever is higher.

22. Nitric Acid

(emission oxides of nitrogen)

Emission of Oxides

of Nitrogen

3 kilogramme of

oxides of nitrogen per tone of weak acid (before concentration)

produced

23. [Sulphuric

Acid Plant]

Emission standards

Limiting

concentration in mg/Nm³, unless stated

Plant capacity for

100% concentration of Sulphuric

Existing

Unit

New

Unit

Acid (tonne/day)

Sulphur dioxide

(SO₂)

up to 300

1370

1250

Above 300

1250

950

Acid Mist/Sulphur

Up to 300

90

70

Trioxide

Above 300

70

50

Note. - (i)

Scrubbing units shall have on-line pH meter with auto recording

facility.

(ii) The height of the stack emitting sulphur-dioxide or acid mist shall be of minimum of 30 metre or as per the formula $H = 14 (Q)^{0.3}$ (whichever is more).

Where "H"

is the Height of stack in metre; and "Q" is the maximum quantity of SO_2 expected to be emitted through the stack at 110 per cent rated capacity of the plants and calculated as per the norms of gaseous emission.

(iii) Plants

having more than one stream or unit of sulphuric acid at one location, the combined capacity of all the streams and units shall be taken into consideration for determining the stack height and applicability of emission standards.

(iv) Plants having separate stack for gaseous emission for the scrubbing unit, the height of this stack shall be equal to main stack.]

S.No.

Industry

Parameter

Standard

24.

Integrated Iron

and Steel Plant

A-Coke oven

(by-product type)

a. Effluent

Standards

Limiting

concentration in mg/l, except for pH

pH

6.0-8.50

Suspended solids

100

BOD, 3 days at

27°C

30

COD

250

Oil and grease

10

Ammonical nitrogen

as N

50

Cyanide (as CN^-)

0.2

Phenol

1.0

b. Emission

Standards

New Batteries (at

green field site)
Rebuild Batteries
Existing Batteries
(i) Fugitive
Visible Emission
Leakage from door
5(PLD)*
10(PLD)*
10(PLD)*
Leakage from
charging lids
1(PLL)**
1(PLL)**
1(PLL)**
Leakage from AP
Covers
4(PLO)+
4(PLO)+
4(PLO)+
Charging emission
(Second/charge)
16(with HPLA)
50(with HPLA)
75
*PLD- Percent
leaking doors; **PLL-Percent leaking lids;
+PLO- Percent
Leaking off takes and HPLA- Aspiration through high pressure
liquor injection in gooseneck.
(ii) (Stack
Emission Standards)
SO₂ (mg/Nm³)
800
800
800
NO_x (mg/Nm³)
500
500
500
Particulate matter
(mg/Nm³)
50
50
50
Particulate matter
during charging of stamp charging batteries (mg/Nm³)
25
25
25
Sulphur in COke
Oven gas used for heating (mg/Nm³)

800

-

-

(iii) Fugitive

Emission: Benzo (a) Pyrene (BaP)

Battery area (top
of the battery)($\mu\text{g}/\text{m}^3$)

5

5

5

Other units in

Coke oven plant ($\mu\text{g}/\text{m}^3$)

2

2

2

B.-Sintering Plant

a. Effluent

Standards

Limiting

concentration in mg/l , except of pH

pH

6.0-8.50

Suspended solids

100

Oil and grease

10

b. Emission

Standards

Particulate matter

(mg/Nm^3)

150

C.- Blast Furnace

a. Effluent

Standards

Limiting

concentration in mg/l , except for pH

pH

6.0-8.05

Suspended solids

(mg/l)

50

Oil and grease

(mg/l)

10

Cyanide as CN

(mg/l)

0.2

Ammonical Nitrogen

50

as $\text{NH}_3\text{-N}$ (mg/l)

b. Emission

Standards

(i) Stack

Emissions

New

Units

Existing

Units

BF Stove

Particulate matter

(mg/Nm³)

50

30

SO₂ (mg/Nm³)

250

200

Nox (mg/Nm³)

150

150

CO (Vol/Vol)

1%

(max)

1%

(max)

(ii) Space

Deducting/Other Stacks of BF area

Particulate matter

(mg/Nm³)

100

50

(iii) Fugitive

Emission

Existing

Units

New

Units

Particulate matter

(Size less than 10 microns) PM₁₀ (µg/m³)

4000

3000

SO₂ (µg/m³)

200

150

NO_x (µg/m³)

150

120

Carbon monoxide

(µg/m³)- 8 hours

5000

5000

1hours

10,000

10,000

Lead, as Pb in

fugitive dust ($\mu\text{g}/\text{m}^3$) at Cast House

2

2

D.- Steel Making

Shop-Basic Oxygen Furnace

a. Effluent

Standards

pH (mg/l)

6.0-8.5

Suspended solids

(mg/l)

100

Oil and grease

(mg/l)

10

(i) Stack Emission

Existing Units

New Units

Converters

Particulate matter

(mg/Nm³)

- Blowing/Lancing

operation

300

Should be with gas

recovery

- Normal operation

150

Should be with gas

recovery

**Secondary

Emission Stack : De-dusting of de-sulphurisation, Secondary

refining etc.

Particulate matter

(mg/Nm³)

100

50

(ii) Fugitive

Emission

Existing

Units

New

Units

Particulate matter

(size less than 10 microns) PM₁₀ ($\mu\text{g}/\text{m}^3$)

4000

3000

SO₂ ($\mu\text{g}/\text{m}^3$)

200

150

NO_x (µg/m³)

- 8 hours

5,000

5,000

1 hours

10,000

10,000

Lead, as Pb in

dust at Converter floor (µg/m³)

2

2

E- Rolling Mills

a. Effluent

Standards

pH

6.0-9.0

Suspended solids

(mg/l)

100

Oil and grease

(mg/l)

10

b. Emission

Standards

Particulate matter

(mg/Nm³)

150

Re-Heating

Sensitive

area

Other

area

Particulate matter

(mg/Nm³)

150

250

F.- Arc Furnaces

Emission Standards

Particulate matter

(mg/Nm³)

150

G.- Induction

Furnaces

Emission Standards

Particulate matter

(mg/Nm³)

150

H.- Cupola

Foundry

Emission Standards

melting capacity
less than 3 tonne/hr
melting capacity 3
tonne/hr and above

Particulate matter
(mg/Nm³)

450

150

SO₂ (mg/Nm³)

300, corrected at

12% CO₂

I.- Calcination

Plant/Lime Kiln/Dolomite Kiln

Emission Standards

capacity upto

40t/day

capacity above

40t/day

Particulate matter

(mg/Nm³)

500

150

J.- Refractory

Unit

Emission Standards

Particulate matter

(mg/Nm³)

150

Note :1.

The height of the each process stack shall be a minimum of 30 metres or as per the formula $H = 14(W)^{0.3}$ (Whichever is more), where "H" is the height of stack in metre; and "Q"

is the maximum quantity of SO₂ in kg/hr expected to be emitted through the stack at rated capacity of the plant(s) and

calculated as per the norms of gaseous emission.2. The Plants

having separate stack for gaseous emission for the scrubbing

unit, the height of this stack shall be equal to main stack of

the plant or 30 metres, whichever is higher. 3. It is essential

that stack constructed over the cupola beyond the charging door

and emissions shall be directed through the stack which should be

at least six times the diameter of cupola.4. In respect of

Arc Furnaces and induction Furnaces provision shall be made for

collecting the fumes before discharging the emissions through the

stack.5. Foundries shall install scrubber, followed by a

stack of height atleast six times the diameter of the Cupola

beyond the charging door.6.Recovery type converters shall be

installed in new plants or expansion projects.

Note:(i)

Stormwater shall not be allowed to mix with effluent, scrubber water and/or floor washings.(ii) Stormwater shall be

channellized through separate drains as per natural gradient,

passing through High Density Polyethylene (HDPE) lined pits, each having holding capacity of 10 minutes (hourly average) of rainfall.

25. [Thermal Power Plants]

Particulate Matter

Emissions :

-generation

capacity 210 MW or more

150 milligramme

per normal cubic meter

-generation

capacity less than 210 MW

350 milligramme

per normal cubic meter

*Depending upon the requirement of local situation, such as protected area, the State Pollution Control Boards and other implementing agencies under the Environment (Protection) Act, 1986, may prescribe a limit of 150 milligramme per normal cubic metre, irrespective of generation capacity of the plant.

26.[***]

[Deleted by Notification No. G.S.R. 221 (E) dated 18.3.2011]

[***]

[Deleted by Notification No. G.S.R. 221 (E) dated 18.3.2011]

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[illegible]

Emissions

[-Pure Asbestos material

[Substituted by Notification No. G.S.R. 46(E) dated 3.2.2006 (w.e.f. 19.11.1986)]

0.5 fibre */cc for one year from the date of notification

0.2 fibre */cc after one year from the date of notification]

-Total dust

4 Fibre */cc

2 mg/m³(normal)

28. Chlor Alkali (Caustic soda)

Emissions

Concentrations in mg/m³(normal)

(a) mercury Cell

Mercury (from hydrogen gas holder stack)

0.2

(b) All processes

Chlorine (from hypo tower)

15.0

(c) All processes

Hydrochloric acid vapours and mist

(from hydrochloric acid plant)

35.0

29. Large pulp and paper

Emissions

Concentrations in mg/m³(normal)

Particulate matter

250**

H₂S

10

*Fibre of length more than 5 micrometre and diameter less than 3 micrometre with an aspect ratio of 3 or more.

**This standard of 250 mg/m³ (normal) shall apply only for a period of 3 years with effect from the date on which the Environment (Protection) Second Amendment Rules, 1989, came into force. After three years the standard to be applicable is 15 Mg/m³ (normal).

30. Integrated Iron and Steel Plants:

Emissions

(a) Coke oven

Particulate matter

50

(b) Refractory material plant

Particulate matter

150

II. Effluents

Concentration in mg/litre (except for pH)

(a) Coke oven By-product plant :

pH

6.0-8.5

Suspended solids

100

Phenol

1.0

Cyanide

0.2

[BOD (3 days at 27°C)]

30

COD

250

Ammonical nitrogen

50

Oil Grease

10

(b) Other plants such as sintering plant, blast furnace, steel melting and rolling mill :

pH

6.0-9.0

Suspended solids

100

Oil Grease

10

31. Re-heating (Reverberatory)

Emissions

Concentrations in mg/m³(normal)

Furnaces:

Capacity: All sizes

150

Sensitive area

Particulate matter

Other area

Particulate matter

450]

32. [Foundries]

[Inserted by G.S.R. 742(E), dated 30-8-1990 (w.e.f. 30-8-1990).]

Emissions

(a) Cupola

Capacity (Melting rate):

Less than 3 Mt/hr.

Particulate matter

450

3 MT/hr. and above

Particulate matter

150

Note. :- It is essential that stack is constructed over the cupola beyond the Charging door and emissions are directed through the stack which should be at least six times the diameter of cupola.

(b) Arc Furnaces

Capacity: All sizes

Particulate matter

150

(c) Induction Furnaces

Capacity: All sizes

Particulate matter

150

Note. - In respect of arc furnaces and induction furnaces, provision has to be made for collecting the fumes before discharging the emissions through the stack.

33. Thermal Power

Plants

Stack

Height /Limits

Power generation

capacity:

- 500 MW and

above

275

- 200 MW/210 MW

and above to less than 500 MW

220

- Less than 200

MW /210 MW

$H=14(Q)^{0.3}$ where Q is emission rate of SO₂ in kg/hr. and H Stack height in meters.

Stream generation

capacity:

- Less than 2

ton/hr.

2-1/2 times the

neighboring building height or 9 metres (whichever is more).

- More than 2

ton/hr. to 5 ton/hr.

12

- More than 5

ton/hr. to 10 ton/hr.

15

- More than 10

ton/hr.

18

- More than 15

ton/hr. to 20 ton/hr.

15

- More than 20

ton/hr. to 25 ton/hr.

24

- More than 25

ton/hr. to 30 ton/hr.

27

- More than 30

ton/hr.

30 or using

formula $H=14(Q)^{0.3}$ (whichever is more) where Q is emission rate of SO₂ in kg/hr. and H is Stack height in metres.

34. Small Boilers

Emissions*

Capacity of Boiler

Particulate matter

-less than 2

ton/hr.

1600

-2 to 15 ton/hr.

1200

-More than 15

ton/hr.

150

*All emissions

normalized to 12 per cent carbon dioxide.

35. [Coffee

Industry]

Instant/Dry

Processing

Limiting value for

concentration in mg/1 except for pH

pH

6.5-8.5

BOD3days, 27°C

100

Total Dissolved

Solids

2100

Wet/Parchment

Coffee Processing

pH

6.5-8.5

BOD3days, 27°C

1000

Notes. -

(i) Coffee growers

having plantation area less than 10 ha with wet processing shall store primary treated effluent in lined lagoons for solar evaporation with a non-permeable system at the base and sides of lagoon.

(ii) Coffee

growers having plantation are between 10-25 ha with wet processing shall store primary (equalisation and neutralisation) treated effluent in lined lagoons for solar evaporation with a non-permeable system at the base and sides of lagoon.

(iii) Coffee

growers having plantation area 25 ha or above with wet processing shall store secondary treated effluent in conformity with above norms in lined lagoons with a non-permeable lining system at the base and sides of lagoon and use the effluent for irrigation after dilution so as BOD of diluted effluent for land application is less than 100 mg/1.

(iv) The minimum

liner specifications for a non-permeable lining system shall be a composite barrier having 1.5 mm High Density Polyethylene (HDPE) geomembrane or equivalent, overlying 90 cm of soil (clay or amended soil) having permeability coefficient not more than 1×10^{-5} cm/sec.

(v) The effluent

storage facilities/lagoons/solar evaporation ponds shall be located above high flood level mark of the nearby stream, revulet, etc., with below mentioned free board and away from any water body/stream at a distance.

Grower ?

Small

Medium

(10 ha)

(10 ? 25 ha)

Free Board (cm) ?

30

60

Distance (m) ?

50

100

(vi) Raw, Treated

and/or diluted effluent shall not be discharged into surface water body or used for recharging ground water under any circumstances what

36. Aluminium

Plants

Emissions

(a) Alumina Plant

:

(i) Raw Material

Handling

Primary and

Secondary Crusher

150

Particulate matter

(ii) Precipitation

Area

Particulate Matter

250

-Calcinations

Carbon Monoxide

1% max.

Stack height

$H = 14 (Q)^{0.3}$ where Q is emission rate of SO₂ in kg/hr. and H is Stack height in metres

(b) Smelter Plant

(i) Green Anode

Shop

Particulate matter

150

(ii) [Anode Bake

Oven

[Substituted by Notification No. G.S.R. 46(E) dated 3.2.2006 (w.e.f. 19.11.1986)]

-do-

50 mg/Nm³

(iii) Pot room

Total Fluoride

2.8 kg/ton by 31st December 2006

For Soderberg*

Technology

For Pre-baked

Technology

0.8 kg/ton by 31st December 2006

*Separate

standards for VSS, HSS, PBSW and PBCW as given in column 4 stands abolished.]

(c) Standards for forage fluoride -

-Twelve

consecutive months average

40 ppm

-Two consecutive

months average

60 ppm

-One month average

80 ppm]

*37. Stone

Crushing Unit

Suspended

particulate matter (SPM)

The Standards

consist of two parts:

(i) Implementation

of the following Pollution Control measures:

(a) Dust

containment cum suppression system for the equipment.

(b) Construction

of wind breaking walls.

(c) Construction

of the metalled roads within the premises.

(d) Regular

cleaning and wetting of the ground within the premises.

(e) Growing of a

green belong along the periphery.

(ii) Quantitative

standard for the SPM:

The

Suspended Particulate Matter contribution value at a distance of

40 metres from a controlled isolated as well as from a unit

located in a cluster should be less than 600 mg/Nm³.

The measurements are to be conducted at least twice a month for

all the 12 months in a year.

38. Petrochemicals

(Basic intermediates)

[A. Effluents]

PH

6.5-8.5

[*BOD (3days at

27°C)]

50

**Phenol

5

Sulphide (as S)

2

COD

250

Cyanide (as CN)

0.2

***Fluoride (as F)

15

Total suspended

solids

1000

Hexavalent

0.1

Chromium (as

CR+6)]

****Total Chromium

(as CR)

2.0

B.

Emission from Chimney/Stack

Limiting

concentration in mg/Nm³, unless stated

“(Furnace,

Boiler, Heater, Vaporiser)

Fuel

Type

Existing

plants

New

Plant/Expansion of Existing Plant

Sulphur Dioxide

(SO₂)

Gas

50

50

Liquid

1700

850

Oxides of Nitrogen

(NO_x)

Gas

350

250

Liquid

450

350

Particulate Matter

(PM)

Gas

10

05

Liquid

100

50

Carbon Monoxide

(CO)

Gas

150

100

Liquid

200

150

Note,-

(i)

All values shall be corrected to 3% Oxygen.

(ii)

Wet scrubber shall necessarily be operated at the time of decoking.

(iii)

Norms for CO shall be monitored only in case of Phthalic Anhydride (PA), Maleic Anhydride (MA), Terephthalic Acid (PTA) and Dimethyl Terephthalate (DMT) Plants. Norms for CO emissions shall not be applicable to PA/MA manufacturing standalone existing plants with an installed capacity of less than 30,000 metric tonnes per annum, provided that such units have a chimney/stack of minimum 30 metres height for emitting Carbon Monoxide.

Process

Emission (Specific Pollutant)

Source

Limiting

concentration in mg/Nm³

Existing

Plants

New

Plants

Chlorine

EDC/VCM Plant and

Incinerator

10

10

Hydrochloric Acid

Mist

EDC/VCM Plant and

Incinerator

30

30

Ammonia

Wastewater
stripper, acrylonitrile plant, carpolacturm plant
75
75
Hydrogen Sulphide
Naphtha
pre-treatment plant, olefin plant
05
05
Phosgene
(TDI) and (MDI)
plant
01
01
Hydrogen
Cyanide(HCN)
Acrylonitrile
plant
10
10
VOC (HAPs)-TDI and
MDI
TDI,
Methylenediphenyl Di-isocyanate (MDI) Plants
0.1
0.1
VOC (HAPs),
Benzene and Butadiene
Benzene, Butadiene
Plants
5.0
5.0
VOC
(HAPs), EO, VCM, EDC, ACN and PO
EO, VCM, EDC, ACN,
PO Plants
20.0
10.0
Organic
Particulate
PA, MA and TDI
Plants
50
25
Process
Emission (General Pollutant)
Source
Limiting
concentration in mg/Nm³
VOC (MA, PA and
Phenol)

MA, PA, Phenol

Plants

20

VOC (EB, Styrene,

Toluene, Xylene, Aromatics, EG and PG)

Ethyl benzene

(EB), Styrene, Toluene, Xylene, Aromatics, EG, PG Plants

10

VOC (Paraffin,

Acetone and Olefins)

Non-methane, HC

(paraffin), Acetone, Olefins Plants

150

Note. - In respect of are furnaces and induction furnaces, provision has to be made for collecting the fumes before discharging the emissions through the stack.

*State Boards may prescribe the BOD value of 30 mg/l if the recipient system so demands.

**The limit for phenol shall be conformed to at the outlet of effluent treatment of phenol plant. However, at the final disposal point, the limit shall be less than 1 mg/l.

***The limit for fluoride shall be confirmed to at the outlet of fluoride removal unit. However, at the disposal point fluoride concentration shall be lower than 5 mg/l.

****The limits for total and hexavalent chromium shall be conformed to at the outlet of the chromate removal unit. This implies that in the final treated effluent, total and hexavalent chromium shall be lower than prescribed therein.

[NOTE.- HAP - Hazardous Air Pollutants are those pollutants that cause cancer or other serious health effects, or adverse environmental and ecological effects.

C. Standards for Fugitive Emission

Storage of Volatile Liquids: General Petrochemical/ Petroleum Products.

(1)

Storage tanks with capacity between 4 to 75m³ and total vapour pressure (TVP) of more than 10 kpa should have fixed roof with pressure value vent.

(2)

Storage tanks with capacity between 75 to 500 m³ and total vapour pressure (TVP) of 10 to 76 kpa should have internal floating roof or external floating roof or fixed roof with vapour control or vapour balancing system.

(3)

Storage tanks with the capacity of more than 500 m³ and total vapour pressure (TVP) of 10 to 76 kpa should have internal floating roof or external floating roof or fixed roof with vapour control system.

(4)

The tanks with the capacity of more than 75 m³ and total vapour pressure (TVP) of more than 76 kpa should have fixed roof with vapour control system.

(5)

Requirement for seals in Floating Roof Tanks-

(i)

(a)

Internal Floating Roof Tank (IFRT) and External Floating Roof Tank (EFRT) shall be provided double seals with minimum vapour recovery of 96%.

(b)

Primary seal shall be liquid or shoe mounted for EFRT and vapour mounted for IFRT. Maximum seal gap width will be 4 cm and maximum gap area will be 200 cm²/m of tank diameter.

(c)

Secondary seal shall be rim mounted. Maximum seal gap width will be 1.3 cm and maximum gap area will be 20 cm²/m of tank diameter.

(d)

Material of seal and construction shall ensure high performance and durability.

(ii)
Fixed roof tanks shall have vapour control efficiency of 95% and vapour balancing efficiency of 90%.

(iii)
(a)
inspection and maintenance of storage tanks shall be carried out under strict control;

(b)
for the inspection, API RP 575 may be adopted;

(c)
In-service inspection with regard seal gap should be carried out once in every six months and repair to be implemented in short time; and

(d)
the possibility of on-stream repair of both shall be examined.

(iv)
Storage tanks shall be painted with white colour shade, except for derogation of visually sensitive area.D. Storage of Benzene, VCM and ACN

D. Storage of Benzene, VCM and ACN

(i)
FRT with vapour for incineration with 99.9% of removal efficiency for volatile organic compounds (VOC) shall be provided; or

(ii)
IFRT/EFRT with double seals, emission-reducing roof fitting and fitted with fixed roof with vapour removal efficiency of at least 99% shall be provided; or

(iii)
Internal floating roof and nitrogen blanketing in between fixed and floating roofs shall be provided.

(Emission control
for Road tank, truck/Rail tank, wagon loading)

Loading of
Volatile Products

Naphtha:(i)
VOC reduction,%or(ii) Emission, gm/m3

(i) _
99.5or(ii) _ 5

Benzene and
Butadiene:(i) VOC reduction,%or(ii) Emission, mg/m3

(i) _
99.99or(ii) _ 20

Toluene/Xylene:(VOC)
reduction,%or(ii) Emission, mg/m3

(i) _
99.98or(ii) _ 150

] [Inserted by Notification No. G.S.R. 820 (E) dated 9.11.2012 (w.e.f. 19.11.1986)]

[***]
[Serial number 39 omitted by G.S.R. 512(E), dated 9-7-2009 (w.e.f. 9-7-2009).]

[
Sr.No.
Industry
Parameter
Standards

1

2

3

4

39

Hotel

Industry

Effluent

Standards

(i) Hotel with atleast 20 bedrooms

limiting concentration in mg/l, except for pH

Inland Surface Water

On land for Irrigation

pH

5.5-9.0

5.5-9.0

BOD3days, 270C

30

100

Total Suspended Solids

50

100

Oil Grease

10

10

Phosphate as P

1

-

(ii) Hotel with less than 20 bedrooms or a

Banquet Hall with minimum floor area of 100 m² or a Restaurant
with minimum seating capacity of 36

pH

5.5-9.0

5.5-9.0

BOD3days, 270C

100

100

Total Suspended Solids

100

100

Oil Grease

10

10

Notes:

i. Hotels, banquet

halls, restaurants, etc. located in coastal area shall also
comply with the provisions of the Coastal Regulation Zone, as
applicable

ii. If, the effluent

is discharged into a municipal sewer leading to a Sewage
Treatment Plant, the hotel or restaurant or banquet hall, as the
case may be, shall provide a proper Oil and Grease Trap for

effluent arising from its kitchen and laundry and shall have to comply with the 'General Standards for Discharge of Environmental Pollutants Part-A: Effluents' notified under Schedule-VI

]
[Inserted by Notification No. G.S.R. 794 (E) dated 4.11.2009 (w.e.f. 19.11.1986)]
Prior to its omission, serial number 39 reads as under :-

{
"39. Pharmaceutical Manufacturing and Formulation Industry
Effluents

1. pH
5.5-9.0
2. Oil and Grease
10
3. Total Suspended solids
100
4. [BOD (3 days at 27°C)]
30
5. Bio-assay test
90% survival of fish after 96 hrs. in 100% effluent
6. Mercury
0.01
7. Arsenic Chromium
0.20
8. Chromium (Hexavalent)
0.10
9. Lead
0.10
10. Cyanide
0.10
11. Phenolics (as C₆H₅OH)
1.00
12. Sulphides (as S)
2.00
13. Phosphates (as P)
5.00

Note.- (1) Parameters listed as 1 to 13 are compulsory for formulators. However, the remaining parameters (6 to 13) will be optional for others.

}
S.No.
Industry
Parameter
Standard
(1)
(2)
(3)
(4)
A.
Emission Standards
40
Pesticide Industry
Limiting

concentration in mg/Nm³

HCl

20

Cl₂

5

H₂S

5

P₂O₅ as H₃PO₄

10

NH₃

30

Pesticides

compounds in the form of particulate matter

20

CH₃Cl

20

HBr

5

B.

Effluent Standards

Limiting

concentration in mg/l, except for pH and Bioassay test

(i)

Compulsory Parameters

pH

6.5-8.5

BOD, 3 days, 27°C

Formulation unit

30

Technical grade

unit

100

Oil

and Grease

10

Suspended

Solids

100

Bioassay

Test

90

percent survival of fish after 96 hours in 100% effluent*

(ii)

Additional Parameters

Arsenic

(as As)

0.2

Copper

1.0

Manganese

1.0

Mercury

0.01

Antimony

(as Sb)

0.1

Zinc

1.0

Nickel, etc.

(heavy metals individually)

Shall not exceed

individually 5 times the drinking water standards as per Bureau
of Indian Standards

Cyanide

(as CN)

0.2

Nitrate

(as NO₃)

50

Phosphate

(as P)

5.0

Phenol

and Phenolic Compounds as C₆H₅OH

1.0

Sulphur

0.03

Benzene

Hexachloride (BHC)

0.01

Carbonyl

0.01

Copper

Oxychloride

9.6

DDT

0.01

Dimethoate

0.45

24D

0.4

Endosulfan

0.01

Fenitrothion

0.01

Malathion

0.01

Methyl

parathion

0.01

Paraquat

2.3

Phenathoate

0.01

Phorate

0.01

Proponil

7.3

Pyrethrums

0.01

Ziram

1.0

Other

Pesticide (individually)

0.10

*Bioassay Test

shall be carried out as per IS: (6582-1971). Note: 1. The concerned State Pollution Control Board/Pollution Control Committee shall prescribe limits of Total Dissolved Solids (TDS), Sulphates and Chlorides depending on the usages of recipient water body in down stream, in which effluent shall be disposed off. 2. No limit for chemical Oxygen Demand (COD) is prescribed but, COD in the treated effluent shall be monitored. If COD is persistently reported more than 250 mg/l, the industrial units discharging such an effluent shall be required to identify chemical causing the same. In case, these are found to be toxic, as defined in Schedule I of the Manufacture, Storage and Import of Hazardous Chemicals Rules, 1989, the concerned State Pollution Control Board/Pollution Control Committee in such cases shall direct the industries to install tertiary treatment system by 31st March, 2012. 3. Parameters listed as "Additional parameters" shall be prescribed depending upon the process and product, on a case basis.

C.

Emission Standards for Incinerator

Limiting

concentration in mg/Nm³, unless stated

Sampling Duration

in minutes, unless stated

Particulate

Matter

50

30 or more (for

sampling of 300 litres of emission)

HCl

50

30

SO₂

200

30

CO

100

Daily

average

Total
organic Carbon
20
30
Total Dioxins and
Furans*
Existing
Incinerator

0.2
ng TEQ//Nm³
8
hours

New Incinerator
0.1
ng TEQ/NM³
8
hours

Sb+As+Pb+Cr+Co+Cu+Mn+Ni+V
and their compounds
1.5
2
hours

*The existing

plant shall comply with norms for Dioxins and Furans as 0.1 ng TEQ/Nm³ by 18th August, 2013. Note :i. All monitored values shall be corrected to 11% oxygen on dry basis.ii. The CO₂ concentration in tail gas shall not be less than 7%. iii. In case, halogenated organic waste is less than 1% by weight in input waste, all the facilities in single chamber incinerators shall be designed so as to achieve a minimum temperature of 1100°C in the incinerator. For fluidized bed technology based incinerator, temperature shall be maintained at 950°C.iv. In case, halogenated organic waste is less than 1% by weight in input waste, waste shall be incinerated only in twin chamber incinerators and all the facilities shall be designed to achieve a minimum temperature of 850±25°C in primary chamber and 1100°C in secondary combustion chamber with a gas residence time in secondary combustion chamber not less than two seconds.v. Scrubber meant for scrubbing emissions shall not be used as quencher. vi. Incineration plants shall be operated (combustion chambers) with such temperature, retention time and turbulence, as to achieve Total Organic Carbon (TOC) content in the incineration ash and residue less than 3%, and their loss on ignition is less than 5% of the dry weight. In case of non-conformity, ash and residue as the case may be, shall be re-incinerated.vii. The incinerator shall have a chimney of atleast thirty metres height.

D.
Effluent from Incinerator

Note:(i)

Stormwater shall not be allowed to mix with scrubber water and/or

floor washings.(ii) Stormwater shall be channelized through separate drains passing through a HDPE lined pit having holding capacity of 10 minutes (hourly average) of rainfall.

E.

Stormwater

Note:(i)

Stormwater shall not be allowed to mix with scrubber water and/or floor washings.(ii) Stormwater shall be channelized through

separate drains passing through a HDPE lined pit having holding capacity of 10 minutes (hourly average) of rainfall.

Substituted by Notification No. G.S.R. 446(E) dated 13.6.2011 (w.e.f. 19.11.1986)

{|

40. Pesticide manufacturing and Formulation Industry

Effluents

1. Temperature

Shall not exceed 5°C above the receiving water temperature

2. pH

6.5-8.5

3. Oil Grease

10

4.[BOD (3 days at 27°C)]

[Substituted by G.S.R. 176(E), dated 2-4-1996 (w.e.f. 3-4-1996).]

30

5. Total suspended solids

100

6. Bio-assay test

90% survival of fish after 96% hrs. in 100% effluent

7.(a) Specific Pesticides:

Benzene hexachloride

10

Carbonyl

10

DDT

10

Endosulfan

10

Diamethoate

450

Fenitrothion

10

Malathion

10

PhorateMethyl

10

Parathion

10

Phenthoate

10

Pyrethrums

10

Copper Oxychloride

9600
Copper Sulphate
50
Ziram
1000
Sulphur
30
Paraquat
2300
Proponil
7300
Nitrogen
780
(b) Heavy Metals:
Copper
1.00
Manganese
1.00
Zinc
1.00
Mercury
0.01
Tin
0.10
Any other metal like
Shall not exceed 5
Nickel, etc.
Times the drinking water standards of BIS
(2)
State Board may prescribe limit for chemical oxygen demand (COD) correlated with BOD limit.
(3)
State Board may prescribe limit for total dissolved solids depending upon uses of recipient waterbody.
(4)
Limits should be complied with at the terminal of the treatment unit before letting out of the factory boundary limits.
(5)
For the compliance of limits, analysis should be done in the composite sample collected every hour for a period of 8 hours.
(c) Organics
Phenol and phenolic compound as C_6H_5OH
1.0
(d) Inorganics :
Arsenic (as As)
0.2
Cyanide (as CN)
0.2
Nitrate (as NO_3)
50.0
Phosphate (as P)
5.0
[Emissions]
[Inserted by G.S.R. 46(E), dated 3-2-2006 (w.e.f. 3-2-2006).]

Not to exceed mg/NM3

HCL

20

Cl₂

5

H₂S

5

P₂O₅(as H₃PO₄)

10

NH₃

30

Particulate matter with pesticides compounds

20

CH₃Cl

20

HBr

5]

Note.- 1. Limits should be complied with at the end of the treatment plant before any dilution.

2. Bio-assay test should be carried out with available species of fish in receiving water.

3. State Boards may prescribe limits of total dissolved solids (TDS) sulphates and chlorides depending on the use of recipient water body.

4. State Board may prescribe COD limit correlated with BOD limit.

5. Pesticides are known to have metabolites and isomers. If they are found in significant concentration, standards may be prescribed for those in the list by Central or State Board.

6. Industries are required to analyse pesticides in waste-water by advanced analytical method such as GLC/HPLC.

7. All the parameters will be compulsory for formulators, for others, the 7th will be optional.

l}

41. Tannery (after primary treatment)

Effluents

Disposal: Channel/Conduit

Carrying waste-waters to Secondary treatment plants

Type of Tanneries

-Chrome tanneries/combined chrome and vegetable tanneries

pH

6.5-9.0

SS

Not to exceed 600

Chromium concentration after treatment in the chrome waste-water stream

45

-Vegetable tanneries

pH

6.5-9.0

SS

Not to exceed 600

Note. - The above standards will apply to those tannery units which have made full contribution to a Common Effluent Treatment Plant(CETP) Comprising secondary treatment. Those who have not contributed will be governed by earlier Notification No. S.O. 64(E), dated January 18,1988:

42. Paint Industry (Waste-water discharge)

Effluents

pH

6.0-8.5

Suspected Solids

100

[BOD (3 days at 27°C)]

[Substituted by G.S.R. 176(E), dated 2-4-1996 (w.e.f. 3-4-1996).]

50

Phenolics as C₆H₅OH

1.0

Oil Grease

10.0

Bio-assay test

90% survival in 96 hours

Lead as Pb

0.1

Chromium as Cr.

0.1

Hexavalent

Total

2.0

Copper as Cu

2.0

Nickel as Ni

2.0

Zinc as Zn

5.0

Total heavy metals

7.0

43. Inorganic Chemical Industry

(Waste-Water discharge) Part I (metal compounds of Chromium, Manganese, Nickel, Lead and Mercury)

Effluents

pH

6.0-8.5

Chromium as Cr

Hexavalent

0.1

Total

2.0

Manganese as Mn

2.0

Nickel as Ni

2.0

Copper as Cu

2.0

Zinc as Zn

5.0

Cadmium as Cd

0.2

Lead as Pb

0.1

Mercury as Hg

0.01

Cyanide as CN

0.2

Oil Grease

10.0

Suspended solids

30.0

Note. - In addition to the above, total heavy metals are to be limited to 7mg/l.

44. Bullion Refining (Waste-water discharge)

Effluents

pH

6.5-8.5

Cyanide as CN

0.2

Sulphide as S

0.2

Nitrate as N

10.0

FreeCl₂ as Cl

1.0

Zinc as Zn

5.0

Copper as Cu

2.0

Nickel as Ni

2.0

Arsenic as As

0.1

Cadmium as Cd

0.2

Oil and Grease

10.0

Suspended Solids

100

[* * *]

[Omitted by Notification No. G.S.R. 485 (E) dated 9.6.2010 (w.e.f. 19.11.1986)]

[* * *]

[Omitted by Notification No. G.S.R. 485 (E) dated 9.6.2010 (w.e.f. 19.11.1986)]

[* * *]

[Omitted by Notification No. G.S.R. 485 (E) dated 9.6.2010 (w.e.f. 19.11.1986)]

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[* * *]

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[* * *]

[Omitted by Notification No. G.S.R. 485 (E) dated 9.6.2010 (w.e.f. 19.11.1986)]

[* * *]

[illegible]

Sl. No.

Category

Standards, Db (A)

1

2

3

46.

Noise Limits for Automobiles (Free Field at one meter in dB(A) at the Manufacturing Stage) to be Achieved by the Year 1992.

(a) Motorcycle, Scooters Three Wheelers

80

(b) Passenger Cars

82

(c) Passenger or Commercial Vehicles upto 4 MT

85

(d) Passenger or Commercial Vehicles above 4 MT and upto 12 MT

89

(e) Passenger or Commercial Vehicles exceeding 12 MT

91

47.

Domestic Appliances and Construction Equipments at the manufacturing Stage to be achieved by the Year, 1993

(a) Window Air Conditioners of 1 ton to 1.5 ton

68

(b) Air Coolers

60

(c) Refrigerators

46

[***]

[Item (d) omitted by G.S.R. 371(E), dated 17-5-2002 (w.e.f. 17-5-2002).]

(e) Compactors (rollers) Front loaders, Concrete mixers, Cranes (movable), Vibrators and Saws.

75]

Sl. No.

Industry

Parameter

Standards

1

2

3

4

48. []

[Inserted by G.S.R. 93(E), dated 21-2-1991 (w.e.f. 27-2-1991)]

Glass

Industry

Emissions

A.

Sodalime

Borosilicate and other special Glass (other than Lead)

(a) Furnace:

Capacity

(i) Upto a

product draw capacity of 60 MT/Day

Particulate
matter

2.0 kg/hr.

(ii) Product

draw capacity more than 60 MT/Day

Particulate

matter

0.8 kg./Mt.

of product drawn

(iii) For all

capacities emission and H is stack height in metres.

Stack height

$H=14(Q)^{0.3}$ where Q is the rate of SO₂ in Kg/hr.

Total

fluorides

5.0 mg/NM³

NOX

Use of low

nox burners in new plants.

(b)

Implementation of the following measures for fugitive emission

control from other sections :-

(i) Raw

materials should be transported in leak proof containers.

(ii) Cullet

preparation should be dust-free using water spraying.

(iii) Batch

preparation section should be covered.

B.

Lead Glass

(a) Furnace:

All

capacities

Particulate

matter

50 mg/NM³

Lead

20 mg/NM³

(b)

Implementation of the following measures for fugitive emission

control from other sections :-

(i) Batch

mixing, proportioning section and transfer points should be

covered and it should be connected to control equipments to meet

the following standards:

Particulate

matter

50 mg/NM³

Lead

20 mg/NM³

(ii) Minimum

Stack height should be 30 metres in lead glass units.

(c) Pot

Furnace at Firozabad

Furnace :

Particulate
matter

1200 mg/NM3

Note :-

Depending upon local' environmental conditions, State/ Central
Pollution Control Board can Particulate matter prescribe more
stringent standards than those prescribed above.

C.

Glass

Industry (For all categories)

Effluents

pH

6.5-8.5

Total

Suspended solids

100 mg/l

Oil

Grease

10 mg/l

49

Lime Kiln

Stack height

Capacity:-

Stack height

Upto 5T/day

A Hood should

be provided with a stack of 30 meter height from ground level
(including kiln height).

Above 5T/day

Stack height

$H=14(Q)^{0.3}$ Where Q is emission rate of SO₂ in kg/hr. and H is
Stack Height in meters.

More than

5T/day

Particulate
matter

500 mg/NM3

And upto

40T/day

Above 40T/day

Particulate
matter

150 mg/NM3

[50

[Substituted by Notification No. G.S.R. 1016(E), dated 28.10.2016 (w.e.f. 19.11.1986).]

A.

Slaughterhouses or Meat Processing

Units or Both*

Effluents

Maximum

Concentration values are in mg/l except
for pH

pH

6.5 to 8.5

Bio-chemical

Oxygen Demand

(BOD) [3 days

at 27°C]

30

Chemical

Oxygen Demand

(COD)

250

Suspended

Solids

50

Oil and

Grease

10

B.Sea Food

Industry*

Bio-chemical

Oxygen Demand

(BOD) [3 days

at 27°C]

30

Suspended

Solids

50

Oil and

Grease

10

*The

emission standards from Boiler House of Slaughterhouses or Meat Processing Units or both and Sea Food Industry shall conform to the standards prescribed vide notification No. G.S.R. 742 (E), dated 30.08.1990 as amended from time to time under the Environment (Protection) Act, 1986.

Note:

(i)

For Slaughterhouses operating in local bodies/ municipalities, where the treated effluent is discharged into municipal sewers leading to full-fledged Sewage Treatment Plant, the BOD may be relaxed to 100mg/l.

(ii)

All Slaughterhouses/ meat processing units shall ensure safe and proper disposal of solid waste {Type I (Vegetable matter such as rumen, stomach and intestinal contents, dung, agriculture

residues etc) and Type II (Animal matter such as inedible offal, tissues, meat trimmings, waste and condemned meat, bones etc.)) through suitable technology approved by SPCBs/ PCCs.]

Note :-(i) TLWK-Total Live Weight Killed. (ii) In case of disposal into municipal sewer where sewage is treated the industries shall install screen and oil grease separation unitS. (iii) The industries having slaughter house along with meat processing units will be considered in meat processing category as far as standards are concerned.

*The emission standards from Boiler House shall conform to the standards already prescribed under Environment (Protection) Act, 1986 vide Notification No. G.S.R. 742 (E), dated 30th August, 1990.

Sl. No.

Industry

Parameter

Standards

1

2

3

4

51.

*Food and Fruit Processing Industry:

Effluents

Concentration not to exceed mg/l

except pH

Quantum gm/MT of product

Category :

A.

Soft Drinks

(a) Fruit based/Synthetic(More than 0.4 MT/Day) bottles and tetra pack

pH

6.5-8.5

-

Suspended solids

100

Oil and grease

10

[BOD (3 days at 27°C)]

[Substituted by G.S.R. 176(E), dated 2-4-1996 (w.e.f. 3-4-1996).]

30

(b) Synthetic (Less than 0.4 MT/Day)

Disposal via septic tank

-

B.

Fruit

Vegetables

(a) Above 0.4 MT/Day

pH

6.5-8.5

-

Suspended solids

50

Oil and grease

10

[BOD (3 days at 27°C)]

30

(b) 0.1-0.4 MT/Day

Disposal via septic tank

-

(10 MT/Yr)

C.

Bakery

(a) Bread and Biscuit

(i) Continuous process

(more than 20T/Day)

pH

6.5-8.5

[BOD (3 days at 27°C)]

200

25

(ii) Non-continuous process (less than 20MT/Day)

Disposal via septic tank

(b) Biscuit Production

(i) 10 T/Day above

pH

6.5-8.5

[BOD (3 days at 27°C)]

[Substituted by G.S.R. 176(E), dated 2-4-1996 (w.e.f. 3-4-1996).]

300

35

(ii) Below 10T/Day

Disposal via septic tank

D.

Confectioneries

Effluents

(a) 4 T/Day and above

pH

6.5-8.5

Suspended solids

50

Oil and grease

10

[BOD (3 days at 27°C)]

30

(b) Below 4 T/Day

Disposal via septic tank

Note. - To ascertain the category of "unit fails" the average of daily production and waste-water discharge for the preceding 30 operating days from the date of sampling shall be considered.

*The emission from the Boiler House shall conform to the standards already prescribed under Environment (Protection) Act, 1986 vide Notification NO. G.S.R. 742(E), dated 30th August, 1990.

Sl. No.

Industry

Parameter

Standards

1

2

3

4

52.

*Jute Processing Industry:

Effluents

Concentration in mg/l

except pH and water consumption.

pH

5.5-9.0

[BOD (3 days at 27°C)]

30

Suspended solids

100

Oil and grease

10

Water consumption

1.60 Cum/Ton of product produced.

Note. - 1. Water Consumption for the jute processing industry will be 1.5 Cum/Ton of product from January, 1992.

2. At the present no limit for colour is given for liquid effluent. However, as far as possible colour should be removed.

*Stack emissions from boiler house shall conform to the standards already prescribed under Environment (Protection) Act, 1986 vide Notification-No. GSR 742(E), dated 30th August, 1990.

53. Large Pulp Paper/News Print/Rayon Grade Plants of [Capacity above 24,000 MT per annum]

[Substituted by G.S.R. 546(E), dated 30-8-2005 (w.e.f. 30-8-2005).]

Effluents

Concentration in mg/l

except pH and TOCL

pH

7.0-8.5

[BOD (3 days at 27°C)]

30

COD

350

Suspended solids

50

[Absorbable Organic Halogens (AOX) in effluent discharge

[Substituted by G.S.R. 546(E), dated 30-8-2005 (w.e.f. 30-8-2005).]

1.5

kg/ton of product with effect from the date of publication of this notification 1.0 kg/ton of product with effect from the 1st day of March, 2008]

Flow (Total waste-water discharge)

** (i) Large pulp and paper

200 Cum/Ton of paper produced

(ii) Large rayon grade/News Print

150 Cum/Ton of paper produced

*The Standards for Total Organic Chloride (TOCL) will be applicable from January, 1992.

**The Standards with respect of total waste-water discharge for the large pulp and paper mills be established from 1992, will meet the standards of 100 Cum/Ton of paper produced.

54. Small Pulp and Paper, Paper Plant of Capacity upto 24,000 MT/Annum:

Effluents

Category :

A.* Agro-based

Total Waste-water discharge

200 Cum/Ton of paper produced

B. **Waste-paper based

-do-

75 Cum/Ton of paper produced

*The agro-based mills to be established from January, 1992 will meet the standards of 150 cum/Ton of paper produced.

**The waste-paper mills to be established from January, 1992 will meet the standards of 50 cum/Ton of paper produced.

55. Common Effluent Treatment Plants:

Effluents

A. Primary Treatment

(Inlet effluent quality for CETP)

(Concentration in mg/l)

pH

5.5-9.0

Temperature °C

45

Oil Grease

20

Phenolic Compounds (as C₆H₅OH

)

5.0

Ammonical Nitrogen (as N)

50

Cyanide (as CN)

20

Chromium hexavalent (as Cr+6)

2.0

Chromium (total) (as Cr)

2.0

Copper (as Cu)

3.0

Lead (as Pb)

1.0

Nickel (as Ni)

3.0

Zinc (as Zn)

15

Arsenic (as As)

0.2

Mercury (as Hg)

0.01

Cadmium (as Cd)

1.0

Selenium (as Se)

0.05

Fluoride (as F)

15

Boron (as B)

2.0

Radioactive Materials

Alpha emitters, ?c/ml

10-7

Beta emitters, $\mu\text{Ci/ml}$

10-8

Note.- 1. These standards apply to the small scale industries, i.e., total discharge upto 25 KL/Day.

2. For each CETP and its constituent units, the State Board will prescribe standards as the local needs and conditions; these can be more stringent than those prescribed above. However, in case of clusters of units, the State Boards with the concurrence of CPCB in writing may prescribe suitable limits.

Into inland surface waters

On land for Irrigation

Into Marine Coastal areas

(a)

(b)

(c)

B. Treated Effluent Quality of common effluent treatment plant

Concentration in mg/l except pH and Temperature

pH

5.5-9.0

5.5-9.0

5.5-9.0

[BOD (3 days at 27°C)]

30

100

100

Oil Grease

10

10

[10]

[Substituted for the words "20" by Notification No. G.S.R. 739 (E) dated 9.9.2010 (w.e.f. 19.11.1986)]

Temperature

Shall not exceed 40°C in any Section of the stream within 15 metres downstream from the effluent outlet.

45°C at the point of discharge

Suspended solids

100

200

(a) For process waste-waters 100

(b) For cooling water effluents 10 per cent. above total suspended matter of in effluent cooling water

Dissolved Solids (Inorganic)

2100

2100

-

Total residual chlorine

1.0

-

1.0

Ammonical nitrogen (as N)

50

-

50

Total Kjeldahl Nitrogen (as N)

100

-

100
Chemical Oxygen Demand
250
-
250
Arsenic (as As)
0.2
0.2
0.2
Mercury (as Hg)
0.01
-
0.01
Lead (as Pb)
0.1
-
1.0
Cadmium (as Cd)
1.0
-
2.0
Total Chromium (as Cr)
2.0
-
2.0
Copper (as Cu)
3.0
-
3.0
Zinc (as Zn)
5.0
-
15
Selenium (as Se)
0.05
-
0.05
Nickel (as Ni)
3.0
-
5.0
Boron (as B)
2.0
2.0
-
Percent Sodium
-
60
-
Cyanide (as CN)
0.2

0.2

0.2

Chloride (as Cl)

1000

600

-

Fluoride (as F)

2.0

-

15

Sulphate (as SO₄)

1000

1000

-

Sulphide (as S)

2.8

-

5.0

Pesticides

Absent

Absent

Absent

Phenolic compounds C₆H₅OH

1.0

-

5.0

Note.

- All efforts should be made to remove colour and unpleasant odour as far as possible.]

56. Dairy

Effluents

Concentration in mg/l, except pH

Quantity per product processed

pH

6.5-8.5

-

*[BOD (3 days at 27°C)]

100

-

**Suspended Solids

150

-

Oil and Grease

10

-

Waste-water generation

-

3m³/kl of milk

Note.- *BOD may be made stringent upto 30 mg/l if the recipient fresh water body is a source for drinking water supply. BOD shall be upto 350 mg/l for the chilling plant effluent for applying on land provided the land is designed and operated as a secondary treatment system with suitable monitoring facilities. The drainage water from the land after secondary treatment has to satisfy a limit of 30 mg/l of BOD and 10 mg/l of nitrate expressed as "N". The net addition to the

groundwater quality should not be more than 3 mg/l of BOD and 3 mg/l of nitrate expressed as "N". This limit for applying on land is allowed subject to the availability of adequate land for discharge under the control of the industry, BOD value is relaxable upto 350 mg/l, provided the waste-water is discharged into a town sewer leading to secondary treatment of the sewage.

** Suspended solids limit is relaxable upto 450 mg/l, provided the waste-water is discharged into town sewer leading to secondary treatment of the sewage.

57. Tanneries Effluents

Effluents

Concentration in mg/l, except pH

Quantum per raw hide processed

pH

6.5-9.0

[BOD (3 days at 27°C)]

100

Suspended Solids

100

-

Sulphides (as S)

1

-

Total Chromium (as Cr)

2

-

Oil and Grease

10

-

Waste-water generation

-

28m³/T

*For effluent discharge into inland surface waters BOD limit shall be made stricter to 30 mg/l by the concerned State Pollution Control Board.

58.[***]

[Deleted by Notification No. G.S.R. 221 (E) dated 18.3.2011]

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59. [Agriculture residue fired Boilers (including bagasse)]

[Substituted by Notification No. G.S.R. 647 (E) dated 13.9.2002 (w.e.f. 19.11.1986)]

Emissions

(Concentration in mg/1)

(a) Step grate

Particulate matter

250

(b) Horse shoe/pulsating grate

Particulate matter

500(12% CO₂)

(c) Spreader stroker

Particulate matter

800(12% CO₂)

Note.- In the case of horse shoe and spreader stroker boilers, if more than one boiler is attached to a single stack, the Standard shall be fixed based on added capacity of all the boilers connected with the stack.

60. Man-made fibre industry

(Semi-Synthetic).

Effluent

Concentration in mg/1 except for pH

pH

5.5-9.0

Suspended Solids

100

[BOD (3 days at 27°C)]

30

Zinc (as Zn)

1

61. Ceramic Industry

Emissions

(Concentration in mg/Nm³)

A. Kilns

(a) Tunnel, Top Hat, Chamber

Particulate matter

150

Fluoride

10

Chloride

100

Sulphurdioxide

**

(b) Down-draft

Particulate matter

1200

Fluoride

10

Chloride

100

Sulphurdioxide

**

(c) Shuttle

Particulate matter

250

Fluoride

10

Chloride

100

Sulphurdioxide

**

(d) Vertical shaft Kiln

Particulate matter

250

Fluoride

10

Sulphurdioxide

**

(e) Tank furnace

Particulate matter

150

Fluoride

10

Sulphurdioxide

**

B. Raw Material handling, Processing and operations

(a) Dry raw material handling and processing operations

Particulate matter

150

(b) Basic raw materials and processing operations

Particulate matter

*

(c) Other sources of air pollution generation

Particulate matter

*

C. Automatic spray unit

(a) Dryers

(i) Fuel fired dryers

Particulate matter

150

(ii) For heat recovery dryers

Particulate matter

*

(b) Mechanical finishing operation

Particulate matter

*

(c) Lime/plaster of Paris manufacture

Particulate matter

*

Capacity

Stack height

Upto 5T/day

-do-

A. Hood should be provided with a stack of 30 metre height from ground level (including kiln height)

Above 5T/day

-do-

$H = 14(Q)^{0.3}$ Where Q is emission rate of SO₂ in/kg/hr and H is Stack height in metres

More than 5T/day and upto 40T/day

Particulate matter

500 mg/Nm³

-do-

150 mg/Nm³

Note. - Oxygen reference level for particulate matter concentration calculations for Kilns mentioned at A(c) is 18% and for those at A(b), A(d), and A(e) is 8%.

*All possible preventive measures should be taken to control pollution as far as practicable.

**The standard for sulphur dioxide in terms of stack height limits for kilns with various capacities of coal consumption shall be as indicated below :-

Coal Consumed per day

Stack Height

Less than 8.5 mt.

9 m.

More than 8.5 to 21 mt.

12 m.

More than 21 to 42 mt.

15 m.

More than 42 to 64 mt.

18 m.

More than 64 to 104 mt.

21 m.

More than 104 to 105 mt.

24 m.

More than 105 to 126 mt.

27 m.

More than 126 mt.

30 m. or using formula $H = 14(Q)^{0.3}$ (whichever is more)

Note. - In this notification, H-Physical height of the stack Qg-Emission of sulphur dioxide in Kg/hr., MT-Metric tones m-meters.

62. [Viscose Filament Yarn (Sub-sector of man-made fibre semi-Synthetic Industry)]

[Added by G.S.R. 801(E), dated 31-12-1993 (w.e.f. 31-12-1993).]

Effluents

(Concentration in mg/l except for pH)

pH

5.5-9.0

Suspended solids

100

[BOD(3 days at 27°C)]

[Substituted by G.S.R. 176(E), dated 2-4-1996 (w.e.f. 3-4-1996).]

30

Zinc (as Zn)

5

63. [Starch Industry (Maize products)]

[Inserted by G.S.R. 176(E), dated 2-4-1996 (w.e.f. 3-4-1996).]

Effluents

Concentration not to exceed mg/l (except pH and waste-water discharge)

pH

6.5-8.5

BOD

100

(3 days at 27°C)

Suspended solids

150

Waste-water discharge

8 m³/tonne of maize processed

Note. - The prescribed limits for BOD and suspended solids shall be made more stringent or less stringent depending upon the conditions and local requirements as mentioned below:-

(i)

BOD shall be made stringent upto 30 mg/l if the recipient fresh water body is a source for drinking water supply.

(ii)

BOD shall be allowed upto 350 mg/l for applying on land, provided the land is designed and operated as a secondary treatment system with the requisite monitoring facilities. The drainage water from the land after secondary treatment has to satisfy a limit of 30 mg/l of BOD and 10 mg/l of nitrate expressed as "N". The net addition to ground water quality should not be more than 3 mg/l of BOD and 10 mg/l of nitrate expressed as "N".

(iii)

BOD shall be allowed upto 350 mg/l for discharge into a town sewer, if sewer such leads to a secondary biological treatment system.

(iv)

Suspended solids shall be allowed upto 450 mg/l for discharge into a town sewer, if such sewer leads to a secondary biological treatment system.

(v)

In the event of bulking of sludge, the industry shall immediately apprise the respective State Pollution Control Board.

64. Beehive hard coke oven

Emission:

150 mg/Nm³

(i) New unit

Particulate matter (corrected to 6%

CO₂)

Hydrocarbons

25 ppm

(ii) Existing units

Particulate matter (corrected to 6%

CO₂)

350 mg/Nm³

Note. - For control of emissions and proper dispensation of pollutants the following guidelines shall be followed:-

(i)

Units set up after the publication of this notification shall be treated as new units.

(ii)

A minimum stack height of 20 meters shall be provide by each unit.

(iii)

Emissions from coke ovens shall be channelised through a tunnel and finally emitted through a stack. Damper adjustment techniques shall be used to have optimum heat utilisation and also to control the emission of unburnt carbon particles and combustible flue gases.

(iv)

Wet scrubbing system or waste heat utilisation for power generation or by-product recovery systems should be installed preferably to achieve the prescribed standards.

(v)

After four years from the date of this notification, all the existing units shall comply with the standards prescribed for the new units.

65. Briquette Industry (Coal)

Emission:

(a) Units having capacity less than 10 tonnes.

Particulate matter (corrected to 6% CO₂)

350 mg/Nm³

(b) Units having capacity 10 tonnes or more.

Particulate matter (corrected to 6% CO₂)

150 mg/Nm³

Note. - For control of emissions/and proper dispersal of pollutants, the following guidelines shall be followed by the industry :-

(i)

A minimum stack height of 20 metres shall be provided.

(ii)

All ovens shall be modified to single chimney multi-oven systems.

(iii)

Emissions from ovens shall be channelised through in-built draft stack. Optimum heat utilization shall be used.

(iv)

In case of units having capacity 10 tonnes and above, wet scrubbing system shall be provided to control air pollution.

66.

Soft Coke Industry

Particulate matter (corrected to 6% CO₂)

350

mg/Nm³

Note. - Wet scrubbing systems alongwith by-product recovery system shall be provided.

Guidelines for Emission Control to Improve Work Zone Environment (applicable for industries at serial numbers 64, 65 and 66):

(a)

Water used for quenching and wet scrubbing shall be recirculated and reused through catch-pits.

(b)

Leakages in the oven shall be sealed by bentonite or by any suitable paste and by proper maintenance to avoid fugitive emission.

Guidelines for Coal Handling and Crushing Plant (applicable to industries at Serial Numbers 64,65 and 66):

(a)

Unloading of coal trucks shall be carried out with proper care avoiding dropping of the materials from height. It is

advisable to moist the material by sprinkling water while unloading.

(b)

Pulverisation of coal shall be carried out in an enclosed place and water sprinkling arrangement shall be provided at coal heaps, crushing area and on land around the crushing unit.

(c)

Work area surrounding the plant shall be asphalted or concreted.

(d)

Green belt shall be developed along the boundary of the industry.

(e)

Open burning of coal to manufacture soft coke shall be stopped.

67. Edible Oil Vanaspati Industry

Effluents

Temperature

Not more than 5°C above ambient temperature of the recipient water body.

pH

6.5-8.5

Suspended solids

150 mg/l

Oil and grease

20 mg/l

BOD(3 days at 27°C)

100 mg/l

COD

200 mg/l

Waste-water discharge

(i) Solvent extraction

2.0 cum/tonne of product (oil)

(ii) Refinery/Vanaspati

(refined oil Vanaspati)

2.0 cum/tonne of product

(iii) Integrated unit of solvent extraction and refinery/vanaspati

4.0 cum/tonne of refined oil/vanaspati produced

(iv) Barometric cooling water/De-odoriser water

15.0 cum/tonne of refined oil/vanaspati

Note. - (i) The above standards shall be applicable to waste-water from processes and cooling.

(ii)

BOD shall be made stringent upto 30 mg/l if the recipient fresh water body is source of drinking water supply.

(iii)

The standards for boiler emissions shall be applicable as prescribed under Schedule I of these rules.

[

Sl.No.

Industry

Parameter

Standard

(1)

(2)

(3)

(4)

68.

Organic Chemicals Manufacturing Industry

A. Effluent Standards

Limiting concentration in mg/1, except for pH and Bioassay test

Compulsory parameters

pH

6.5-8.5

BOD 3 days, 27°C

100

Oil Grease

10

Bioassay test +

Minimum 90% survival after 96 hours in 100%

effluent Additional parameters

Nitrate (as N)

10

Arsenic (as As)

0.2

Chromium Total

1.0

Lead (as pb)

0.1

Cyanide (as CN)

0.2

Zinc (as Zn)

5.0

Mercury (as Hg)

0.01

Copper (as Hg)

2.0

Nickel (as Nil)

2.0

Phenolics (as C₆ H₅ OH)

5.0

Sulphide

2.0

+ The Bioassay test shall be conducted as per IS: 6582+-1971

Note:(i) Industries covered under

this group include halo aliphatics, plasticizers, aromatics (alcohols, phenols, esters, acids and salts, aldehydes and ketones), substituted aromatics, aliphatics (alcohols, esters, acids, aldehydes, ketones amines and amides) and detergents.(ii)

Though norms for COD are not mentioned here but, COD shall be monitored. If the COD in treated effluent exceeds 250 mg/1, the concerned industrial units discharging such effluent shall be required to identify chemicals responsible for high COD in effluent shall be required to identify chemicals responsible for high COD in effluent. In case, these are found to be toxic as defined under the Manufacture, Storage and Import of Hazardous Chemicals Rules, 1989, the concerned industry shall install tertiary treatment system.(iii) The above mentioned standards shall not be applicable to small scale detergent formulating units.

B. Emission Standards for Incinerator

Limiting concentration in mg/Nm³, unless otherwise stated

Sampling Duration in minutes unless otherwise stated

Particulate Matter

50

30 or more (for sampling about 300 litres of emission)

HCl

50

30

SO₂

200

30

CO

100

daily average

Total Organic Carbon

20

30

Total Dioxins Existing and Furans* Incinerator-

0.2 ng TEQ/Nm³

8 hours

New Incinerator

0.1 ng TEQ/Nm³

8 hours

Sb+As+Pb+Cr+Co+Cu+Mn+Ni+V+Cd+Th+Hg and their compounds

1.5

2 hours

*The existing plant shall comply with norms for Dioxins and Furans as 0.1 ng TEQ/Nm³ by 1st January, 2014.

Note:(i) All monitored values shall

be corrected to 11% oxygen on dry basis.(ii) The CO₂

concentration in tail gas shall not be less than 7%.(iii) In

case, halogenated organic waste is less than 1% by weight in

input waste all the facilities in twin chamber incinerator shall

be designated so as to achieve a minimum temperature of 850±25°

in primary chamber and 950°C in secondary combustion chamber

and with a gas residence time in secondary combustion chamber not

less than two or all the facilities in single chamber

incinerator for gaseous hazardous waste shall be designated so as

to achieve a minimum temperature of 950° in the combustion

chamber with a gas residence time not less than two seconds.(iv)

In case halogenated organic waste is more than 1% by weight in

input waste, waste shall be incinerated only in twin chamber

incinerators and all the facilities shall be designated to

achieve a minimum temperature of 850±25° C in primary

chamber and 1100°C in secondary combustion chamber with a gas

residence time in secondary combustion chamber not less than two

seconds.(v) Scrubber meant for scrubbing emissions shall not

be used as quencher.(vi) incineration plants shall be

operated (i.e. combustion chambers) with such temperature, retention time and turbulence, as to achieve Total Organic Carbon (TOC) content in the incineration ash and residue less than 3% and their loss on ignition is less than 5% of their weight. In case of non-conformity, ash and residue, as the case may be, shall be re-incinerated. (vii) The incinerator shall have a chimney of at least thirty meters height.

C. Effluent Standards for Incinerator

Note: (i) Effluent from scrubber (s)

and floor washing shall flow through closed conduit or pipe network and be treated to comply with the effluent standards mentioned at 'A' above. (ii) The built up in Total Dissolved

Solids (TDS) in waste water of floor washings shall not exceed 1000 mg/l over and above the TDS of raw water used.

D. Storm water

Note: (i) Storm water shall not be allowed to mix with scrubber water and/or floor washings. (ii) Storm water shall be channelized through separate drains passing through a HDPE lined pit having holding capacity of 10 minutes (hourly average) of rainfall.

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[Substituted by Notification No. G.S.R. 608 (E) dated 21.7.2010 (w.e.f. 19.11.1986)]

Substituted by Notification No. G.S.R. 608 (E) dated 21.7.2010 (w.e.f. 19.11.1986)

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68. Organic Chemicals manufacturing industry

Effluents

(a) Compulsory parameters

pH

6.5-8.5

BOD (3 days at 27°C)

100

mg/l

Oil and grease

10

mg/l

Bio-assay test

Minimum 90%

survival after 96 hours with fish at 100% effluent)

(b) Additional parameters

(mg/l)

Nitrate (as N)

10

Arsenic

0.2

Hexavalent Chromium

0.1

Total Chromium

1.0

Lead

0.1

Cyanide as CN

0.2

Zinc

0.5

Mercury

0.01

Copper

2.0

Nickel

2.0

Phenolics as C_6H_5OH

5.0

Sulphide

2.0

Note.- (i) No limit for COD is prescribed but it shall be monitored. If the COD in a treated effluent is persistently greater than 250 mg/l, such industrial units are required to identify chemicals causing the same. In case these are found to be toxic as defined in Hazardous Chemicals Rules, 1989 in Part I of Schedule I, the State Boards in such cases may direct the industries to install tertiary treatment system stipulating time-limit. This may be done on case-to-case basis.

(ii)

These standards are not applicable to small-scale detergent (formulating units).

(iii)

The standards for boiler emissions shall be applicable as per the existing emission regulations.

(iv)

Industry covered under this group are halo-aliphatics, plasticizers, aromatics (alcohols, phenols, esters, acids and salts, aldehydes and ketone), substituted aromatics, aliphatic (alcohols, esters, acids, aldehydes, ketones, amines and amides) and detergents.

l}

Sl.No.

Industry

Parameter

Standards

(1)

(2)

(3)

(4)

69.

Grain

Processing, Flour Mills, Paddy Processing, Pulse Making or

Grinding Mills

A-Emission

Standards

Capacity

(tonne per hour)

Limiting

Concentration in mg/Nm³

1 to

3

150

Particular matter

More

than 3

100

Notes:-(i)

All dust generating equipments or processes shall be provided with dust extraction arrangement.(ii) The bag houses, etc., shall be connected to chimneys/stacks of 12 metres height or at least 02 metres above the top most point of the building or shed or plant in the industry.(iii) The unit shall channelise shop floor/fugitive emissions through a stack of 12 metres height or at least 02 metres above the top most point of the building or shed or plant in the industry.

B-Effluent

Standards

Limiting

value for concentration in mg/l except for pH

Inland

surface water

Public

sewer

Land

for irrigation

pH

5.5-9.0

5.5-9.0

5.5-9.0

Suspended Solids

100

600

200

Oils and Grease

10

20

10

BOD 3 days at 27°C

30

350

100

COD

250

-

-

C-Stormwater

Standards

(I) Stormwater for

a unit (having plot size atleast 250 square meters) shall not be allowed to mix with scrubber water, effluent and/or floor washings.(ii) Stormwater within the battery limits of a unit shall be channelized through separate drain/pipe passing through a HDPE lined pit having holding capacity of 10 minutes (hourly average) of rainfall.

Notes. - (i) BOD shall be made stringent upto 30 mg/l if the recipient fresh water body is a source for drinking water supply.

(ii)

BOD shall be allowed upto 350 mg/l for (applying on land, provided the land is designed and operated as a secondary treatment system with the requisite monitoring facilities. The drainage water from the land after secondary treatment has to satisfy a limit of 30 mg/l of BOD and 10 mg/l of nitrate expressed as "N". The net addition to ground water quality should not be more than 3 mg/l of BOD and 10 mg/l of nitrate expressed as "N".

(iii)

BOD shall be allowed upto 350 mg/l for discharge into a town sewer, if such sewer leads to a secondary biological treatment system.

(iv)

Suspended solids shall be allowed upto 450 mg/l for discharge into a town sewer, if such sewer leads to a secondary biological treatment system.

70. Boilers(Small)

Steam generation capacity (ton/hour)

Particulate Emission matter

mg/NM³

less than 2

1200*

2 to less than 10

800*

10 to less than 15

600*

15 and above

150**

*to meet the respective standards, cyclone/multicyclone is recommended as control equipment with the boiler.

**to meet the standard, bag filter/ESP is recommended as control equipment with the boiler.

Note. - (i) 12% of CO² correction shall be the reference value for particulate matter emission standards for all categories of boilers.

(ii)

These limits shall supersede the earlier limits notified under Schedule I at serial number 34 of Environment (Protection) Act, 1986 vide notification G.S.R. 742(E), dated 30th August,1990.

(iii)

Stack Height for small Boilers.

For the small boilers using coal or liquid fuels, the required stack height with the boiler shall be calculated by using the formula -

$$H=14 Q^{0.3}$$

Where H-Total stack height in metres from the ground level.

Q=SO² emission rate in kg/hr.

In no case the stack height shall be less than 11 metres.

Where providing all stacks are not feasible using above formula the limit of 400 mg/Nm³ for Q=SO² emission shall be met by providing necessary control equipment with a minimum stack height of 11 metres.

71. Pesticides industry

(i)

Compulsory Parameters

mg/l except pH

pH

6.5-8.5

BOD (3 days at 27oC)

100

Oil Grease

10

Suspended solids

100

Bio-assay test

Minimum 90%

survival of fish after 96 hours with 90% effluent and 10% dilution water. Test shall be carried out as per IS : 6502-1971.

(ii) Additional Parameters

mg/l

(a) Heavy metal

Copper

1.0

Manganese

1.0

Zinc

1.0

Mercury

0.01

Tin

0.1

Any other like Nickel

shall not exceed 5 times the drinking water standards (BIS) individually.

(b) Organics

Phenol Phenolic Compounds as C₆H₅OH

1.0

(c) Inorganics

Arsenic as AS

0.2

Cyanide as CN

0.2

Nitrate as NO₃

50

Phosphate as P

5.0

(d) Specific pesticide

(microgram/litre)

Benzene

10

Hexachloride

DDT

10

Dimethoate

450

Copper oxychloride

9600

Ziram

1000

2,4D

400

Paraquat

23000

Propanil

7300

Nitrogen

780

Other/below mentioned Pesticides individually

100

Other pesticides :

(i)

Insecticides :

Aluminium Phosphide

Lindane

Pyrethrum extract

Dichlorvos

Malathion

Quinalphos

EDTC Mixer

Methyl Bromide

Monocrotophos

Ethylene Dibromide

Nicotine Sulphate

Carbaryl

Ethion

Oxydemeton Methyl

Endosulfan

Fenitrothion

Methyl Parathion

Fenvalerate

Lime-sulphur

Phosphamidon

Phorate

Temephos

(ii) Fungicides:

Aureofungin

Organomercurials (MEMC PMA)

Barium Polysulphide

Sulphur (Colloidal, Wettable Dust)

Cuprous Oxide

Streptocycline

Ferbam

Thiram

Mancozeb

Zineb

Manab

Carbendazim

Nickel Chloride

Tridemorph

(iii) Rodenticides

Comafuryl

Warfarin

Zinc Phosphide

(iv) Nematicides:

MethamN-Sodium

(v) Weedicides:

Fluchloralin

Isoproturon

Butachlor

Anilphos

(vi) Plant Growth Regulants

:

ChloromequatChloride

NemphaleneAcetic Acid

(vii) Any other pesticide not specified above.

Note. - (1) Limits shall be complied with at the end of the treatment plant before any dilution.

(2)

From the "Additional Parameters" specified in 71 (ii), only the relevant parameters (based on the raw-materials used and products manufactured) may be the concerned State Board on a case-to-case basis .

(3)

No limit for COD is prescribed. If the COD in a treated effluent is persistently more than 250 mg/l, such industrial units are required to identify the chemicals causing the same. In case, there are found to be toxic as defined in Schedule I of the Hazardous Chemicals Rules, 1989, the State Boards in such cases may direct the industries to install tertiary treatment, stipulating time-limit. This may be done on a case-to-case basis.

(4)

Solar evaporation followed by incineration is a recognised practice, provide the guidelines of solar evaporation as given below are followed:

Guidelines on solar evaporation system or waste-water from pesticide industry.

(i)

Solar evaporation pans shall be constructed in such a way that the bottom is at least one metre above the ground level.

Tripura State Pollution Control Board 89

(ii)

Solar evaporation pans shall be leak proof and construction and designed as per IS: 90.

(iii)

The solar evaporation pans shall be designed on the basis of evaporation rate matching to the out put of waste-water.

(iv)

Waste-water must be pre-treated as below before subjecting to solar evaporation:

(a)

Oil and grease and floating organics shall be removed so that the rate of evaporation is not affected.

(b)

Acidic/Alkaline waste must be neutralised before solar evaporation to maintain pH in the range of 6.5 to 8.5.

(c)

Toxic volatile matter shall be removed so as not to cause air pollution.

(v)

During the rainy season, storm water shall not be allowed to mix with process waste and enter the pans. The waste-water shall in no case outflow from the evaporation pans. Alternative arrangements shall be made to hold the waste-water in proper impervious tanks and if necessary, force evaporated.

(vi)

In no circumstances, the liquid effluent shall be discharged without conforming to the minimal national standards or stored in a holding arrangement which is likely to cause pollution.

(vii)

The sludge from the solar evaporation pans shall be incinerated or disposed as per the guidelines for management and handling of hazardous waste, published by the Ministry of Environment and Forests, Government of India, after obtaining authorisation from the State Pollution Control Board under the Hazardous Wastes (Handling and Management) Rules, 1989.

(viii)

The facility shall be protected from flood and storm to prevent embankments from erosion or any other damage which may render any portion inoperable.

(ix)

Facilities shall be protective enclosure to keep wildlife, domestic animals, unauthorised persons, etc., away.

72. Oil Drilling and Gas Extraction Industry

A.

Standards for Liquid Effluent

1.0

On-Shore facilities (For Marine Disposal)

pH

5.5-9.0

Oil Grease

10 mg/l

Suspended solids

100 mg/l

BOD

(3 days at 27°C)

30 mg/l

Notes. - (i) For on-shore discharge of effluents, in addition to the standards prescribed above, proper marine outfall has to be provided to achieve the individual pollutant concentration level in sea water below their toxicity limits as given below, within a distance of 50 metre from the discharge point, in order to protect the marine aquatic life:

Parameter

Toxicity limit, mg/l

Chromium as Cr

0.1

Copper, as Cu

0.05

Cyanide, as CN

0.005

Fluoride, as F

1.5

Lead, as Pb

0.05

Mercury, as Hg

0.01

Nickel, as Ni

0.1

Zinc, as Zn

0.1

(ii)

Oil and gas drilling and processing facilities, situated on land and away from saline water sink, may opt either for disposal of treated water by on-shore disposal or by re-injection in abandoned well, which is allowed only below a depth of 1000 metres from the ground level. In case of re-injection in abandoned well the effluent have to comply only with respect to suspended solids and oil and grease at 100 mg/l and 10 mg/l, respectively. For on-shore disposal, the permissible limits are given below:

Sl. No.

Parameter

On-shore discharge standards (Not to exceed)

1

2

3

1.

pH

5.5-9.0

2.
Temperature
40°C
3.
Suspended Solids
100 mg/l
4.
Zinc
2 mg/l
5.
BOD
30 mg/l
6.
COD
100mg/l
7.
Chlorides
600 mg/l
8.
Sulphates
1000 mg/l
9.
TDS
2100 mg/l
10.
%Sodium
60 mg/l
11.
Oil and Grease
10 mg/l
12.
Phenolics
1.2 mg/l
13.
Cyanides
0.2 mg/l
14.
Fluorides
1.5 mg/l
15.
Sulphides
2.0 mg/l
16.
Chromium (Cr+6)
0.1 mg/l
17.
Chromium (Total)
1.0 mg/l
18.
Copper
0.2 mg/l

19.

Lead

0.1 mg/l

20.

Mercury

0.01 mg/l

21.

Nickel

3.0 mg/l

2.0

Off-shore facilities :

For off-shore discharge of effluents, the oil content of the treated effluent without dilution shall not exceed 40 mg/l for 95% of the observation and shall never exceed 100 mg/l. Three 8-hourly grab samples are required to be collected daily and the average value of oil and grease content of the three samples shall comply with these standards.

B. Guidelines for Discharge of Gaseous Emission:

1.0

DG Sets .

1.1

DG Sets at drill site as well as production station shall conform with the norm notified under the Environment (Protection) Act, 1986.

2.0

Elevated/ground flares.

2.1

Cold Venting of gases shall never be resorted to and all the gaseous emissions are to be flared.

2.2

All flaring shall be done by elevated flares except where there is any effect on crop production in adjoining areas due to the flaring. In such cases, one may adopt ground flaring.

2.3

In case of ground flare, to minimise the effects of flaring, the flare pit at Group Gathering Station(GGS)/Oil Collecting Station(OCS) and Group Collection Station(GCS) shall be made of RCC surrounded by a permanent wall (made of refractory brick) of minimum 5m height, to reduce the radiation and glaring effects in the adjoining areas.

2.4

A green belt of 100m width may be developed around the flare after the refractory wall in case of ground flaring.

2.5

If the ground flaring with provision of green belt is not feasible, enclosed ground flare system shall be adopted, and be designed with proper enclosure height, to meet the ground level concentration(GLC) requirement.

2.6

In case of elevated flaring, the minimum stack height shall be 30m. Height of the stack shall be such that the maximum GLC never exceeds the prescribed ambient air quality limit.

3.0

Burning of effluent in the pits shall not be carried out at any stage.

[C. Guidelines for Disposal of Solid Waste, Drill Cutting and Drilling Fluids for Off-shore and On-shore Drilling Operation]
[Substituted by G.S.R. 546(E), dated 30-8-2005 (w.e.f. 30-8-2005).]

:-

1. Disposal of Drill Cuttings and Drilling Fluids for On-shore Installations:

(a)

Drill Cuttings (DC) originating from on-shore or locations close to shore line and separated from Water Base Mud (WBM) should be properly washed and unusable drilling fluids (DF) such as WBM, Oil Base Mud (OBM), Synthetic Base Mud (SBM) should be disposed off in a well designed pit lined with impervious liner located off-site or on-site. The disposal pit should be provided additionally with leachate collection system.

Design aspects of the impervious waste disposal pit; capping of disposal pit should be informed by the oil industry to

State Pollution Control Board (SPCB) at the time of obtaining consent.

(b)

Use of diesel base mud is prohibited. Only WBM should be used for on-shore oil drilling operations.

(c)

In case of any problem due to geological formation for drilling, low toxicity OBM having aromatic content 1% should be used. If the operators intend to use such OBM to mitigate specific whole problem/SBM it should be intimated to Ministry of Environment and Forests/State Pollution Control Board.

(d)

The chemical additives used for the preparation of DF should have low toxicity, i.e., 96 hr LC₅₀ 30,000 mg/l as per mysid toxicity or toxicity test conducted on locally available sensitive sea species. The chemicals used (mainly organic constituents) should be biodegradable.

(e)

DC separated from OBM after washing should have oil content at 10 mg/kg for disposal into disposal pit.

(f)

The waste pit after it is filled up shall be covered with impervious liner, over which, a thick layer of native soil with proper top slope is provided.

(g)

Low toxicity OBM should be made available at installation during drilling operation.

(h)

Drilling wastewater including DC wash water should be collected in the disposal pit evaporated or treated and should comply with the notified standards for on-shore disposal.

(i)

Barite used in preparation of DF shall not contain Hg 1 mg/kg and Cd 3 mg/kg.

(j)

Total material acquired for preparation of drill site must be restored after completion of drilling operation leaving no waste material at site. SPCB should be informed about the restoration work.

(k)

In case, environmentally acceptable methods for disposal of drill waste such as (a) injection to a formation through casing annulus, if conditions allow (b) land farming at suitable location (c) bio-remediation (d) incineration or (e) solidification can be considered, in such cases oil industry is required to submit proposal to Ministry of Environment and Forests/State Pollution Control Board (MoEF/SPCB) for approval.

2. Disposal of Drill Cutting and Drilling Fluids for Off-shore Installations:

(a)

Use of diesel base mud is prohibited. Only WBM is permitted for off-shore drilling. If the operator intend to use low toxicity OBM or SBM to mitigate specific hole problems in the formation, it should be intimated to MoEF/SPCB. The low toxicity OBM should have aromatic content 1%.

(b)

The toxicity of chemical additives used in the DF (WBM or OBM or SBM) should be biodegradable (mainly organic constituents) and should have toxicity of 96 hr LC₅₀ Value 30,000 mg/l as per mysid toxicity or toxicity test conducted on locally available sensitive sea species.

(c)

Hexavalent chromium compound should not be used in DF. Alternative chemical in place of chrome lignosulfonate should be used in DF. In case, chrome compound is used, the DF, DC should not be disposed off-shore.

(d)

Bulk discharge of DF in off-shore is prohibited except in emergency situations.

(e)

WBM/OBM/SBM should be recycled to a maximum extent. Unusable portion of OBM should not be discharged into sea and shall be brought to on-shore for treatment and disposal in an impervious waste disposal pit.

(f)

Thoroughly washed DC separated from WBM/SBM and unusable portion of WBM/SBM having toxicity of 96 hr LC₅₀ 30,000 mg/l shall be discharged off-shore into sea intermittently, at an average rate of 50 bbl/hr/well from a platform so

as to have proper dilution and dispersion without any adverse impact on marine environment.

(g)

Drill cutting of any composition should not be discharged in sensitive areas notified by the Ministry of Environment and Forests.

(h)

In case of specific hole problem, use of OBM will be restricted with zero discharge of DC. Zero discharge would include re-injection of the DC into a suitable formation or to bring to shore for proper disposal. In such a case, use of OBM for re-injection should be recorded and made available to the regulatory agency. Such low toxic OBM having aromatic content 1% should be made available at the installation.

(i)

In case, DC is associated with high oil content from hydrocarbon bearing formation, then disposal of DC should not have oil content 10 gm/kg.

(j)

The DC wash water should be treated to confirm limits notified under EPA, before disposal into sea. The treated effluent should be monitored regularly.

(k)

Discharge of DC from the installation located within 5 km away from shore should ensure that there is no adverse impact on marine Eco-system and on the shore. If, adverse impact is observed, then the industries have to bring the DC on-shore for disposal in an impervious waste disposal pit.

(l)

If any, environmental friendly technology emerges for substitution of DF and disposal technology, it may be brought to the notice of MoEF and regulatory agencies. If the operator desires to adopt such environment friendly technology a prior approval from Ministry of Environment and Forests is required.

(m)

Barite used in preparation of DF shall not contain Hg 1 Mg/kg and Cd 3 mg/kg.

(n)

Oil drilling operators are required to record daily discharge of DC and DF to off-shore and also to monitor daily the effluent quality, and submit the compliance report once in every six month to Ministry of Environment and Forests.

73. [Pharmaceutical (Manufacturing and Formulation) Industry]

[Substituted by G.S.R. 512(E), dated 9-7-2009 (w.e.f. 9-7-2009).]

Effluent Standards

(i)

Compulsory Parameters

Limiting concentration in mg/l, except for pH

pH

6.0-8.5

Oil grease

10

BOD (3 days 27°C)

100*

Total suspended solids

100

Bioassay Test

90% survival of fish after first 96

hours in 100% effluent**

(ii) Additional Parameters

Mercury

0.01

Arsenic

0.20

Chromium (Cr6+)

0.10

Lead

0.10

Cyanide

0.10

Phenolics(C₆H₅SOH)

1.0

Sulphides(as S)

2.0

Phosphate (as P)

5.0

Note. -

*The BOD and COD limits shall be 30 mg/1 and 250 mg/1 respectively, if treated effluent is discharged directly into a fresh water body, i.e., stream, canal, river or lake.

**The Bioassay Test shall be conducted as per IS: 6582-1971.

(i)

Parameters listed as 'Additional Parameters' shall be prescribed depending upon the process and product.

(ii)

Limits for total dissolved solids in effluent shall be prescribed by the concerned pollution control board/pollution control committee depending upon the recipient water body.

[A. Emission from Incinerator]

Limiting concentration in mg/Nm³, Unless stated

Sampling duration in (minutes) unless stated

Particulate Matter

50

30 or more (for sampling about 300 litre emission)

HCl

50

30

SO₂

200

30

CO

100

30

Total Organic Carbon

20

30

Total Dioxins and Furans*

Existing Incinerator

0.2 ng TEQ/Nm³

8 hours

New Incinerator

0.1 ng TEQ/Nm³

8 hours

Sb+As+Pb+Cr+Co+Cu+Mn+Ni+V+Cd+Th+Hg and their compounds

1.5

2 hours

*The existing plant shall comply with norms for dioxins and furans as 0.1 ng/TEQ/ Nm³ within 5 years from the date of notification.

Notes. - (i) All monitored values shall be corrected to 11% oxygen on dry basis.

(ii) The CO₂ concentration in tall gas shall not be less than 7%.

(iii) In case, halogenated organic waste is less than 1% by weight in input waste, all the facilities in twin chamber incinerator shall be designed so as to achieve a minimum temperature of $850 \pm 25^{\circ}\text{C}$ in primary chamber and 950°C in secondary combustion chamber and with a gas residence time in secondary combustion chamber not less than 2 (two) seconds.

or

all the facilities in single chamber incinerator for gaseous hazardous waste shall be designed so as to achieve a minimum temperature of 950°C in the combustion chamber with a gas residence time not less than 2 (two) seconds.

(iv) In case halogenated organic waste is more than 1% by weight in input waste, waste shall be incinerated only in twin chamber incinerators and all the facilities shall be designed to achieve a minimum temperature of $850 \pm 25^{\circ}\text{C}$ in primary chamber and 1100°C in secondary combustion chamber with a gas residence time in secondary combustion chamber not less than 2 (two seconds).

(v) Scrubber meant for scrubbing emissions shall not be used as quencher.

(vi) Incineration plants shall be operated (combustion chambers) with such temperature, retention time and turbulence, as to achieve Total Organic Carbon (TOC) content in the incineration ash and residue less than 3%, and their loss on ignition is less than 5% of the dry weight. In case of non-conformity, ash and/or residue shall be re-incinerated.

(vii) The incinerator shall have a chimney of at least thirty metre height.

B. Effluent from Incinerator

(i) Effluent from scrubber (s) and floor washing shall flow through closed conduit/pipe network.

(ii) Storm water shall not be allowed to mix with scrubber water and/or floor washings.

(iii) Storm water shall be channelized through separate drains passing through a HDPE lined pit having holding capacity of 10 minutes (hourly average) of rainfall.

(iv) The built up in Total Dissolved Solids (TDS) in wastewater of floor washings shall not exceed 1000 mg/l over and above the TDS of raw water used.

(v) Effluent shall not be stored in holding tank(s) in such manner which may cause pollution to groundwater."

(vi) Effluent (scrubber water and floor washings) shall be discharged into receiving water conforming to the norms prescribed under Schedule VI: General Standards for Discharge of Environment Pollutants (Part A: Effluents) notified under the Environment (Protection) Rules, 1986.

74. [Brick Kilns]

[Substituted by Notification No. G.S.R. 543 (E) dated 22.7.2009 (w.e.f. 19.11.1986)]

Emission Standards

(i) Bull's Trench Kiln (BTK)

Category*

Limiting concentration in mg/Nm³

Particular matter

small

1000

medium

750

large

750

Minimum (metre)

Stack height

small

22 or induced draft fan operating with minimum draft of 50 mm

WG with 12 metre stack height.

medium

27 or induced draft fan operating with minimum draft of 50 mm

WG with 15 metre stack height.

large

30 or induced draft fan operating with minimum draft of 50 mm

WG with 17 metre stack height.

*Category

Trench width (m)

Production (bricks/day)

small BTK

4.50

Less than 15,000

medium BTK

4.50-6.75

15,000-30,000

large BTK

above 6.75

above 30,000

(ii) Down-Draft Kiln (DDK)

Category++

Limiting concentration in mg/ Nm³

Particular matter

small/large/medium

1200

Minimum (metre)

Stack height

small

12

medium

15

large

18

Category++

Production (bricks/day)

Small DDK

less than 15,000

medium DDK

15,000-30,000

large DDK

above 30,000

(iii) Vertical Shaft Kiln (VSK)

Category**

Limiting concentration in mg/ Nm³

Particular matter

small/large/medium

250

Minimum (metre)

Stack height

small

11 (at least 5.5 m from loading platform)

medium

14 (at least 7.5 m from loading platform)

large

16 (at least 8.5 m from loading platform)

**Category

No. of shaft

Production (bricks/day)

small VSK

1-3

Less than 15,000

medium BTK

4-6

15,000-30,000

large BTK

7 or more

above 30,000

Notes. - 1. Gravitational Settling Chamber alongwith fixed chimney of appropriate height shall be provided for all Bull's Trench kilns.

2. One chimney per shaft in Vertical Shaft Kiln shall be provided. The two chimneys emanating from a shaft shall either be joined (at the loading platform in case of brick chimney or at appropriate level in case of metal chimney) to form a single chimney.

3. The above standards shall be applicable for different kilns if coal, firewood and/or agricultural residues are used as fuel.

Sl.No.

Industry

Parameter

Standard

(1)

(2)

(3)

(4)

[75]

[Substituted by Notification No. G.S.R. 424 (E) dated 1.6.2011 (w.e.f. 19.11.1986)]

Soda Ash Industry

Effluent

standardsA. Solvay Proces

Limiting

Concentration mg/1 except for pH, Temperature and Bio-assay

Creek

Marine Coastal

Zone

Estuary Area

Inland Surface

Water

Suspended Solids

500*

1000*

200

100

Ammonical Nitrogen

as N

50

50

50

50

Oil and Grease

5

5

5

5

Bio-assay***

Minimum 90%

survival of fish after 96 hours in 100% effluent

pH

6.5-9.0

Tempertaure

Not ot exceed 5°C

above the ambient temperature of the reveiving water body

*The effluent

discharge point in creek shall be beyond low tide line.

**The diffuser

system shall be located in conformity with the Coastal Regulation

Zone Notification , 2011 at a minimum depth of 5 metres below low

tide level and with exit velocity for effluent more than 3

metres/sec

***The Bio-assay

test shall be conducted as per IS :6582-1971

B.

Dual Process

Inland

Surface Water

ph

6.5-9.0

Ammonical

Nitrogen, as N

50

Nitrate Nitrogen,
as N

10

Cyanide, as CN

2

Hexavalent

Chromium

0.1

Total Chromium

2

Suspended Solids

100

Oil and Grease

10

C.

Stormwater

Note :-

(i) Stormwater

shall not be allowed to mix with effluent and/or floor washings.

(ii) Stormwater

within battery limit of industry shall be channelized through
separate drain (s) passing through HDPE lined pit(s) each having
holding capacity of 10 minutes (hourly average) of rainfall for
its catchment area

76.

Emission Standard for SO₂ from Cupola furnace:

Standard for Sulphur Dioxide emission from Cupola Furnace :

Characteristics

Emission limit

Sulphurdioxide (SO₂) emission

300 mg/Nm³at 12% CO₂corrections

To achieve the standard, foundries may install scrubber, followed by a stack of height six times the diameter of the
Cupola beyond the charging door.

Note. - In case due to some technical reasons, installation of scrubber is not possible, then value of SO₂ to the ambient
air has to be effected through the stack height.

77.

Specifications of Motor Gasoline for Emission Parameters:

Sl. No.

Characteristics

Requirement

Method of test ref. to P: of IS: 1448

(i).

Reid Vapour Pressure at 38°C, Kpa

35 to 70

P : 39

(ii).

Benzeno, Percent by volume, Max

5.0(1)

P: 104

(iii).

Lead Content(as Pb)

g/l, Max

0.15(low leaded)(2) 0.013 unleaded)

P: 38

(iv).

Sulphur, percent by mass, Max

0.10 (unleaded) 0.20 (leaded)

P : 34

(v).

Potential Gum, g/m³, Max

50

ASTM 873 : 8

(vi).

Gum (Solvent Washed)g/m³Max

40

P : 29

(vii).

Oxygenates Content Ether (MTBE, ETBE)

Alcohol,percent by volume, Max

15

(viii).

Phosphorus

See Foot Note(3)

ASTMD 3231

(1)

3.0 per cent by volume maximum in metro cities by 2000 AD.

(2)

0.15 g/l by 31st December, 1996 (for entries country). 0.013 g/l by April 1995 (in four metro cities);

by 1st December,1998(for all State capitals/UTs and major metro cities) and by 1st April, 2000 for the entire country.

(3)

Phosphorous containing additives shall be absent.

Note. - (a) Above specifications applies to leaded as well as unleaded petrol except lead content.

(b)

For new refineries coming up during or after 1997 the specification applicable by 2000 for existing refineries shall be applicable by 1997.

78.

Specification of Diesel Fuel for Emission Related Parameters :

Sl. No.

Characteristics

Requirement

Method of Test Ref. to P : or IS : 1448

(i)

Density at 15°C, Kg/m³

820 to 880 (1)

P : 32

(ii)

CetaneNumber, Min

45°0(2)

P : 9

(iii)

Distillation 85 percent by volume recovery

350

P : 18
 at °C Max 95 per cent by volume recovery at °C Max
 370
 (iv)
 Sulphur, percent by mass
 0.50(3)
 P : 33
 (1)
 820 to 860 by 2000 AD
 (2)
 48 by 31st December, 1998 (except in the refineries- Digboi, Gauhati and Bongaigaon Refineries Petrochemicals Ltd.).
 (3)
 (i)
 0.50 per cent by mass by 1st April 1996 in four metros and Taj Trapezium;
 (ii)
 0.25 per cent by mass by 1st October, 1996 in Taj Trapezium;
 (iii)
 0.25 per cent by mass by 1st April, 1996 throughout the country.
 Note. - (a) Above specifications apply to HSD only.
 (b)
 For new refineries coming during or after 1997 specification applicable by 2000 for existing refineries shall be applicable by 1997.
 (c)
 "P" refers to parts of IS:1448.
 Sl. No.
 Industry
 Parameter
 Standards
 New Batteries (at Green Field Site)
 Rebuild Batteries
 Existing Batteries
 79. []
 [Substituted by G.S.R. 46(E), dated 3-2-2006 (w.e.f. 3-2-2006).]
 Coke oven plants (by product recovery type)
 Fugitive Visible Emissions
 (a) Leakage from door
 5 (PLD)*
 10 (PLD)*
 10 (PLD)*
 (b) Leakage from charging lids
 1 (PLL)*
 1 (PLL)*
 1 (PLL)*
 (c) Leakage from AP Covers
 4 (PLO)*
 4 (PLO)*
 4 (PLO)*
 (d) Charging emission (second/charge)
 16 (with HPLA)*
 50 (with HPLA)*
 75

Stack Emission of Coke Oven

(a) SO₂(mg/Nm³)

800

800

800

(b) Nox(mg/Nm³)

500

500

500

(c) SPM (mg/Nm³)

50

50

50

(d) SPM emission during-charging-for stamp charging batteries (stack emission (mg/Nm³))

25

25

25

(e) SPM emission during coke pushing (stack emission) gm/ton of coke

5

5 (applicable to stationary land based system)

-

(f) Sulphur in Coke Oven gas used for heating (mg/Nm³)

800

-

-

Emission for quenching operation

(a) Battery area (top of the battery)

5

5

5

(b) Other units in coke oven plant

2

2

2

(c) Ambient air standards (mg/Nm³)

10

10

10

For control of emissions and to maintain environmental quality in work zone area, the following guidelines shall be followed, namely:-

(i)

New coke oven units shall follow any of the low-emission procedures, such as, coke dry cooling, non-recovery coke-ovens. Indirect Quenching Process, Jumbo coke oven reactor, modified Wet Quenching System with appropriate environmental controls (e.g., baffles, filtering media, collection and treatment of residual water from quench tower and recycling; Treated effluent conforming to the effluent discharge standards can be used for quenching, Use of untreated process water as quenching water shall not be permissible).

(ii)

Effective pollution control measures (for e.g., Extensive maintenance and cleaning of oven doors and frame seals, ascension pipes, charging holes and lids and other equipment; On-main charging system(HPLA); Luting charging holes with clay-suspension; Modified guide/transfer car with emission control system, etc., shall be used to reduce coal charging and coke pushing emissions.

(iii)

During rebuilding or installing new coke oven batteries, the following clean technology and pollution control measures be adopted:

(a)

air-cooled self-sealing dorrs;

(b)

the hydro-jet cleaning system shall be provided for the door and door frame cleaning with a facility of hydro jet pressure of 600kg/cm²;

(c)

the charging should be accomplished with hermetically sealed charging sleeves and screw feeder in charging car. The charging car. The charging car should also be equipped with magnetic lid lifter and lid an frame cleaning mechanism (applicable to top charging batteries);

(d)

to provide aspiration through high-pressure ammonia liquor (HPLA) injection in goose neck and emission should be transferred directly to gas collecting mains;

(e)

water sealed AP covers should be provided;

(f)

computerised combustion control and moisture control systems.

(iv)

In addition to the above the new coke oven batteries, which will be installed after the date of publication of this notification at green field site and rebuild batteries wherever technically feasible should also be equipped to treat their pushing emission with stationary land-based system with collection hood and wet scrubbing units for gas cleaning.

(v)

In the case of existing coke ovens with wet quenching, the new procedures as in (i) and (ii) shall be adopted.

(vi)

The fugitive visible emission standards, i.e., PLD*, PLL* and PLO*, charging emission (second/charge).

Note. - Units set up after the publication of this notification shall be treated as new units.

*HPLA

Aspiration through high pressure liquor injection in goose neck.

*PLD

Percent leaking doors;

*PLL

Percent leaking lids; and

*PLO

Percent leaking off takes.

80.

[SPECIFICATIONS OF TWO-STROKE ENGINE OIL:

[Inserted by G.S.R. 504(E), dated 20.8.1998 (w.e.f. 21.8.1998).]

Specification

Standard

Test Procedure

Two-stroke engine oil grade JASO-FC as per

Minimum smoke Index of 85

JASO-M 342-92 for JASO-FC and ASTM D-4857 for APITC

JASO M-345-93

specification and APITC as per specification No. ASTM D 4859

The above specification shall be effective from the 1st day of April, 1999.

81.

[Battery manufacturing industry

(i)

Lead Acid Battery Manufacturing Industries. Emission Stanadrds.]

Source

Pollutant

Standards Concentration based (mg/Nm³)

Grid casting

Lead

10

Particulate matter

25

Oxide manufacturing

Lead

10

Particulate matter

25

Paste mixing

Lead

10

Particulate matter

25

Assembling

Lead

10

Particulate matter

25

PVC Section

Particulate matter

150

-To comply with the respective standards, all the emissions from above-mentioned sources shall be routed through stack connected with hood and fan in addition to above, installation of control equipment viz., bag filter/ventury scrubber, is also recommended.

-The minimum stack height shall be 30 m.

Liquid Effluent Discharge Standards :

Pollutants

Concentration based standards

pH

6.5-8.5

Suspended solids

50 mg/l

Lead

0.1 mg/l

(ii)

Dry Cell Manufacturing Industry: Emission Standards

Pollutant

Standards Concentration-based (mg/Nm³)

Particulate matter

50

Manganese as Mn

5

-To Comply with the respective standards, all the emissions from above-mentioned sources shall be routed through stack connected with hood and fan. In addition to above, installation of control equipment, viz., bag filter/ventury scrubber, is also recommended.

-The minimum stack height shall be 30m.

Effluent Standards

Pollutant

Concentration Based standards

pH

6.5-8.5

Total suspended solids

100

mg/l

Manganese as Mn

2

mg/l

Mercury as Hg

0.02

mg/l

Zinc as Zn

5 mg/l

(iii)

Secondary Lead Smelters :

Pollutant

Concentration Based standards

Lead as Pb

10 mg/Nmp3

Particulate matter

50 mg/Nm3

Minimum stack height

30m

82.

Environmental Standards for Gas/Naptha-based Thermal Power Plants

(i)

Limit for emissions of NOx

(a)

For existing units-150ppm (v/v) at 15 Per cent excess oxygen.

(b)

For new units with effect from 1.6.1999.

Total generation of gas turbine

Limit for Stack NOx emission [v/v), at 15% excess oxygen]

(a)400 MW and above

(i) 50 ppm for the units burning natural gas

(ii) 100 ppm for the units burning naphtha.

(b) Less than 400

MW but Upto 100 MW

(i) 75 ppm for the units burning natural gas.

(ii) 100 ppm for the units burning naphtha

(c) Less than 100 MW

100 ppm for units burning natural gas or naphtha as fuel

(d) For the plants burning gas in a conventional boiler.

100 ppm

(ii)

Stack height H in m should be calculated using the formula $H=14Q^{0.3}$, where Q is the emission rate of SO₂ in kg/hr, subject to a minimum of 30 mts.

(iii)

Liquid waste discharge limit

Parameter

Maximum limit of concentration (mg/l except for pH and temperature)

2

3

pH

6.5-8.5

Temperature

As applicable for other thermal power plants.

Free available chlorine

0.5

Suspended Solids

100.0

Oil and grease

20.0

Copper (total)

1.0

Iron (total)

1.0

Zinc

1.0

Chromium (Total)

0.2

Phosphate

5.0

[***]

[Serial No. 83 omitted by G.S.R. 371(E), dated 17.5.2002 (w.e.f. 17.5.2002). Earlier it was inserted by G.S.R. 7, dated 22.12.1998 (w.e.f. 2.1.1999).]

84.

Temperature limit for discharge of condenser cooling Water From Thermal Power Plant :-

A: New thermal power plants commissioned after June 1, 1999:

New thermal power plants, which will be using water from rivers/lakes/reservoirs shall install cooling towers-irrespective location and capacity. For thermal power plants which will use sea water for cooling purposes, the condition below will apply.

B: New projects in coastal areas using sea water:

The thermal power plants using sea water should adopt suitable system to reduce water temperature at the final discharge point so that the resultant rise in the temperature of receiving water does not exceed 7°C over and above the ambient temperature of the receiving water bodies.

C: Existing thermal power plants:

Rise in temperature of condenser cooling water from inlet to the outlet of condenser shall not be more than 10°C.

D: Guidelines for discharge point:

(1)

The discharge point shall preferably be located at the bottom of the water body at midstream for proper dispersion of thermal discharge.

(2)

In case of discharge of cooling water into sea, proper marine outfall shall be designed to achieve the prescribed standards. The point of discharge may be selected in consultation with concerned State Authorities/NIO.

(3)

No cooling water discharge shall be permitted in estuaries or near ecologically sensitive areas such as mangroves, coral reefs/spawning and breeding grounds of aquatic flora and fauna.

85.

Environmental Standards for Coal Washeries :-

1. Fugitive emission standards.

-The difference in the value of suspended particulate matter, delta (?), measured between 25 and 30 metre from the enclosure of coal-crushing plant in the downward and leeward wind direction shall not exceed 150 microgram per cubic meter. Method of measurement shall be High Volume Sampling and Average Flow Rate, not less than 1.1 m³ per minute, using upwind downwind method of measurement:

2. Effluent discharge standards:

-The coal washeries shall maintain the close-circuit operation with zero effluent discharge.

-If in case due to some genuine problems like periodical cleaning of the system, heavy rainfall, etc., it becomes necessary to discharge the effluent to sewer/land/stream then the effluent shall conform to the following standards at the final outlet of the coal washery :

Sl. No

Parameter

Limits

1.

pH

5.5-9.0

2.

Total suspended solids

100 mg/l

3.

Oil Grease

10 mg/l

4.

B.O.D (3 days 27

deg C)

30 mg/l

5.

COD

250 mg/l

6.

Phenolics

1.0 mg/l

3. Noise level standards :

-Operational/Working zone-not to exceed 85 dB(A) Leq for 8 hours' exposure.

-The ambient air quality standards, in respect of noise as notified under the Environmental (Protection) Rules, 1986, shall be followed at the boundary line of the coal washery.

4. Code of practice for Coal Washery:

-Water or Water mixed chemical shall be sprayed at all strategic coal transfer points such as conveyors, loading/unloading points, etc. As far as practically possible conveyors, transfer points, etc., shall be provided with enclosures.

-The crushers/pulverisers of the coal washeries shall be provided with enclosures, fitted with suitable air pollution control measures and finally emitted through a stack of minimum height of 30m, conforming particulate matter emission standard of 150 mg/Nm or provided with adequate water sprinkling arrangement.

-Water sprinkling by using fine atomizer nozzle arrangement shall be provided on the coal heaps and on around the crushers/pulverisers.

-Area, in and around the coal washery shall be pucca either asphalted or concreted.

-Water consumption in the coal washery shall not exceed 1.5 cubic meter per tonne of coal.

-The efficiency of the settling ponds of the waste-water treatment system of the coal washery shall not be less than 90 per cent.

-Green belt shall be developed along the roadside, coal-handling plants, residential complex, office building and all around the boundary line of the coal washery.

-Storage bunkers, hoppers, rubber decks in chutes and centrifugal chutes shall be provided with proper rubber linings.

-Vehicles movement in the coal washery area shall be regulated effectively to avoid traffic congestion. High-pressure horn shall be prohibited. Smoke emission from heavy duty vehicles operating in the coal washeries should conform the standards prescribed under Motor Vehicle Rules, 1989.

86.

Water quality standards for coastal waters marine outfalls.

In a coastal segment marine water is subjected to several types of uses. Depending on the types of uses and activities, water quality criteria have been specified to determine its suitability for a particular purpose. Among the various types of uses there is one use that demands highest level of water quality/purity and that is termed a "designated best use" in that stretch of the coastal segment. Based on this, primary water quality criteria have been specified for following five designated best uses :-

Class

Designated best use

SW-I (See Table 1.1.)

Salt pans, Shell fishing, Mariculture and ecologically Sensitive Zone.

SW-II (See Table 1.2)

Bathing, Contact Water Sports and Commercial fishing.

SW-III (See Table 1.3)

Industrial cooling, Recreation (non-contact)

and Aesthetics

SW-IV (See Table 1.4)

Harbour

SW-V (See Table 1.5)

Navigation and Controlled Waste Disposal.

The Standards along with rationale/remarks for various parameters for different designated best uses, given in Table 1.1 to 1.5

TABLE 1.1

PRIMARY WATER QUALITY CRITERIA FOR CLASS SW-I WATERS

(For Salt Pans, Shell Fishing, Mariculture and Ecologically Sensitive Zone)

S.No.

Parameter

Standards

Rationale/Remarks

1

2

3

4

1.

pH range

6.5-8.5

General broad range, Conductive for propagation of aquatic lives is given. Value largely dependant upon soil-water interaction.

2.

Dissolved Oxygen

5.0 mg/l or 60 per cent saturation value whichever is higher

Not less than 3.5

mg/l at any time of the year for protection of aquatic lives.

3.

Colour and Odour

No noticeable colour or offensive odour.

Specially caused by chemical compound like creosols, phenols, naphtha pyridine benzene, toluene, etc., causing visible colouration of salt crystal and tainting fish flesh.

4.

Floating Matters

Nothing obnoxious or detrimental for use purpose.

Surfactants should not exceed an upper limit of 1.0 mg/l and the concentration not to cause any visible foam.

5.

Suspended Solids

None from sewage or industrial waste origin

Settleable inert matters not in such concentration that would impair any usages specially assigned to this class.

6.

Oil and Grease

(including Petroleum Products)

0.1 mg/l

Concentration should not exceed 0.1 mg/l as because it has effect on fish eggs and larvae.

7. []

[Substituted by G.S.R. 682(E), dated 5.10.1999 (w.e.f. 5.10.1999).]

Heavy Metals:

Values depend on :

Mercury (as Hg)

0.001 mg/l

(i) Concentration in salt, fish and shell fish.

Cadmium (as Cd)

0.001 mg/l

(ii) Average per capita consumption per day.

Lead (as Pb)

0.01 mg/l

(iii) Minimum ingestion rate that induces symptoms of resulting diseases.

Note. - SW-I is desirable to be safe and relatively free from hazardous chemicals like pesticides, heavy metals and radionuclide concentrations. Their combined (synergistic or antagonistic) effects on health and aquatic lives are not yet clearly known. These chemicals undergo bio-accumulation, magnification and transfer to human and other animals through food chain. In areas where fisheries, salt pans are the governing considerations, and presence of such chemicals apprehended/reported, bio-assay test should be performed following appropriate methods for the purpose of setting case specific limits.

TABLE 1.2

PRIMARY WATER QUALITY CRITERIA FOR CLASS SW-II WATERS

(For Bathing, Contact Water Sports and Commercial Fishing)

S.No.

Parameter

Standards

Rationale/Remarks

1

2

3

4

1.

pH range

6.5-8.5

Range does not cause skin or eye irritation and is also conducive for propagating aquatic lives.

2.

Dissolved Oxygen

4.0 mg/1 or 50 percent saturation value whichever is higher.

Not less than 3.5

mg/1 at any time for protection of aquatic lives.

3.

Colour and Odour

No noticeable colour or offensive odour.

Specially caused by chemical compound like creosols, phenols, naphtha, benzene, pyridine, toluene, etc., causing visible, colouration of water and tainting of and odour in fish flesh.

4.

Floating Matters

Nothing obnoxious or detrimental for use purpose

None in concentration that would impair usages specially assigned to this class.

5.

Turbidity

30 NTU (Nephelo Turbidity Unit).

Measured at 0.9m depth.

6.

Fecal Coliform

100/100 ml(MPN)

The average value not exceeding 200/100 ml. in 20 per cent of samples in the year and in 3 consecutive samples in monsoon months.

7.

Biochemical Oxygen Demand (BOD) (3

days at 27°C)

3 mg/1

Restricted for bathing (aesthetic quality of water). Also prescribed by IS : 2296-1974.

TABLE 1.3

PRIMARY WATER QUALITY CRITERIA FOR CLASS SW-III WATERS

(For Industrial Cooling, Recreation (non-contact) and Aesthetics)

S.No.

Parameter

Standards

Rationale/Remarks

1

2

3

4

1.

pH range

6.5-8.5

The range is conducive for propagation of aquatic species and restoring natural system.

2.

Dissolved Oxygen

3.0 mg/1 or 40 per cent saturation value whichever is higher

To protect aquatic lives

3.

Colour and Odour

No noticeable colour or offensive odour assigned to this class.

None in such concentration that would impair usages specifically assigned to this class.

4.

Floating Matters

No visible, obnoxious floating debris, oil slick, scum.

As in (3) above

5.

Fecal Coliform

500/100 ml(MPN)

Not exceeding 1000/100 ml in 20 percent of samples in the year and in 3 consecutive samples in monsoon months.

6.

Turbidity

30 NTU

Reasonably clear water for Recreation, aesthetic appreciation and Industrial Cooling purposes.

*7.

Dissolved Iron (as Fe)

0.5 mg/1 or less

It is desirable to have the collective concentration dissolved Fe and Mn less or equal to 0.5 mg/1 to avoid scaling effect.

*8.

Dissolved Manganese

(as Mn)

0.5 mg/1 or less

*Standards included exclusively for Industrial Cooling purpose. Other parameters same.

TABLE 1.4

PRIMARY WATER QUALITY CRITERIA FOR CLASS SW-IV WATERS

(For Harbour Waters)

S.No.

Parameter

Standards

Rationale/Remarks

1

2

3

4

1.

pH range

6.0-9.0

To minimize corrosive and scaling effect.

2.

Dissolved Oxygen

3.0 mg/1 or 40 per cent saturation value whichever is higher

Considering bio degradation of oil and inhibition to oxygen production - through photosynthesis.

3.

Colour and Odour

No visible colour or offensive odour.

None from reactive chemicals which may corrode paints/ metallic surfaces

4.

Floating materials, Oil, grease and scum (including Petroleum products)

10 mg/1

Floating matter

'Should be free from excessive living organisms which may clog or coat operative parts of marine vessels/equipment.

5.

Fecal Coliform

500/100 ml(MPN)

Not exceeding 1000/100 ml in 20 per cent of samples in the year and in 3 consecutive samples in monsoon months.

6.

Biochemical Oxygen Demand (3 days at 27°C)

5mg/1

To maintain water relatively free from pollution caused by sewage and other decomposable wastes.

TABLE 1.5

PRIMARY WATER QUALITY CRITERIA FOR CLASS SW-V WATERS

(For Navigation and Controlled Waste Disposal)

S.No.	Parameter	Standards	Rationale/Remarks
1.	pH range	6.0-9.0	As specified by New England Inter-state Water Pollution Control Commission.
2.	Dissolved Oxygen	3.0 mg/1 or 40 percent saturation value whichever is higher	To protect aquatic lives
3.	Colour and Odour	None in such concentrations that would impair any usages specially assigned to this class.	As in (i) above
4.	Sludge deposits, Solid refuse floating solids oil grease and scum	None except for such small amount that may result from discharge of appropriately treated sewage and or industrial wastes effluents.	As in (1) above
5.	Fecal Coliform	500/100 ml (MPN)	Not exceeding 1000/100ml in 20 per cent of samples in the year and in 3 consecutive samples in monsoon months
67.	Emission Regulations for Rayon Industry:-		
(a)	Existing Plants	Estimation of Uncontrolled emission quantity (EQ) of CS ₂	For VSF
		EQ=125 Kg of CS ₂ /t of fibre	For VFY,
		EQ=225 Kg of CS ₂ /t of fibre	Stack Height (H) requirement, m
	Remarks	11Q 0.41-3 Vs D/u	A minimum of 80 per cent. of total emission shall pass through stack. If the calculated stack height is less than 30m, a

minimum height of 30m shall be provided.

Where

Q - CS₂ emission rate, kg/hr

VS - stack exit velocity, m/sec.

D - diameter of stack, m.

U - annual average wind speed at top of stack, m/sec.

Multiple Stacks :

1. If there are more than one stack existing in the plant, the required height of all stacks shall be based on the maximum emission rate in any of the stacks. In other words, all the stacks carrying CS₂ emission shall be of same heights (based on the maximum emission rate).
2. Number of stacks shall not be increased from the existing number. However, the number of stacks may be reduced. The existing stacks may be rebuilt and if stacks are to be relocated, condition 3 below applies.
3. Spacing among the stacks (x) at the minimum shall be 3.0 H (in m). If distance, x, between two stacks is less than 3.0 H (in m), emission shall be considered as single point source and height of both the stacks shall be calculated considering all emission is going through one stack.

(b)

Ambient Air Quality Monitoring :

The industry shall instal three air quality monitoring stations for CS₂ and H₂S measurements in consultation with State Pollution Control Board (SPCB) to ensure attainment of WHO recommended ambient air quality norms (CS₂ = 100 ug/m³ and H₂S = 150 ug/m³, 24 = hr. average).

(c)

[For new plants/expansion projects being commissioned on or after 1.6.1999.

[Substituted by G.S.R. 640(E), dated 16.10.2006 (w.e.f. 16.10.2006).]

For new plants or expansion projects, the emission standards for existing plants covered in (a) above shall apply subject to compliance of the ambient air quality norms for CS₂ and H₂S indicated in (b) above. The new plants or expansion projects shall provide adequate space for undertaking retrofittings.

(Note: (a) and (b) above also apply to new plants/expansion projects).]

[Inserted by G.S.R. 7, dated 22.12.1998 (w.e.f. 2.1.1999)]

88.

[Generator Sets Run on Petrol and Kerosene

A. Emission Standards- The emission standards for Generator sets on Petrol and Kerosene shall be as follows:-

Class

Displacement

(CC)

CO

(g/kw-hr)

HC+NO_x

(G/kw-hr)

1

Upto

99

—

250

—

12

2

99

and upto 225

—

250

—

10

3

225

—

250

—

8

(i)

Test method shall be as specified in SAE J 1088 and the measurement mode shall be 01-3 mode cycle specified under ISO 8178: Part 4 (Weighting Factor of 0.3 for 100 percent load, 0.5 for 75 percent load and 0.2 for 50 percent load);

(ii)

Any of the following institutions shall test and certify emission standards for the petrol and kerosene based generator sets, at manufacturing stage, namely: -

(a)

The Automotive Research Association of India, Pune (Maharashtra);

(b)

The International Centre for Automotive Technology, Manesar (Haryana);

(c)

The Indian Oil Corporation, Research and Development Centre, Faridabad (Haryana);

(d)

The Indian Institute of Petroleum, Dehradun (Uttarakhand); and

(e)

The Vehicle Research Development Establishment, Ahmednagar (Maharashtra),

(iii)

, Type Approval or Conformity of Production certificates in respect of emission standards, issued prior to the date of publication of this notification and valid upto the 31st May 2014 or beyond, shall be re-issued considering above revised norms by the respective certification agency,

B. Noise Limits.- (i) The noise limit for new generator sets run with petrol and kerosene shall be as follows:-

Noise

Limits

Sound

Power Level Lwa

86

dBA

(ii)

Any of the following institutions shall undertake 'type approval' and for 'verification of conformity of production' for noise norms for petrol and kerosene gensets, namely :-

(a)

The Automotive Research Association of India, Pune (Maharashtra);

(b)

The International Centre for Automotive Technology, Manesar (Haryana);

(c)

The Fluid Control Research Institute, Palghat (Kerala);

(d)

The National Test House, Ghaziabad (Uttar Pradesh);

(e)

The National Aerospace Laboratory, Bangalore (Karnataka); and

(f)

The Naval Science and Technology Laboratory, Visakhapatnam (Andhra Pradesh).

C. General Conditions

1. Applicability.- The stipulations in respect of emissions and noise referred to in entry A and entry B shall apply to all

new generator sets using petrol and kerosene as fuel, manufactured in or, imported into India:

Provided that this provision shall not apply to,-

(a)

genset manufactured or, imported for the purpose of exports outside India; or,

(b)

genset intended for the purpose of Research and Development and not for sale or, captive use in India.

2. Requirement of certification. - Every manufacturer or importer (hereinafter referred to as manufacturer) of genset (hereinafter referred to as product) to which these conditions apply shall have a separate valid certificate of type approval for all the product models for emission as well as noise norms being manufactured or imported.

3. Verification of conformity of production. - Every manufacturer shall submit its products to the verification for conformity of production for emission and noise, by any of the institutions, as applicable, every conformity of production year.

4. Sale of generator sets not complying with these conditions. - The sale of product model, not having valid type approval certificate, or not complying with the emission or noise norms, as determined by the verification for conformity of production, shall continue to be prohibited in India.

5. Requirement of conformance labeling. - (1) The manufacturer of the product shall affix a conformance label on the product containing the following requirements, namely:-

(i)

the label shall be durable and legible;

(ii)

the label shall be affixed on a part necessary for normal operation of the product and not normally requiring replacement during the product life.

(2)

The conformance label must contain the following information, namely:-

(i)

name and address of the manufacturer (even, if the address is described in the owners manual);

(ii)

statement that this product conforms to the Environment (Protection) Rules, 1986; and

(iii)

type approval certificate number and time phase (namely from the January 2014, the January 2016 or the January 2017).

6. Nodal agency. - (1) The Central Pollution Control Board shall be the nodal agency for implementation of these stipulations.

(2)

In case of any dispute or difficulty in implementation of these rules the matter shall be referred to the nodal agency.

(3)

The nodal agency shall constitute a Standing Committee for emission related issues and a National Committee for noise related issues, respectively to advise it on all matters related to the implementation of these rules including the dispute, if any.

7. Compliance and testing procedure. - (1) The compliance and testing procedure as published from time to time, if reviewed by Central Pollution Control Board shall be followed.

(2)

The Central Pollution Control Board may revise the compliance and testing procedure.

(3)

The institutes referred to in paragraph A and B above shall submit the testing and certification details in respect of emission or, noise, as applicable to the Central Pollution Control Board, annually and the Central Pollution Control Board shall be free to depute its official(s) to oversee the testing.]

89.

[Noise standards for fire-crackers. -

[Inserted by Notification No. G.S.R. 682(E), dated 5.8.1999 (w.e.f. 19.11.1986)]

A. (i) The manufacture, sale or of fire-crackers generating noise level exceeding 125 dB(A) of 145 dB(C) at 4 metres distance from the point of bursting shall be prohibited.

(ii)

For individual fire-cracker constituting the series (joined fire-crackers), the above mentioned limit be reduced by $5 \log_{10}(N)$ dB, where N=number of crackers joined together.

B. The broad requirements for measurement of noise from fire-crackers shall be-

(i)

The measurements shall be made on hard concrete surface of minimum 5 metre diameter or equivalent.

(ii)

The measurements shall be made in free field conditions, i.e, there shall not be any reflecting surface upto 15 metre distance from the point of bursting

(iii)

The measurement shall be made with an approved sound level meter.

C. [Petroleum and Explosives Safety Organization]

[Substituted by Notification No. G.S.R. 535(E) dated 7.8.2013 (w.e.f 19.11.1986)]

shall ensure implementation of these standards.

Note. - dB(AI) : A-weighted impulse Sound Pressure level in decibel dB(C)pk: C - weighted Peak Sound Pressure level in decibel.]

[D. The fire crackers for the purpose of export shall be exempted from the sub-paragraphs A,B and C above, subject to the compliance of the following conditions, namely :-

(i)

the manufacturer shall have an export order;

(ii)

the fire crackers shall conform to the level prescribed in the country to which it is exported;

(iii)

they shall have a different packing colour code; and

(iv)

there shall be a declaration on the box "not for sale in India" or "only for export in other countries.]

90.

[Standards for coal mines]

[Inserted by G.S.R. 742(E), dated 25.9.2000 (w.e.f. 25.9.2000).]

1. Air Quality Standards. - The Suspended Particulate Matter (SPM), Respirable Particulate Matter (RPM), Sulphur Dioxide (SO₂) and Oxides of Nitrogen (NO_x) concentration in downwind direction considering predominant wind direction, at a distance of 500 metres from the following dust generating source shall not exceed the standards specified in the Tables I, II and III given below.

Dust Generating Sources. - Loading or unloading, Haul road, coal transportation road, Coal Handling Plant (CHP), Railway sliding, Blasting, Drilling, Overburden dumps, or any other dust generating external sources like coke ovens (hard as well as soft), briquette industry, nearby road, etc.

TABLE I

Category

Pollutant

Time weighted average

Concentration in Ambient Air

Method of Measurement

1

2

3

4

5

I

Suspended Particulate Matter(SPM)

Annual Average*

360 µg/m³

- High Volume Sampling (Average flow rate not less than 1.1 m³/minute)

New Coal Mines (Coal Mines commenced operation after the date of publication of this notification)

20 hours**

500 µg/m³

Respirable Particulate Matter (size less than 10 µm) (RPM)

Annual Average*

180 µg/ m³

Respirable Particulate Matter sampling and analysis

24 hours*

250 µg/m³

Sulphur Dioxide (SO₂)

Annual Average*

80 µg/m³

1. Improved west and Gaeke method

24 hours**

120 µg/m³

2. Ultraviolet fluorescence

Oxide of Nitrogen as NO₂

Annual Average*

80 µg/m³

1. Jacob and Hochheiser Modified (Na-Arsenic) Method

24 hours**

120 µg/m³

2. Gas phase Chemiluminescence

TABLE II

Category

Pollutant

Time weighted average

Concentration in Ambient Air

Method of Measurement

1

2

3

4

5

II

Suspended Particulate Matter (SPM)

Annual Average*

430 µg/m³

- High Volume Sampling (Average flow rate not less than 1.1 m³/minute)

Existing coal fields/mines given below:

Karanpura, Ramgarh, Giridih, Rajhara, Wardha, Nagpur Silewara, Pench Kanhan, Patharkhera, Umrer, Korba, Chirimiri, Central India Coalfields (including Baikunthpur, Bistrampur), Singrauli, Ib Valley, Talcher, Godavari-Valley and any other

24 hours**

600 µg/m³

Respirable Particulate Matter (size less than 10 µm) (RPM)

Annual Average*

215 µg/ m³

Respirable Particulate Matter sampling and analysis

24 hours*

300 µg/m³

SulphurDioxide (SO₂)

Annual Average*

80* µg/m³

1. Improved west and Gaeke method

24 hours**

120 µg/m³

2. Ultraviolet fluorescene

Oxide of Nitrogen as NO₂

Annual Average*

80 µg/m³

1. Jacob and Hochheiser Modified (Na-Arsenic) Method

24 hours**

120 µg/m³

2. Gas phase Chemiluminescence

TABLE III

Category

Pollutant

Time weighted average

Concentration in Ambient Air

Method of Measurement

1

2

3

4

5

III

Suspended Particulate Matter (SPM)

Annual Average*

500 µg/m³

- High Volume Sampling (Average flow rate not less than 1.1 m³/minute)

Coal mines located in the coal fields of -

- Jharia

-Raniganj

-Bokaro

24 hours**

700 µg/m³

RespirableParticulate Matter (size less than 10 µm) (RPM)

Annual Average*

250 µg/ m³

RespirableParticulate Matter sampling and analysis

24 hours*

300 µg/m³

SulphurDioxide (SO₂)

Annual Average*

80 µg/m³

1. Improved west and Gaeke method

24 hours**

120 µg/m³

2. Ultraviolet fluorescene

Oxide of Nitrogen as NO₂

Annual Average*

80 µg/m³

1. Jacob and Hochheiser Modified (Na-Arsenic) Method

24 hours**

120 µg/m³

2. Gas phase Chemiluminescence

Note. -

*Annual Arithmetic mean for the measurements taken in a year, following the guidelines for frequency of sampling laid down in clause 2.

** 24 hourly/8 hourly values shall be met 92% of the time in a year. However, 8% of the time it may exceed but not on two consecutive days.

Unauthorised construction shall not be taken as a reference of nearest residential or commercial place for monitoring.

In case, any residential or commercial or industrial place falls within 500 metres of any dust generating sources, the National Ambient Air Quality Standards notified under Schedule VII shall be applicable.

2. Frequency of Sampling

-Air quality monitoring at a frequency of once in a fortnight at the dust generating sources given in clause 1 shall be carried out.

-As a result of monthly monitoring, if it is found that the value of the pollutant is less than 50% of the specified standards for three consecutive months, then the sampling frequency may be shifted to two days in a quarter year (3 months).

-In case, the value has exceeded the specified standards, the air quality sampling shall be done twice a week. If the results of four consecutive weeks indicate that the concentration of pollutants is within the specified standards, then fortnight monitoring may be reverted to.

3. Effluent Standards. - The standards for effluent discharge into sewer or stream or land, are given below :

pH

-

5.5-9.0

Chemical Oxygen Demand (COD)

-

250 mg/l

Total Suspended Solids (TSS)

-

100 mg/l

200 mg/l (Land for irrigation)

OilGrease(OG)

-

10mg/l

(Monitoring frequency of these parameters shall be once in a fortnight.)

Optional parameters. - All other parameters indicated in the general standards for discharge of environment pollutants under Schedule VI, shall be in addition to the effluent standards specified under clause 3.

(Monitoring frequency shall be once in a year for the optional parameters.)

4.

Noise Level Standards

6.00 A.M. - 10.00 P.M.

10.00 P.M. - 6.00 A.M.

Noise level

Leq75 dB (A)

Leq70 dB (A)

(Monitoring frequency for noise level shall be once in fortnight).

Occupational exposure limit of noise specified by Director-General of Mines Safety (DGMS) shall be complied with by the coal mines.

[* * *]

[Omitted 91. by Notification No. G.S.R. 535(E) dated 7.8.2013 (w.e.f. 19.11.1986)]

92.

Standards for Effluents from Textile Industry

Parameter

Concentration not to exceed, milligram per litre (mg/l), except pH

pH

5.5-9.0

Total suspended solids

100

Bio-chemical oxygen demand (BOD)

30

Chemical oxygen demand (COD)

250

Total residual chlorine

1

Oil and grease

10

Total chromium as Cr

2

Sulphide as S

2

Phenolic compounds as C₆H₅OH

1

Note. - 1. Where the treated effluent is discharged into municipal sewer leading to terminal treatment plant, the BOD may be relaxed to 100 mg/l and COD to 400 mg/l.

2. The quantity of effluent (litre per kilogram of product) shall not exceed 100, 250 and 80 in composite cotton textile industry, composite woollen textile industry and textile processing industry, respectively.

93.

Primary Water Quality Criteria for Bathing Waters

In a water body or its part, water is subjected to several types of uses. Depending on the types of uses and activities, water quality criteria have been specified to determine its suitability for a particular purpose. Among the various types of uses there is one use that demands highest level of water quality or purity and that is termed as "Designated Best Use" in that stretch of water body. Based on this, water quality requirements have been specified for different uses in terms of primary water quality criteria. The primary water quality criteria for bathing water are specified along with the rationale in Table 1.

TABLE 1

PRIMARY WATER QUALITY CRITERIA FOR BATHING WATER

(Water used for organised outdoor bathing)

CRITERIA

RATIONALE

1. Fecal Coliform MPN/100 ml:

500 (desirable) 2500 (Maximum permissible)

To ensure low sewage contamination. Fecal Coliform and Fecal Streptococci are considered as they reflect the bacterial pathogenicity.

2. Fecal Streptococci MPN/100ml:

100 (desirable) 500 (Maximum permissible)

The desirable and permissible limits are suggested to allow for fluctuation in environmental conditions such as seasonal change, changes in flow conditions, etc.

3. pH:

Between 6.5-8.5

The range provides protection to the skin and delicate organs like eyes, nose, ears, etc., which are directly exposed during outdoor bathing.

4. Dissolved Oxygen:

5 mg/1 or more

The minimum dissolved oxygen concentration of 5 mg/1 ensures reasonable freedom from oxygen consuming organic pollution immediately upstream which is necessary for preventing production of anaerobic gases (obnoxious gases) from sediment.

5. Biochemical Oxygen demand 3 day,

27°C:

3 mg/1 or less

The Biochemical Oxygen Demand of 3 mg/1 or less of the water ensures reasonable freedom from oxygen demanding pollutants and prevents production of obnoxious gases.

94.

[Noise Limit for Generator Sets Run with Diesel]

[Inserted by G.S.R. 371(E), dated 17.5.2002 (w.e.f. 17.5.2002).]

1. Noise limit for diesel generator sets (upto 1000 KVA) manufactured on or after the [1st January, 2005]

[Substituted by G.S.R. 448(E), dated 12.7.2004 (w.e.f. 12.7.2004).]

The maximum permissible sound pressure level for new diesel generator (DG) sets with rated capacity upto 1000 KVA, manufactured on or after [the 1st January, 2005]

[Substituted by G.S.R. 448(E), dated 12.7.2004 (w.e.f. 12.7.2004).]

shall be 75 dB(A) at 1 metre from the enclosure surface.

The diesel generator sets should be provided with integral acoustic enclosure at the manufacturing stage itself.

The implementation of noise limit for these diesel generator sets shall be regulated as given in paragraph 3 below.

2. Noise limit for DG sets not covered by paragraph 1. Noise limits for diesel generator sets not covered by paragraph 1, shall be as follows:-

2.1

Noise from DG set shall be controlled by providing an acoustic enclosure or by treating the room acoustically, at the users end.

2.2

The acoustic enclosure or acoustic treatment of the room shall be designed for minimum 25 dB(A) insertion loss or for meeting the ambient noise standards, whichever is on the higher side (if the actual ambient noise is on the higher side, it may not be possible to check the performance of the acoustic enclosure/acoustic treatment. Under such circumstances the performance may be checked for noise reduction upto actual ambient noise level, preferably, in the night time). The measurement for Insertion Loss may be done at different points at 0.5m from the acoustic enclosure/room, and then averaged.

2.3

The DG set shall be provided with proper exhaust muffler with insertion loss of minimum 25 dB(A).

2.4

These limits shall be regulated by the State Pollution Control Boards and the State Pollution Control Committees.

2.5

Guidelines for the manufacturers/users of Diesel Generator sets shall be as under:

01.

The manufacturer shall offer to the user a standard acoustic enclosure of 25 dB(A) insertion loss and also a suitable exhaust muffler with insertion loss of 25 dB(A).

02.

The user shall make efforts to bring down the noise levels due to the DG set, outside his premises, within the ambient noise requirements by proper siting and control measures.

03.

Installation of a DG set must be strictly in compliance with the recommendations of the DG set manufacturer.

04.

A proper routine and preventive maintenance procedure for the DG set should be set and followed in consultation with the DG set manufacturer which would help prevent noise levels of the DG set from deteriorating with use.

3. Limits of Noise for DG sets (upto 1000 KVA) manufactured on or after the [1st January, 2005]
[Substituted by G.S.R. 448(E), dated 12.7.2004 (w.e.f. 12.7.2004).]

3.1

Applicability

01. These rules apply to DG sets upto 1000 KVA rated output, manufactured or imported in India, on or after [the 1st January, 2005]

[Substituted by G.S.R. 448(E), dated 12.7.2004 (w.e.f. 12.7.2004)]

02. These rules shall not apply to-

(a)

DG sets manufactured or imported for the purpose of exports outside India; and

(b)

DG sets intended for the purpose of sample and not for sale in India.

3.2

Requirement of Certification

[Every manufacturer or assembler or importer (hereinafter referred to as the "manufacturer")]

[Substituted by G.S.R. 752(E), dated 24.10.2008 (w.e.f. 24.10.2008).]

of DG set (hereinafter referred to as "product") to which these regulations apply must have valid certificates of Type Approval and also valid certificates of Conformity of Production for each year, for all the product models being [manufactured or assembled or imported]

[Substituted by G.S.R. 752(E), dated 24.10.2008 (w.e.f. 24.10.2008).]

from [the 1st January, 2005]

[Substituted by G.S.R. 448(E), dated 12.7.2004 (w.e.f. 12.7.2004).]

with the noise limit specified in paragraph 1.

3.3

Sale, import or use of DG sets not complying with the rules prohibited

No person shall sell, import or use of a product model, which is not having a valid Type Approval certificate and Conformity of Production certificate.

3.4

Requirement of conformance labelling

(i)

The [manufacturer]

[Substituted by G.S.R. 752(E), dated 24.10.2008 (w.e.f. 24.10.2008).]

of the "product" must affix a conformance label on the product meeting the following requirements:-

(a)

The label shall be durable and legible.

(b)

The label shall be affixed on a part necessary for normal operation of the "product" and not normally requiring replacement during the "product" life.

(ii)

The conformance label must contain the following information:-

(a)

Name and address of the [manufacturer]

[Substituted by G.S.R. 752(E), dated 24.10.2008 (w.e.f. 24.10.2008).]

(if the address is described in the owner's manual, it may not be included in the label).

(b)

Statement "This product conforms to the Environment (Protection) Rules, 1986".

(c)

Noise limit, viz., 75 dB(A) at 1m.

(d)

Type approval certificate number.

(e)

Date of manufacture of the product.

3.5

Nodal Agency

(i)

The Central Pollution Control Board shall be the nodal agency for implementation of these regulations.

(ii)

In case of any dispute or difficulty in implementation of these regulations, the matter shall be referred to the nodal agency.

(iii)

The nodal agency shall constitute a Committee to advise it on all matters; including the disputed matters, related to the implementation of these regulations.

3.6

Authorised agencies for certification

The following agencies are authorised to carry out such tests as they deem necessary for giving certificates for Type Approval and Conformity of Production testings of DG sets and to give such certificates :-

(i)

Automotive Research Association of India, Pune

(ii)

National Physical Laboratory, New Delhi

(iii)

Naval Science Technology Laboratory, Visakhapatnam

(iv)

Fluid Control Research Institute, Palghat

(v)

National Aerospace Laboratory, Bangalore

3.7

Compliance and Testing Procedure

The compliance and testing procedure shall be prepared and published by the Central Pollution Control Board, with the help of the certification agencies.

4. [0 Exemption from the provisions of paragraph 1 and 3, for the products (diesel generator sets upto 30 KVA) purchased by the Ministry of Defence, Government of India.]

[Inserted by G.S.R. 315(E), dated 16.5.2005 (w.e.f. 16.5.2005).]

The products manufactured in or imported into India till [30th April, 2007]

[Substituted by G.S.R. 464(E), dated 7.8.2006 (w.e.f. 7.8.2006).]

for the purpose of supplying to the Ministry of Defence, shall be exempted from the regulations given in paragraphs 1 to 3 above, subject to the following conditions, namely :-

(i)

The [manufacturer]

[Substituted by G.S.R. 752(E), dated 24.10.2008 (w.e.f. 24.10.2008).]

shall manufacture or import the products only after getting purchase order from the ministry of Defence and shall maintain the record of receipts, production/import, dispatch, etc., for inspection by the Central Pollution Control Board.

(ii)

The special dispensation for noise norms shall be only for the mobile Defence vehicles which, with the present design/configuration, cannot carry the gensets with acoustic enclosures.

(iii)

Director, Ministry of Defence shall ensure and maintain the serial number of all gensets for the Army and he shall also direct the manufacturers of these gensets to emboss on the engine and the main body of the gensets, the words "For the use of Army only".

(iv)

The genset serial number shall be specially assigned by Ministry of Defence with the request for proposal and contract purchase order and this information shall be forwarded to the Central Pollution Control Board for inspection as and when required.

(v)

Registers shall be maintained at the manufacturers premises and in the Ministry of Defence to ensure that the number of gensets manufactured under special dispensation are not misused.

(vi)

The gensets procured under this dispensation shall be operated in the remote areas and not in the cities.

(vii)

This shall be a one-time exemption during which the Army shall remodel its vehicles to contain the new gensets and also obtain the necessary Type Approval of the gensets.

5. [0 Exemption from the provisions of paragraph 1 and 3 for sixteen Diesel Generator sets of 45 KVA purchased by the Ministry of Defence, Government of India.]

[Inserted by G.S.R. 566(E), dated 29.8.2007 (w.e.f. 29.8.2007).]

The 45 KVA DG sets manufactured in India for the purpose of their use in Mobile Decontamination System for use by the Ministry of Defence shall be exempted from the regulations given in paragraph 1 to 3 above subject to the following conditions, namely :-

(i)

The special dispensation for the noise norms shall be only for the DG sets to be used in Mobile Decontamination System (MDS) by Army which, with the present design/configuration cannot carry the gensets with acoustic enclosures.

(ii)

The Director, Ministry of Defence shall ensure and maintain the serial numbers for sixteen gensets and he shall also direct the manufacturers of these generator sets to emboss on the engine and main body of the gensets, the words "For the use of Army only in Mobile Decontamination System (MDS)".

(iii)

A register shall be maintained at the manufacturers premises and in the Ministry of Defence to ensure that only sixteen numbers of 45 KVA gensets are manufactured under special dispensation and are not misused elsewhere.

6. [0 Transportation of Diesel Generator Sets (above 250 KVA)

(i)

Diesel Generator set shall be transported after fulfilling the requirement of certification specified in paragraph 3.2 as a complete unit with acoustic enclosure, or dismantled, with relevant genset number specified on acoustic enclosure and silencer for reassembling at the site of its operation.

(ii)

Compliance with the noise norms shall be monitored after reassembling the D.G. set at the location of the installation by the concerned State Pollution Control Board or, as the case may be, the Union Territory Pollution Control Committee.]

95.

[Emission limits for new diesel engine up to 800 kW for generator set (Genset) application.- The emission limits for new diesel engine upto 800 kW for generator set (hereinafter referred to as Genset) application [shall, in respect of the power category 'upto 19 kW', as specified in column (I) of the table below, be effective from [1st May, 2015]

[Substituted by Notification No. G.S.R. 771(E) dated by 11.12.2013 (w.e.f. 19.11.1986)]

, and in respect of the power categories 'more than 19 KW upto 75 KW' and 'more than 75 KW upto 800 KW, as specified in the said Table, be effective from 1st July, 2014,]

[Inserted by G.S.R. 752(E), dated 24.10.2008 (w.e.f. 24.10.2008).]

subject to the general conditions contained therein, namely:-

TABLE

Power Category

Emission

Limits (g/k W-hr)

Smoke Limit

(light absorption coefficient, M-1)

NOx+HC

CO
PM
Upto 19 KW
—
7.5
_3.5
_0.3
_0.7
More than 19 KW
upto 75 KW
—
4.7
—
3.5
—
0.3
—
0.7
More than 75 KW
upto 800 KW

—
4.0
—
3.5
—
0.2
_0.7

Note:

1. The abbreviations used in the Table shall mean as under: NO_x - Oxides of Nitrogen; HC - Hydrocarbon; CO - Carbon Monoxide; and PM - Particulate Matter.
2. Smoke shall not exceed above value throughout the operating load points of the test cycle.
3. The testing shall be done as per D2 - 5 mode cycle of ISO: 8178- Part 4.
4. The above mentioned emission limits shall be applicable for Type Approval and Conformity of Production (COP) carried out by authorised agencies.
5. Every manufacturer, importer or, assembler (hereinafter referred to as manufacturer) of the diesel engine (hereinafter referred to as 'engine') for genset application manufactured or imported into India or, diesel genset (hereinafter referred to as 'product'), assembled or imported into India shall obtain Type Approval and comply with COP of their product(s) for the emission limits which shall be valid for the next COP year or, the date of implementation of the revised norms specified above, whichever earlier.

Explanation- The term 'COP year' means the period from 1st April to 31st March.

6. Stack height (in metres), for genset shall be governed as per Central Pollution Control Board (CPCB) guidelines.

General Conditions

1. Applicability - These conditions shall apply to all new engines for genset application and products manufactured, assembled or, imported into India, as the case may be :

Provided that these rules, shall not apply to, -

- (a) any engine or, product, assembled or manufactured or imported, as the case may be, for the purpose of export outside India, or;
- (b) any engine or product intended for the purpose of sample limited to four in number and to be exported back within three months, and not for sale in India.

2. Requirement of certification - Every manufacturer of engine or product, as the case may be, shall have valid certificate(s) of Type Approval and COP for each COP year, for all engine models being manufactured or, for all engine or product models being imported, after the effective date for the emission limits, as specified above and the COP for the genset [of the power category 'upto 19KW' sold on or after [1st May, 2015]

[Substituted for the words "shall be effective from 1st July, 2014 as specified in the Table below" by Notification No. G.S.R. 789 (E) dated 11.11.2014 (w.e.f. 19.11.1986)]

and of the power categories 'more than 19kW upto 75 KW' and 'more than 75 KW upto 800 KW' sold on or after 1st July, 2014 shall be effective and in force as per revised emission norms with effect from [1st May, 2016]

[Substituted for the words "1st January, 2016" by Notification No. G.S.R. 105(E) dated 17.2.2015 (w.e.f. 19.11.1986)] and 1st July, 2015, respectively].

3. Sale, import or use of engine or product not complying with these rules - No person shall sell, import or use an engine for genset application or, a product which is not having a valid Type Approval certificate and certificate of COP referred to in condition 2.

4. Requirement of conformance labeling. - (1) All the engines, individually or as part of the product shall be clearly engraved 'Genset Engine' on the cylinder block.

(2)

the engine or the product shall be affixed with a conformance label meeting the following requirements,namely:-

(a)

the label shall be durable and legible;

(b)

the label shall be affixed on a part necessary for normal operation of the engine or the product and not normally requiring replacement during the life of the engine or the product.

(3)

The conformance label shall contain the following information, namely:-

(a)

name and address of the manufacturer of engine or product, as the case may be;

(b)

statement that the engine or product conforms to the Environment (Protection) Rules, 1986;

(c)

Type Approval certificate number;

(d)

date of manufacture of engine and the product or in case of import, the date of import of the engine and the product; and

(e)

rated speed and corresponding gross power in kW.

5. Nodal Agency. - (1) The Central Pollution Control Board shall be the nodal agency for implementation of these rules.

(2)

In case of any dispute or difficulty in implementation of these rules, the matter shall be referred to the nodal agency.

(3)

The nodal agency shall constitute a Committee to advise it on all matters, including the disputed matters, related to the implementation of these rules.

6. Authorised agencies for certification. - The following institutions are authorised to carry out such tests as they may deem necessary, for giving certificates of Type Approval and Conformity of Production tests for diesel engines or products and to give such certificates, namely:-

(i)

the Automotive Research Association of India, Pune (Maharashtra);

(ii)

the International Centre for Automotive Technology, Manesar (Haryana);

(iii)

the Indian Oil Corporation, Research and Development Centre, Faridabad (Haryana);

(iv)

the Indian Institute of Petroleum, Dehradun (Uttarakhand); and

(v)

the Vehicle Research Development Establishment, Ahmednagar (Maharashtra).

7. Compliance and testing procedure. - (1) The Compliance and Testing Procedure, as published by the Central Pollution Control Board shall be followed by all concerned.

(2)

The authorised agencies for certification shall submit the testing and certification details in respect of the emission to the Central Pollution Control Board annually.

8. Fuel Specification - The specification of commercial fuel applicable for diesel gensets shall be the same as applicable for commercial High Speed Diesel applicable for diesel vehicles in the area where product would be operated, from time to time, as per policy of Government of India.

9. Engine component or parts identification - All the details of engine components or parts responsible for the emission performance shall be clearly marked in English language.]

[Exemption from the Emission Norms for twenty Diesel Gensets of 113.2 kW to be procured by Bharat Electronics Limited for 3D Tactical Control Radar for Indian Army.

[Inserted by Notification No. G.S.R. 54 (E) dated 23.1.2015 (w.e.f. 19.11.1986)]

The 113.2 kW diesel gensets manufactured by M/s Kirloskar Oil Engine Limited, Pune for the purpose of their use in 3D Tactical Control Radars System for use by the Indian Army shall be exempted from the regulations specified for new diesel engine upto 800 kW for generator set, under this entry, subject to following conditions, namely:-

(i)

the said gensets would comply with emission norms as existing on 30th June 2013, as per G.S.R. 371(E), dated 17th May, 2002;

(ii)

the special dispensation for the emission norms shall be only for the diesel gensets, not exceeding twenty in number, to be used in 3D Tactical Control Radars System, with the present design or configuration which shall be procured and used on or before 30th June, 2015;

(iii)

the General Manager (Radar) and Unit Head, Bharat Electronics Limited, Sahibabad shall ensure and maintain the serial numbers for twenty gensets and shall direct the manufacturer of these generator sets to emboss on the engine and main body of the gensets, the words "For the use of Army only in 3D Tactical Control Radars System".

(iv)

a register shall be maintained at the premises of the manufacturer or assembler of gensets and in the Bharat Electronics Limited, Sahibabad to ensure that only twenty diesel gensets of 113.2 kW are manufactured under special dispensation and this exemption shall not be misused elsewhere, for similar or other purpose.]

[95A. Genset run on dedicated Natural Gas (NG) or Liquid Petroleum Gas (LPG) :-

A. Emission Limits. - The emission limits for dedicated NG or LPG driven engine for genset application or genset (upto 800 kW) shall be effective from 1st July, 2016 as specified in the Table below subject to the general conditions contained therein, namely:-

Table

Power Category

Emission Limits (g/kW-hr)

NOX + NMHC Or NOX + RHC

CO

Upto 19 kW

? 7.5

? 3.5

More than 19 kW upto 75 kW

? 4.7

? 3.5

More than 75 kW upto 800 kW

? 4.0

? 3.5

Notes. - The abbreviations used in the Table shall mean as under: 1. NOX - Oxides of Nitrogen; CO - Carbon Monoxide; NMHC - Non-Methane Hydrocarbon; and RHC - Reactive Hydrocarbon.

2. Dedicated NG or LPG genset engine shall mean a mono-fuel engine starting and operating with only one fuel, i.e., NG or LPG.

3. NOX + NMHC or NOX + RHC shall be measured in case of dedicated NG or LPG genset engine. NMHC shall be equal to $0.3 \times \text{Total Hydrocarbon (THC)}$ in case of NG, and RHC is equal to $0.5 \times \text{THC}$ in case of LPG.

4. These norms shall be applicable to Original Equipment Manufacturer (OEM) built dedicated NG or LPG genset engines.

5. The above mentioned emission limits shall be applicable for Type Approval and Conformity of Production (COP) carried out by authorised agencies.

6. Any of the following institutions shall undertake Type Approval and for verification of Conformity of Production for emission standards for engine products and to issue such certificates on compliance of the prescribed norms, namely:-

(a)

The Automotive Research Association of India, Pune (Maharashtra);

(b)

The International Centre for Automotive Technology, Manesar (Haryana);

(c)

The Indian Oil Corporation, Research and Development Centre, Faridabad (Haryana);

(d)

The Indian Institute of Petroleum, Dehradun (Uttarakhand); and

(e)

The Vehicle Research Development Establishment, Ahmednagar (Maharashtra).

7. Stack height (in metres), for genset shall be governed as per Central Pollution Control Board (CPCB) guidelines.

B. Noise Limits. - 1. The maximum permissible sound pressure level for genset, with rated capacity upto 800 kW shall be 75 dB(A) at 1 metre from the enclosure surface. Gensets should be provided with integral acoustic enclosure at the manufacturing stage itself. The noise norms shall be effective from the 1st January, 2017.

2. Noise limit for gensets not covered under paragraph (1) shall be as follows:-

(a)

Noise from gensets shall be controlled by providing an acoustic enclosure or by treating the room acoustically, at the users end.

(b)

The acoustic enclosure shall be designed for minimum 25 dB(A) insertion loss or for complying with the ambient noise standards, whichever is on the higher side (if the actual ambient noise is on the higher side, it may not be possible to check the performance of the acoustic enclosure or acoustic treatment. Under such circumstances the performance may be checked for noise reduction upto actual ambient noise level, preferably, in the night time between 10.00 pm-6.00 am). The measurement for insertion loss may be done at different points at 0.5m from the acoustic enclosure or room, and then averaged.

(c)

The genset shall be provided with proper exhaust muffler with insertion loss of minimum 25 dB(A).

(d)

These limits shall be regulated by the State Pollution Control Boards and the State Pollution Control Committees.

(e)

The manufacturer shall offer to the user a standard acoustic enclosure of 25 dB(A) insertion loss and also a suitable exhaust muffler with insertion loss of 25 dB(A).

(f)

The user shall make efforts to bring down the noise levels due to the genset, outside his premises, within the ambient noise requirements by proper siting and control measures.

(g)

Installation of a genset shall be strictly in compliance with the recommendation of the genset manufacturer.

(h)

A proper routine and preventive maintenance procedure for the genset shall be set and followed in consultation with the

genset manufacturer.

3. Any of the following institutions shall undertake Type Approval and for verification of Conformity of Production for noise norms for dedicated NG or LPG gensets and issue such certificates on compliance of the prescribed norms, namely :-

(a)

The Automotive Research Association of India, Pune (Maharashtra);

(b)

The International Centre for Automotive Technology, Manesar (Haryana);

(c)

The Fluid Control Research Institute, Palghat (Kerala);

(d)

The National Test House, Ghaziabad (Uttar Pradesh);

(e)

The National Aerospace Laboratory, Bangaluru (Karnataka); and

(f)

The Naval Science and Technology Laboratory, Visakhapatnam (Andhra Pradesh).

(g)

[National Physical Laboratory, New Delhi.]

C. General Conditions. - 1. Every manufacturer, importer or assembler (hereinafter referred to as the 'manufacturer') of the dedicated NG or LPG engine (hereinafter referred to as 'engine') for genset application manufactured or imported into India or dedicated NG or LPG genset (hereinafter referred to as 'product'), assembled or imported into India shall obtain Type Approval and comply with the COP of their products for the emission limits which shall be valid for the next COP year or, the date of implementation of the norms specified above, whichever is earlier. Thereafter, the manufacturer shall obtain COP approval every COP year. NG or LPG kit shall also have independent Type Approval and shall independently comply with COP requirement, as and when notified.

2. These conditions shall apply to all new engines for genset application and products manufactured, assembled or imported into India, as the case may be:

Provided that these rules, shall not apply to, -

(a)

any engine or, product, assembled or manufactured or imported, as the case may be, for the purpose of export outside India; or

(b)

any engine or product intended for the purpose of sample limited to four in number and to be exported back within three months, and not for sale in India.

3. Every manufacturer of engine or product, as the case may be, shall have valid certificates of Type Approval and COP for each COP year, for all engine models being manufactured or, for all engine or product models being imported, after the effective date of the emission limits, as specified above and CPCB shall develop system and procedure to monitor the norms and COP year.

4. Every manufacturer shall submit its engine or products, as the case may be, for the verification for conformity of production for emission and noise, by any of the institutions, as applicable, every COP year.

Note. - The term COP year, duty cycle and any other requirement for compliance of Type Approval and COP shall be prescribed in the system and procedure to be developed by the Central Pollution Control Board.

5. No person shall manufacture, sell, import or use an engine for genset application or any product which is not having a valid Type Approval certificate and certificate of COP referred to in sub - paragraph (3) above, as applicable.

6. All the engines, individually or as part of the product shall be clearly engraved as NG genset Engine or LPG genset engine on the cylinder block, as the case may be.

7. The engine or the product shall be affixed with a conformance label meeting the following requirements, namely:-

(a)

the label shall be durable and legible;

(b)

the label shall be affixed on a part necessary for normal operation of the engine or the product and not normally

requiring replacement during the life of the engine or the product.

8. The conformance label shall contain the following information, namely:-

- (a)
Name and address of the manufacturer of engine or product, as the case may be;
- (b)
Statement that the engine or product conforms to the Environment (Protection) Rules, 1986;
- (c)
Type Approval Certificate number;
- (d)
Date of manufacture of engine and the product or in case of import, the date of import of the engine and the product;
and
- (e)
Rated speed and corresponding gross power in kW.

9. (a) The Central Pollution Control Board shall be the nodal agency for implementation of these rules.

- (b)
In case of any dispute or difficulty in implementation of these rules, the matter shall be referred to the nodal agency.
- (c)

The nodal agency shall constitute a Standing Committee for emission related issues and a National Committee for noise related issues, respectively, to advise it on all matters related to the implementation of these rules including disputes, if any.

10. (a) All genset engines operating on dedicated NG or LPG shall be tested for Type Approval and COP of emission and noise limits compliance as per system and procedure published from time to time by the Central Pollution Control Board.

- (b)
The Central Pollution Control Board may develop or as the case may be, revise the compliance and testing procedure allowing a time of six months for all concern.
- (c)

The institutes referred to in paragraph A and B above shall submit the testing and certification details in respect of emission and, or, noise, as applicable, to the Central Pollution Control Board, annually and the Central Pollution Control Board shall be free to depute its officials to oversee the testing.

11. All genset engines operating on dedicated NG or LPG shall comply safety requirements.

12. The specification of commercial fuel applicable for NG or LPG shall be applicable for vehicles run on NG or LPG in the area where product shall be operated, from time to time, as per policy of Government of India.

13. All the details of engine components or part responsible for the emission performance shall be clearly marked in English language.

95B.

Genset run on Petrol and Natural Gas (NG) or Petrol and Liquid Petroleum Gas (LPG):-

A. Emission Limits. - The emission limits for Petrol and NG or Petrol and LPG genset (upto 19 kW) powered by SI engine (upto 400 cc displacement) (hereinafter referred to as Genset) shall be effective from the 1st August, 2016 as specified in the following Table:-

Table

Class

Engine Displacement (cc)

CO (g/kWh)

NOX + THC/ NOX + NMHC/ NOX + RHC (g/kWh)

1.

Upto 99

? 250

? 12

2.

99 and upto 225

? 250

? 10

3.

?225? 400

? 250

? 8

Notes. - The abbreviations used in above table shall mean as under: 1. SI - Spark Ignition, NG - Natural Gas, LPG - Liquid Petroleum Gas, NOX - Oxides of Nitrogen, THC - Total Hydrocarbon, CO - Carbon Monoxide, NMHC - Non-Methane Hydrocarbon and RHC - Reactive Hydrocarbon.

2. Dual fuel engine operation shall mean a two - fuel system having petrol as a primary combustion fuel and NG or LPG as supplementary fuel, both in a certain proportion, throughout the engine operating zone. Such dual fuel Genset engine may operate on petrol stand-alone mode in absence of gaseous fuel i.e. NG or LPG.

3. NMHC shall be equal to $0.3 \times \text{THC}$ in case of Natural Gas and RHC shall be equal to $0.5 \times \text{THC}$ in case of LPG.

4. These norms shall be applicable to Original Equipment Manufacturer (OEM) built petrol and NG or petrol and LPG genset (upto 19 kW) powered by SI engine (upto 400 cc displacement). Conversion or Retrofitment of the existing petrol or Kerosene generator sets to run on petrol and NG or petrol and LPG shall not be permitted.

5. The above emission limits shall be applicable for Type Approval and Conformity of Production (COP) undertaken by authorized agencies and shall be complied with petrol alone or petrol and NG or LPG fuel mode separately.

6. Any of the following institutions shall undertake Type Approval and for verification of Conformity of Production for emission standards for engine products and to issue such certificates on compliance of the prescribed norms, namely:-

(a)

The Automotive Research Association of India, Pune (Maharashtra);

(b)

The International Centre for Automotive Technology, Manesar (Haryana);

(c)

The Indian Oil Corporation, Research and Development Centre, Faridabad (Haryana);

(d)

The Indian Institute of Petroleum, Dehradun (Uttarakhand); and

(e)

The Vehicle Research Development Establishment, Ahmednagar (Maharashtra).

7. NOX + THC shall be measured as emissions from petrol alone in bi-fuel fuel mode of operation. NOX + NMHC or NOX +RHC shall be measured in case of petrol and NG or petrol and LPG fuel mode of operation, respectively.

B. Noise Limits. - 1. The noise limit for gensets (upto 19 kW) powered by an SI engine (upto 400 cc displacement) run on petrol and NG or petrol and LPG shall be effective from the 1st September, 2016 as specified in the following table :-

Table

Noise Parameter

Noise Limits

Sound Power Levelwa

86 dB(A)

2. Any of the following institutions shall undertake Type Approval and for verification of Conformity of Production for noise norms for dedicated petrol or petrol and NG or LPG gensets and issue such certificates on compliance of the prescribed norms, namely :-

(a)

The Automotive Research Association of India, Pune (Maharashtra);

(b)

The International Centre for Automotive Technology, Manesar (Haryana);

(c)

The Fluid Control Research Institute, Palghat (Kerala);

(d)

The National Test House, Ghaziabad (Uttar Pradesh);

(e)

The National Aerospace Laboratory, Bangaluru (Karnataka); and

(f)

The Naval Science and Technology Laboratory, Visakhapatnam (Andhra Pradesh).

(g)

[National Physical Laboratory, New Delhi.]

[Inserted by Notification No G.S.R. 97(E), dated 29.1.2018 (w.e.f. 19.11.1986).]

C. General Conditions. - 1. Every manufacturer, importer or assembler (hereinafter referred to as the 'manufacturer') of the petrol and NG or LPG engine (hereinafter referred to as 'engine') for genset application manufactured or imported into India or petrol and NG or LPG genset (hereinafter referred to as 'product'), assembled or imported into India shall obtain Type Approval and comply with the COP of their products for the emission limits which shall be valid for the next COP year or, the date of implementation of the norms specified above, whichever is earlier. Thereafter, the manufacturer shall obtain COP approval every COP year. Petrol and NG or LPG kit shall also have independent Type Approval and shall independently comply with COP requirement, as and when notified.

2. These conditions shall apply to all new engines for genset application and products manufactured, assembled or imported into India, as the case may be:

Provided that these rules, shall not apply to, -

(a)

any engine or, product, assembled or manufactured or imported, as the case may be, for the purpose of export outside India; or

(b)

any engine or product intended for the purpose of sample limited to four in number and to be exported back within three months, and not for sale in India.

3. Every manufacturer of engine or product, as the case may be, shall have valid certificates of Type Approval and COP for each COP year, for all engine models being manufactured or, for all engine or product models being imported, after the effective date of the emission limits, as specified above and CPCB shall develop system and procedure to monitor the norms and COP year.

4. Every manufacturer shall submit its engines or products to the verification for conformity of production for emission and noise, by any of the institutions, as applicable, every COP year.

Note. - The term 'COP year', duty cycle and any other requirement for compliance of Type Approval and COP to be prescribed in the System and Procedure developed by the Central pollution Control Board.

5. No person shall manufacture, sell, import or use an engine for genset application or any product which is not having a valid Type Approval certificate and certificate of COP referred to in sub - paragraph (3) above, as applicable.

6. All the engines, individually or as part of the product shall be clearly engraved as petrol and NG genset Engine or petrol and LPG genset engine on the cylinder block, as the case may be.

7. The engine or the product shall be affixed with a conformance label meeting the following requirements, namely:-

(a)

the label shall be durable and legible;

(b)

the label shall be affixed on a part necessary for normal operation of the engine or the product and not normally requiring replacement during the life of the engine or the product.

8. The conformance label shall contain the following information, namely:-

(a)

Name and address of the manufacturer of engine or product, as the case may be;

(b)

Statement that the engine or product conforms to the Environment (Protection) Rules, 1986;

(c)

Type Approval Certificate number;

(d)

Date of manufacture of engine and the product or in case of import, the date of import of the engine and the product;

and

(e)

Rated speed and corresponding gross power in kW.

9. (a) The Central Pollution Control Board shall be the nodal agency for implementation of these rules.

(b)

In case of any dispute or difficulty in implementation of these rules, the matter shall be referred to the nodal agency.

(c)

The nodal agency shall constitute a Standing Committee for emission related issues and a National Committee for noise related issues, respectively, to advise it on all matters related to the implementation of these rules including disputes, if any.

10. (a) All genset engines operating on petrol and NG or LPG shall be tested for Type Approval and COP of emission and noise limits compliance as per system and procedure published from time to time by the Central Pollution Control Board.

(b)

The Central Pollution Control Board may develop or as the case may be, revise the compliance and testing procedure allowing a time of six months for all concern.

(c)

The institutes referred to in paragraph A and B above shall submit the testing and certification details in respect of emission and, or, noise, as applicable, to the Central Pollution Control Board, annually and the Central Pollution Control Board shall be free to depute its officials to oversee the testing.

11. All genset engines operating on petrol and NG or LPG shall comply safety requirements.

12. The specification of commercial fuel applicable for petrol and NG or LPG shall be applicable for vehicles run on petrol and NG or LPG in the area where product shall be operated, from time to time, as per policy of Government of India.

13. All the details of engine components or part responsible for the emission performance shall be clearly marked in English language.

95C.

Genset run on Diesel and Natural Gas (NG) or Diesel and Liquid Petroleum Gas (LPG) :-

A. Emission Limits. - The emission limits for Diesel and NG or Diesel and LPG driven engine (upto 800 kW) for generator set (hereinafter referred to as Genset) application shall be effective from the 1st July, 2016 as specified in the Table below, subject to the general conditions specified therein, namely:-

Table

Power Category

Emission Limits (g/kW-hr)

Smoke Limit (light absorption coefficient,m-1)

NOX + THC or NOX + NMHC or RHC

CO

PM

Upto 19 kW

? 7.5

? 3.5

? 0.3

? 0.7

More than 19 kW upto 75 kW

? 4.7

? 3.5

? 0.3

? 0.7

More than 75 kW upto 800 kW

? 4.0

? 3.5

? 0.2

? 0.7

Notes. - The abbreviations used in the Table shall mean as under: 1. NOX - Oxides of Nitrogen; THC - Total Hydrocarbon; CO- Carbon Monoxide; PM - Particulate Matter; NMHC - Non - Methane Hydrocarbon; and RHC - Reactive Hydrocarbon.

2. Dual fuel engine operation shall mean a two - fuel system having diesel as a primary combustion fuel and NG or LPG as supplementary fuel, both in a certain proportion, throughout the engine operating zone. Such dual fuel genset engine may operate on diesel stand-alone mode in absence of gaseous fuel i.e. NG or LPG.

3. NOX + THC shall be measured as emission while diesel alone is used as fuel. NOX + NMHC or NOX + RHC shall be measured in case of diesel and NG or diesel and LPG dual fuel operation respectively. NMHC shall be equal to $0.3 \times$ THC in case of NG and RHC as $0.5 \times$ THC in case of LPG.

4. These norms shall be applicable to Original Equipment Manufacturer (OEM) built diesel and NG or LPG Genset engines. Conversion or retro fitment of the existing diesel engines to run on diesel and NG or diesel and LPG shall not be permitted.

5. The above mentioned emission limits shall be applicable for Type Approval and Conformity of Production (COP) carried out by authorised agencies. For Type Approval and COP for diesel and NG or diesel and LPG dual fuel operation engines, the emission and smoke limits prescribed in above Table shall be met in diesel alone or diesel and NG or diesel and LPG dual fuel mode separately.

6. Any of the following institutions shall undertake Type Approval and for verification of Conformity of Production for emission standards for engine products and to issue such certificates on compliance of the prescribed norms, namely:-

(a)

The Automotive Research Association of India, Pune (Maharashtra);

(b)

The International Centre for Automotive Technology, Manesar (Haryana);

(c)

The Indian Oil Corporation, Research and Development Centre, Faridabad (Haryana);

(d)

The Indian Institute of Petroleum, Dehradun (Uttarakhand); and

(e)

The Vehicle Research Development Establishment, Ahmednagar (Maharashtra).

7. Stack height (in metres) for genset shall be governed as per Central Pollution Control Board (CPCB) guidelines.

8. NOX + THC shall be measured as emissions from diesel alone in bi-fuel fuel mode of operation. NOX + NMHC or NOX +RHC shall be measured in case of diesel and NG or diesel and LPG fuel mode of operation, respectively.

9. The emission standards for smoke and particulate matter shall be applicable, when diesel is used as fuel. Smoke limit prescribed in above Table shall not exceed throughout the operating load points of the test cycle.

B. Noise Limits. - 1. The maximum permissible sound pressure level for genset, with rated capacity upto 800 kW shall be 75 dB(A) at 1 metre from the enclosure surface. Gensets shall be provided with integral acoustic enclosure at the manufacturing stage itself. The noise norms shall be effective from the 1st January, 2017.

2. Noise limit for gensets not covered under paragraph (1) shall be as follows:-

(a)

Noise from gensets shall be controlled by providing an acoustic enclosure or by treating the room acoustically, at the users end.

(b)

The acoustic enclosure shall be designed for minimum 25 dB(A) insertion loss or for complying with the ambient noise standards, whichever is on the higher side (if the actual ambient noise is on the higher side, it may not be possible to check the performance of the acoustic enclosure or acoustic treatment. Under such circumstances the performance may be checked for noise reduction upto actual ambient noise level, preferably, in the night time between 10.00 PM-6.00 AM). The measurement for insertion loss may be done at different points at 0.5m from the acoustic enclosure or room, and then averaged.

(c)

The genset shall be provided with proper exhaust muffler with insertion loss of minimum 25 dB(A).

(d)

These limits shall be regulated by the State Pollution Control Boards and Pollution Control Committees.

(e)

The manufacturer shall offer to the user a standard acoustic enclosure of 25 dB(A) insertion loss and also a suitable exhaust muffler with insertion loss of 25 dB(A).

(f)

The user shall make efforts to bring down the noise levels due to the genset, outside his premises, within the ambient noise requirements by proper siting and control measures.

(g)

Installation of a genset shall be strictly made in compliance with the recommendations of the genset manufacturer.

(h)

A proper routine and preventive maintenance procedure for the genset shall be set and followed in consultation with the genset manufacturer.

3. Any of the following institutions shall undertake Type Approval and for verification of Conformity of Production for noise norms for dedicated diesel or diesel and NG or LPG gensets and issue such certificates on compliance of the prescribed norms, namely :-

(a)

The Automotive Research Association of India, Pune (Maharashtra);

(b)

The International Centre for Automotive Technology, Manesar (Haryana);

(c)

The Fluid Control Research Institute, Palghat (Kerala);

(d)

The National Test House, Ghaziabad (Uttar Pradesh);

(e)

The National Aerospace Laboratory, Bangaluru (Karnataka); and

(f)

The Naval Science and Technology Laboratory, Visakhapatnam (Andhra Pradesh).

(g)

[National Physical Laboratory, New Delhi.]

[Inserted by Notification No G.S.R. 97(E), dated 29.1.2018 (w.e.f. 19.11.1986).]

C. General Conditions. - 1. Every manufacturer, importer or assembler (hereinafter referred to as the 'manufacturer') of the diesel and NG or LPG engine (hereinafter referred to as 'engine') for genset application manufactured or imported into India or diesel and NG or LPG genset (hereinafter referred to as 'product'), assembled or imported into India shall obtain Type Approval and comply with the COP of their products for the emission limits which shall be valid for the next COP year or, the date of implementation of the norms specified above, whichever is earlier. Thereafter, manufacturer shall obtain COP approval every COP year. Diesel and NG or LPG kit shall also have independent Type Approval and shall independently comply with COP requirement, as and when notified.

2. These conditions shall apply to all new engines for genset application and products manufactured, assembled or imported into India, as the case may be:

Provided that these rules, shall not apply to, -

(a)

any engine or, product, assembled or manufactured or imported, as the case may be, for the purpose of export outside India; or

(b)

any engine or product intended for the purpose of sample limited to four in number and to be exported back within three months, and not for sale in India.

3. Every manufacturer of engine or product, as the case may be, shall have valid certificates of Type Approval and COP for each COP year, for all engine models being manufactured or, for all engine or product models being imported, after the effective date of the emission limits, as specified above and CPCB shall develop system and procedure to monitor the norms and COP year.

4. Every manufacturer shall submit its engine or products, as the case may be, for the verification of conformity of production for emission and noise, by any of the institutions, as applicable, every COP year.

Note. - The term COP year, duty cycle and any other requirement for compliance of Type Approval and COP shall be prescribed in the system and procedure to be developed by the Central Pollution Control Board.

5. No person shall manufacture, sell, import or use an engine for genset application or any product which is not having a valid Type Approval certificate and certificate of COP referred to in sub - paragraph (3) above.

6. All the engines, individually or as part of the product shall be clearly engraved as diesel and NG genset Engine or diesel and LPG genset engine on the cylinder block, as the case may be.

7. The engine or the product shall be affixed with a conformance label meeting the following requirements, namely:-

(a)

the label shall be durable and legible;

(b)

the label shall be affixed on a part necessary for normal operation of the engine or the product and not normally requiring replacement during the life of the engine or the product.

8. The conformance label shall contain the following informations, namely:-

(a)

Name and address of the manufacturer of engine or product, as the case may be;

(b)

Statement that the engine or product conforms to the Environment (Protection) Rules, 1986;

(c)

Type Approval Certificate number;

(d)

Date of manufacture of engine and the product or in case of import, the date of import of the engine and the product; and

(e)

Rated speed and corresponding gross power in kW.

9. (a) The Central Pollution Control Board shall be the nodal agency for implementation of these rules;

(b)

In case of any dispute or difficulty in implementation of these rules, the matter shall be referred to the nodal agency;

(c)

The nodal agency shall constitute a Standing Committee for emission related issues and a National Committee for noise related issues, respectively, to advise it on all matters related to the implementation of these rules including disputes, if any.

10. (a) All genset engines operating on diesel and NG or LPG shall be tested for Type Approval and COP of emission and noise limits compliance as per system and procedure published from time to time by the Central Pollution Control Board.

(b)

The Central Pollution Control Board may develop or as the case may be, revise the compliance and testing procedure allowing a time for a period of six months for all concern.

(c)

The institutes referred to in paragraph A and B above shall submit the testing and certification details in respect of emission and, or, noise, as applicable, to the Central Pollution Control Board, annually and the Central Pollution Control Board shall be free to depute its officials to oversee the testing.

11. All genset engines operating on diesel and NG or LPG shall comply safety requirements.

12. The specification of commercial fuel applicable for diesel and NG or LPG shall be applicable for vehicles run on diesel and NG or LPG in the area where product shall be operated, from time to time, as per the policy of the Government of India.

13. All the details of engine components or part responsible for the emission performance shall be clearly marked in English language.]

[Inserted by Notification No. G.S.R. 281(E), dated 7.3.2016 (w.e.f. 19.11.1986).]

96.

[

[Inserted by G.S.R. 489(E), dated 9.7.2002 (w.e.f. 11.7.2002).]

Emission Standards for Diesel Engines (Engine rating more than 0.8 MW (800 KW) for Power Plant, Generator Set Applications and other Requirements.]

[Inserted by G.S.R. 640(E), dated 16.10.2006 (w.e.f. 16.10.2006).]

TABLE

Parameter

Area Category

Total engine rating of the plant (includes existing as well as new generator sets)

Generator sets commissioning date

Before 1-7-2003

Between 1-7-2003 and 1-7-2005

On or after 1-7-2005

NO_x(as NO₂) (at 15% O₂), dry basis, in ppmv

A

Upto 75 MW

1100

970

710

B

Upto 150 MW

A

More than 75 MW

1100

710

360

B

More than 150 MW

NMHC (as C) (at 15% O₂), mg/Nm³

Both A and B

150

100

PM (at 15% O₂), mg/Nm³

Diesel Fuels-HSD and LDO

Both A and B

75

75

Furnace Oils-LSHS and FO

Both A and B

150

100

CO (at 15% O₂), mg/Nm³Both A and B

150

150

Sulphurcontent in fuel

A

2%

B

4%

Fuel specification

For A only

Upto5 MW

Only Diesel Fuels (HSD, LDO) shall be used.

Stack height (for generator sets commissioned after 1-7-2003)

Stack height shall be maximum of the following, in metre:

- (i) $14 Q^{0.3}$, Q =Total SO₂emission from the plant in kg /hr.
- (ii) Minimum 6 m. above the building where generator set is installed,
- (iii) 30 m.

Note.-

1. Acronyms used:

MW .

Mega (10⁶) Watt

FO

Furnace Oil

NO_x

Oxides of Nitrogen

HSD

High Speed Diesel

NO₂

Nitrogen Dioxide

LDO

Light Diesel Oil

O₂

Oxygen

LSHS

LowSulphurHeavy Stock

NMHC

Non-Methane Hydrocarbon

kPa

Kilo Pascal

C

Carbon

mm

Milli(10⁻³) metre

PM

Particulate Matter

kg/hr

Kilo (10³) gram per hour

CO

Carbon Monoxide

mg/Nm³

Milli(10⁻³) gram per normal metre cubic

SO₂ppmv

SulphurDioxide part per million (10⁶) by volume

2. Area categories A and B are defined as follows:

Category A: Areas within the municipal limits of towns/cities having population more than 10 lakhs and also upto 5 km beyond the municipal limits of such town/cities.

Category B: Areas not covered by category A.

3. The standards shall be regulated by the State Pollution Control Boards or Pollution Control Committees, as the case may be.

4. Individual units with engine ratings less than or equal to 800 KW are not covered by this notification.

5. Only following liquid fuels, viz., High Speed Diesel, Light Diesel Oil, Low Sulphur Heavy Stock and Furnace Oil or liquid fuels with equivalent specifications shall be used in these power plants and generator sets.

6. For expansion project, stack height of new generator sets shall be as per total Sulphur Dioxide emission (including

existing as well as additional load).

7. For multi-engine plants, fuels shall be grouped in cluster to get better plume rise and dispersion. Provision for any future expansion should be made in planning stage itself.

8. Particulate matter, Non-Metharie Hydrocarbon and Carbon Monoxide results are to be normalised to 25°C, 1.01 Kilo Pascal (760 mm of mercury) pressure and zero percent moisture (dry basis).

9. Measurement shall be performed at steady load conditions of more than 85% of the rated load.

10. Continuous monitoring of Oxides of Nitrogen shall be done by the plants whose total engine capacity is more than 50 Mega Watt. However, minimum once in six month monitoring for other parameters shall be adopted by the plants.

11. Following methods may adopted for the measurement of emission parameters,-

Sl. No.

Emissionparameters

Measurement methods

1.

Particulates

Gravimetric

2.

SO₂

Barium Perchlorate-Thorin indicator method

3.

NO_x

Chemiluminescence, Non-Dispersive Infra Red, Non-Dispersive Ultraviolet (for continuous measurement), Phenol disulphonic method

4.

CO

Non-Dispersive Infra Red

5.

O₂

Paramagnetic, Electrochemical sensor

6.

NMHC

Gas Chromatograph-Flame Ionisation Detector].

97. []

[Added by G.S.R. 546(E), dated 30.8.2005 (w.e.f. 30.8.2005).]

Boilers Using Agriculture Waste as Fuel

Step Great Particulate matter

250

mg/Nm³

Horse Shoe/Pulsating

500

mg/Nm³

Particulate matter

(12%

of CO₂)

Spreader stoker

500mg/Nm³

Particulate matter

(12%

of CO₂);

98.

Guidelines for Pollution Control in Ginning Mills.-

Measures for Noise Control-

(i)
Creating separate soundproof enclosures for the fans within the ginning area

(ii)
Keeping the fans outside the ginning room in separate enclosures

(iii)
Roller gins may be covered by sound proof enclosures and use of pneumatic feeding of raw cotton while suction of ginned cotton is introduced to considerably reduce the dust pollution level

Measures for Dust Control-

(i)
The fugitive emission can be largely controlled by employing mechanical or pneumatic handling of raw material and ginned material through covered ducts and providing overhead hoods connected to exhaust through ducts and filters; use of lifting platforms for bale formers

(ii)
The overhead hoods with exhaust arrangement can be provided at:

(a)
The saw-ginning machine where manual handling to maintain proper feeding in the machine

(b)
At the feeding point of the roller ginning machine when manual feeding is carried out

(c)
At the collection points of ginned cotton from saw ginning condenser.

Sl. No.

Industry

Parameter

Standards

1

2

3

4

99. []

Sponge Iron

Plant (Rotary Kiln)

A. Emission

Standards*

Particulate

matter

Fuel Type

Limiting

value for concentration

coal

100 mg/Nm³

gas

50 mg/Nm³

Carbon

Monoxide (Vol./Vol.)

coal/gas

1%

Stack

Height** (minimum)

coal/gas

30.0m

Note.-*Emission

shall be normalised at 12% CO₂ in stack emission.

****Stack**

height shall be calculated as $H=14.0Q^{0.3}$ where Q is emission of Sulphur Dioxide (SO₂) in kg. /hr. i.e.

SO₂(kg/hr)

Height

(metre)

up to 12.68

30

12.69-33.08

40

33.09-69.06

50

69.07-127.80

60

127.81-213.63

70

(De-dusting

unit)

Particulate

matter

Existing unit

New unit

mg/Nm³

100

50

Note. -(i) Stack attached to de-dusting unit shall have minimum height of 30.0 metre.

(ii) If,

De-dusting unit is connected to After Burner Chamber (ABC), emission shall be emitted through common stack (minimum height 30.0 metre) having separate arrangements for emission monitoring for de-dusting unit.

B. Fugitive

Emission Standards

Rotary Kiln/

De-dusting unit

Particulate

matter

Existing unit

New unit

(µg/m³)

3000

2000

Note.-(i) The existing industry shall comply with a standard of 2000 (µg/m³) after one year from the date of notification.

(ii) Fugitive

emission shall be monitored at a distance 10.0 metre from the source of fugitive emission as per following:

Area

Monitoring

location

Raw material

handling area

Wagon

tipppler, Screen area, Transfer points, Stock bin area

Crusher

area

Crushing

plant, Vibrating screen, Transfer points

Raw material

feed area

Feeder area,

Mixing area, Transfer points

Cooler

discharge area

Oversize

discharge area, Transfer points, Intermediate stock bin area,

Screening plant, Magnetic separation unit, Transfer points, Over

size discharge area, Product separation area, Bagging area

Product

processing area

Intermediate

stock bin area, Screening plant, Magnetic separation unit,

Transfer points, Over size discharge area, Product separation

area, Bagging area

Other areas

As specified

by State Pollution Control Board/Pollution Control Committee

C. Effluent

Standards

pH

5.5-9.0

Total

suspended solids

100 mg/1

Oil and

grease

10mg/1

Chemical

oxygen demand

250 mg/1

Note.-(i) All efforts shall be made to reuse and re-circulate the water and to maintain "Zero discharge".

(ii) Storm

water drain shall be provided within the premises of the industry

so as to avoid mixing with effluent.

100. []

Common

Hazardous Waste Incinerator

A. Emission

Limiting concentration in mg/Nm³, unless stated

Sampling

Duration in (minutes) unless stated

Particulate

Matter

50

30

HCl

50

30

SO₂

200

30

CO

100

30

50

24 hours

Total Organic

Carbon

20

30

HF

4

30

NO_x (NO and

NO₂ expressed as NO₂)

400

30

Total dioxins

and furans

0.1 ng

TEQ/Nm³

8 hours

Cd+Th+their

compounds

0.05

2 hours

Hg and its

compounds

0.05

2 hours

Sb+As+Pb+Co+Cr+Cu+

Mn+Ni+V+their compounds

0.50

2 hours

Notes. -

(i) All

monitored values shall be corrected to 11% oxygen on dry basis.

(ii) The CO₂ concentration in tail gas shall not be less than 7%.

(iii) In

case, halogenated organic waste is less than 1% by weight in input waste, all the facilities in twin chamber incinerators shall be designed to achieve a minimum temperature of 950°C in secondary combustion chamber and with a gas residence time in secondary combustion chamber not less than 2 (two) seconds.

(iv) In case

halogenated organic waste is more than 1% by weight in input waste, waste shall be incinerated only in twin chamber incinerators and all the facilities shall be designed to achieve a minimum temperature of 1100°C in secondary combustion chamber with a gas residence time in secondary combustion chamber not less than 2 (two seconds).

(v)

incineration plants shall be operated (combustion chambers) with such temperature, retention time and turbulence, as to achieve Total Organic Carbon (TOC) content in the slag and bottom ashes less than 3%, or their loss on ignition is less than 5% of the dry weight.

101. []

Incinerator

for Pesticide Industry

A. Emission

Limiting concentration in mg/Nm³, unless stated

Sampling

Duration in (minutes) unless stated

Particulate

Matter

50

30

HCL

50

30

SO₂

200

30

CO

100

daily average

Total Organic

Carbon

20

30

Total Dioxins

and Furans*

Existing

Incinerator

0.2 ng

TEQ/Nm³

8 hours

New

Incinerator

0.1 ng

TEQ/Nm³

8 hours

Sb + As + Pb

+ Cr + Co + Cu + Mn + Ni + V + their compounds

1.5

2 hours

*The existing

plant shall comply with norms for dioxins and furans as 0.1

ng/TEQ/Nm³ within a period of five years from the date of publication of this notification.

Notes.-

(i) All

monitored values shall be corrected to 11% oxygen on dry basis.

(ii) The CO₂ concentration in tail gas shall not be less than 7%.

(iii) In

case, halogenated organic waste is less than 1% by weight in

input waste, all the facilities in single chamber incinerators

shall be designed so as to achieve a minimum temperature of

1100°C, in the incinerator. For fluidized bed technology

Incinerator, temperature shall be maintained at 950°C.

(iv) In case

halogenated organic waste is more than 1% by weight in input

waste, waste shall be incinerated only in twin chamber

incinerators and all the facilities shall be designed to achieve

a minimum temperature of 1100°C in secondary combustion

chamber with a gas residence time in secondary combustion chamber

not less than two seconds.

(v) Scrubber

meant for scrubbing emissions shall not be used as quencher.

(vi)

Incineration plants shall be operated (combustion chambers) with

such temperature, retention time and turbulence, as to achieve

Total Organic Carbon (TOC) content in the slag and bottom ashes

less than 3%, and their loss on ignition is less than 5% of the

dry weight.

(vii) The

incinerators shall have a chimney of at least thirty metre height.

B. Wastewater

(i)

Wastewater (scrubber water and floor washings) shall be

discharged into receiving water conforming to the norms

prescribed under Schedule VI: General Standards for Discharge of

Environment Pollutants (Part A: Effluents) notified under the

Environment (Protection) Rules, 1986.

(ii) The

built up in Total Dissolved Solids (TDS) in wastewater of floor

washings shall not exceed 1000 mg/l over and above the TDS of raw

water used.

102. []

Refractory

Industry

A. Emission

Standards

(Q Down Draft

Kiln (Fuel: Coal)

Category*

Limiting

concentration (mg/Nm³)

Particulate

matter

small/medium/large

350

Minimum

(metres)

Stack height

small

15

medium

18

large

21

(ii) Other

than Down Draft Kiln (Fuel: Coal)

Category*

Limiting

concentration (mg/Nm³)

Particulate

Matter

small

300

medium

200

large

150

Minimum

(metres)

Stack height

small

15

medium

18

large

21

(iii) Box,

Tunnel, Down Draft Kiln, etc. (Fuel: Natural Gas/Producer Gas/LPG
or a combination of Fuels/Furnance Oil as Secondary Fuel)

Category*

Limiting

concentration (mg/Nm³)

Particulate

Matter

small
200
Medium/
150
large
Minimum
(metres)
Stack height
small
12
medium
15
large
18
Category*
Production
(tpa)
small kiln
15,000
medium kiln
15,001-50,000
large kiln
above 50,000
(iv) Rotary
Kiln (Fuel: Furnance Oil)
Category**
Limiting
concentration (mg/Nm3)
Particulate
Matter
small
200
Medium/
150
large
Minimum
(metres)
Stack height
small
35
medium
45
large
60
Category**
Production
(tpa)
small rotary
kiln
50
medium rotary

kiln

51-100

large rotary

kiln

above 100

Notes.-

(i) All

values of particulate matter are to be corrected at 6 per cent Carbon Dioxide.

(ii) Fugitive

emission shall not exceed 10 mg/m³ from any process or plant.

(iii) Each

stack shall be at least 2 metre above the top most point of the building, shed or plant in the industry excluding bucket elevator, mill house and vibrating screen.

(iv) If more

than one kiln is connected to single stack, sum of the production capacity of all the kilns would be considered for determining the capacity of the kiln and accordingly depending upon the total capacity, emission standard and stack height would be implemented.

(v)

Monitoring of stack shall be carried out at the time of charging and after the completion of charging and average of these two results shall be considered as emission level.

B. Effluent

Standards

Limiting

value for concentration

Inland

Surface Water

Public

Sewer

Land

for Irrigation

pH

5.5

to 9.0

5.5

to 9.0

5.5

to 9.0

Oil and

Grease

10

20

10

BOD 3 days,

27°C

30

250
100
COD
250
-
-
Suspended
Solids
100
600
200
Phenols
1.0
5.0
-
Cyanide as CN
0.2
2.0
0.2
Cr
(Hexavalent)
0.1
2.0
1.0
Cr (Total)
2.0
2.0
2.0.
[***]
103. []
Cashew Seed
Processing Industry
A-
Emission Standards
Process
Limiting
concentration (mg/Nm³)
Particulate
matter
Roasting
250
Cooking
(roasted shell/deoiled cake as fuel)
150
Borma Oven
Heater (roasted shell/ deoiled cake as fuel)
150
Stack height
minimum
(metres)
Roasting

20

Cooking

15

Borma Oven

Heater

15

Note:

All

values of particulate matter shall be corrected at 4% Carbon

Dioxide

Each

stack shall be at least 2 metres above the top most point of the building, shed or plant in the industry.

The

emissions from 'Dog-house' shall be channelised alongwith

Roasting-drum emissions and shall pass through wet scrubber

Bio-gasifier

shall be installed if roasted shells are used as fuel in the unit.

B-Effluent

standards

Limiting

concentration in mg/l, except for pH

Inland

surface Water

Public

sewer

Land

For Irrigation

pH

6.5

to 8.5

6.5

to 8.5

6.5

to 8.5

Oil

Grease

10

20

10

BOD 3 days,

270c

30

250

100

Suspended

Solids

100

600

200

Phenols

1

5

Sl. No.

Industry

Parameters

Standards

1

2

3

4

Effluent discharge standards (applicable to all mode of disposal)

105. [

[Inserted by Notification No. G.S.R. 1265(E), dated 13.10.2017 (w.e.f. 19.11.1986).]

Sewage Treatment Plants (STPs)

Location

Concentration not to exceed

(a)

(b)

pH

Anywhere in the country

6.5-9.0

Bio-Chemical Oxygen Demand (BOD)

Metro Cities*, all State Capitals except in the

State of Arunachal Pradesh, Assam, Manipur, Meghalaya Mizoram, Nagaland, Tripura Sikkim, Himachal Pradesh, Uttarakhand, Jammu and Kashmir, and Union territory of Andaman and Nicobar Islands, Dadar and Nagar Haveli Daman and Diu and Lakshadweep

20

Areas/regions other than mentioned above

30

Total Suspended Solids (TSS)

Metro Cities*, all State Capitals except in the

State of Arunachal Pradesh, Assam, Manipur, Meghalaya Mizoram, Nagaland, Tripura Sikkim, Himachal Pradesh, Uttarakhand, Jammu and Kashmir and Union territory of Andaman and Nicobar Islands, Dadar and Nagar Haveli Daman and Diu and Lakshadweep

50

Areas/regions other than mentioned above

100

Fecal Coliform (FC) (Most Probable Number per

100 milliliter, MPN/100ml

Anywhere in the country

1000

*Metro Cities are

Mumbai, Delhi, Kolkata, Chennai, Bengaluru, Hyderabad, Ahmedabad and Pune.

Note:

(i) All values in

mg/l except for pH and Fecal Coliform.

(ii) These standards

shall be applicable for discharge into water bodies as well as for land disposal/applications.

(iii) The standards

for Fecal Coliform shall not apply in respect of use of treated effluent for industrial purposes.

(iv) These Standards

shall apply to all STPs to be commissioned on or after the 1st June, 2019 and the old/existing STPs shall achieve these standards within a period of five years from date of publication of this notification in the Official Gazette.

(v) In case of

discharge of treated effluent into sea, it shall be through proper marine outfall and the existing shore discharge shall be converted to marine outfalls, and in cases where the marine outfall provides a minimum initial dilution of 150 times at the point of discharge and a minimum dilution of 1500 times at a point 100 meters away from discharge point, then, the existing norms shall apply as specified in the general discharge standards.

(vi) Reuse/Recycling

of treated effluent shall be encouraged and in cases where part of the treated effluent is reused and recycled involving possibility of human contact, standards as specified above shall apply.

(vii) Central Pollution Control Board/State

Pollution Control Boards/Pollution Control Committees may issue more stringent norms taking account to local condition under section 5 of the Environment (Protection) Act, 1986;.]

[106. Mandatory Implementation of Dust Mitigation Measures for Construction and Demolition Activities for projects requiring Environmental Clearance :

(i)

No building or infrastructure project requiring Environmental Clearance shall be implemented without approved Environmental Management Plan inclusive of dust mitigation measures.

(ii)

Roads leading to or at construction sites must be paved and blacktopped (i.e. metallic roads).

(iii)

No excavation of soil shall be carried out without adequate dust mitigation measures in place.

(iv)

No loose soil or sand or Construction Demolition Waste or any other construction material that causes dust shall be left uncovered.

(v)

Wind-breaker of appropriate height i.e. 1/3rd of the building height and maximum up to 10 meters shall be provided.

(vi)

Water sprinkling system shall be put in place.

(vii)

Dust mitigation measures shall be displayed prominently at the construction site for easy public viewing.]

[Inserted by Notification No. G.S.R. 94(E), dated 25.1.2018(w.e.f. 19.11.1986)]

[Sl. No.

[Inserted by Notification No. G.S.R. 263 (E), dated 22.3.2018 (w.e.f. 19.11.1986).]

Type of Industrial Sector

Standards

SO₂(mg/Nm³)

NO_x(mg/Nm³)

107

Ceramic*

400

600

108

Foundry Industries ** (Furnaces based on Fuel)

300

400

109

Glass***

500 for natural gas firing 1500 for other fuels

1000

110

Lime Kiln****

400

500

111

Reheating furnace*****

300

1000]

Sl. No.

Industry

Parameters

Standards

1

2

3

4

Ambient Air Quality Standards with respect to

Noise in Airport Noise Zone

[112

[Inserted by Notification No. G.S.R. 568(E), dated 18.6.2018 (w.e.f. 19.11.1986).]

Airports

Type of Airports

Limits in dB (A) Leq*

Day Time

Night Time

Busy Airports

70

65

All other Airports excluding proposed airports

65

60]

[113

[Inserted by Notification No. G.S.R. 5(E), dated 3.1.2019 (w.e.f. 19.11.1986).]

Kerosene standards

Characteristic

Requirement

Grade A

Grade B

Appearance

Clear and bright, Free from un-dissolved water,
foreign matter and other visible impurities

Clear and bright, Free from un-dissolved water,
foreign matter and other visible impurities

Acidity, inorganic

Nil

Nil

Burning quality(2)

(a) Char value, mg/Kg of oil consumed,Max

20

20

(b) Bloom on glass Chimney

Not darker than grey

No darker than grey

Colour

(a) Saybolt (in case of undyed kerosene)(3),Min

10

10

(b) Visual (in case of dyed kerosene)

Blue

Blue

Copper strip corrosion for 3 h at 500C

Not worse than No. 1

Not worse than No. 1

Density at 150C, kg/m³

Not limited, but to be reported

Not limited, but to be reported

Distillation

(a) Percent recovered below 2000C, percent

(v/v),Min

20

20

(b) Final boiling point,0C,Max

300

300

Flash point (Abel),0C,Min

35

35

Smoke point(4), mm,Min

18

18

Total sulphur content, percent(5)m/m,Max

0.10

0.20*

*The Ministry of Petroleum and Natural Gas shall

make efforts to produce and supply Grade A kerosene by 2020

Notes:(1) In case of dispute, this shall be the referee method.(2) This test is to be done at refinery end.(3) Where Saybolt chromo meter is not available Lovibond colour of the sample kept in an 18 cell may be measured according to

IS 1448: P-13 in which case the colour shall not be deeper than standard white (IP 4.0), however, in case of dispute [P:14] shall be referee method.(4) For supplies to Defence and Railway signal lamps the smoke point of the product shall be 22 mm, Minimum.(5) For all other specifications i.e. test methods, scope, references, grades, requirements, packing and marketing and sampling. It is require to meet Indian Standard IS 1459:2018 for Kerosene – Specifications (Fourth Revision), ICS No. 75.160.20, published in July, 2018]

Definitions:

(a)

*dB(A) Leq denotes the time weighted average of the level of sound in decibels on scale A which is relatable to human hearing. A day time from 6.00 a.m. to 10.00 p.m. and night time from 10.00 p.m. to 6.00 a.m. are considered for time weighted average.

(b)

"A", in dB(A) Leq, denotes the frequency weighting in the measurement of noise and corresponds to frequency response characteristics of the human ear (The range of human hearing is 20 Hz to 20 kHz).

(c)

A "decibel" is a unit in which noise is measured.

(d)

Leq: It is energy mean of the noise level over a specified period.

(e)

Busy Airport - For the purpose of noise management at airports, a busy airport shall be defined as "a civil airport which has more than 50,000 aircraft movements per year (a movement being a take-off or a landing)" excluding those purely for training purposes on light aircraft.

(f)

Take-off - A phase of flight from the application of takeoff power to an altitude of final take-off segment.

(g)

Landing - A phase of flight from the beginning of the landing flare until aircraft exits the landing runway comes to a stop on the runway, or when power is applied for takeoff in the case of a touch-and-go landing.

(h)

Lmax is unit for aircraft maximum noise level in units dB(A) which is maximum or peak noise value for aircrafts at the monitoring location in accordance with the noise standards notified by the Directorate General of Civil Aviation for respective airports.

(i)

Other Airports - an airport having more than 15000 but less than 50000 aircraft movement annually.

(j)

Proposed Airports - airport that is not functional yet and is under development.

Note : (i) Day time shall mean from 6.00 a.m. to 10.00 p.m and night time shall mean from 10.00 p.m. to 6.00 a.m.

(ii)

The above specified limits shall have a tolerance limit of 10dB (A) Leq.

(iii)

The specified limit excludes defense aircraft and aircraft landing and take-off noise from all runways and aircraft engine/ground run-ups, helipad locations earmarked by Airport Operator for this purpose.

(iv)

However, the limit for aircraft noise as Lmax will be notified by the airport operator with approval of the Directorate General of Civil Aviation at the aircraft noise monitoring locations installed by the airports as mentioned in paragraph 1 of this notification.

(v)

The noise limits specified in above shall replace and supersede the ambient air quality in respect of noise limits of the following existing zones:

(a)

Silence;

(b)

Residential; and

(c)

Commercial areas;

(vi)

As specified in the Noise Pollution (Regulation and Control) Rules, 2000 in the areas falling directly under Airport Noise Zone.

(vii)

The noise standards within the overall boundary of airports shall be applicable as Industrial Areas i.e. day time 75 dB (A) Leq and night time 70 dB (A) Leq as per the Noise (Regulation and Control) Rules 2000 and shall be measured at different points of airport boundary and then averaged.

(viii)

These standards will not be applicable to a civil airport which has less than 15,000 aircraft movement annually.

1.

(1)

For Airports excluding newly proposed airports. - In addition to dB(A) Leq applicable in the 'airport noise zones' specified above, Lmax value in dB(A) shall be published by the airport operator with approval of the Directorate General of Civil Aviation only for airports having more than 50,000 annual traffic movements. These Lmax values shall be complied by airlines and to be monitored and communicated by Airport Operator to the Directorate General of Civil Aviation. These Lmax value shall be reviewed as and when there is a requirement in future.

(2)

For Proposed Airports (yet to be operationalized):

(i)

For any upcoming/New Airports, noise modeling shall be conducted by the airport operators and results should be submitted to the Ministry of Environment, Forest and Climate Change while seeking Environment Clearance under the Environment Impact Assessment Notification, 2006.

(ii)

The airport operators should also develop airport noise zone as specified in paragraph 4 and share the same with Ministry of Housing and Urban Affairs and concerned State Development Authority for necessary land use planning around airport.

(iii)

The concerned State / Union Territory Development Authorities should not allow any new residential, institutions commercial facilities and other noise sensitive area falling in the airport noise zone area without any noise reduction measure.

2. Compliance of noise levels applicable to Airport Noise Zone as specified above shall lie with the airport operator and overseen by the Directorate General of Civil Aviation.

3. Airport operators shall prepare Noise Management Plan for compliance of the Airport Noise Standards.

4. Airport Noise Zones. - (1) The Airport Noise Zone area for each Airport shall be defined as Noise Contour for day and night period by the respective Airport Operator on the basis of existing GSR 751 (E), issued by the Ministry of Civil Aviation (Height Restrictions for Safeguarding of Aircraft Operations) Rules, 2015 published on 30th September, 2015 as amended from time to time on Height Restriction for Safeguarding of Aircraft Operation considering all approach and departure funnels and Instrument Flight Procedures (i.e. Instrument Approach Procedures, Standard Instrument Departure Standard Terminal Arrival Route) in consultation with airports Air Navigation Service Provider as per the Master Plan of the Airport. The same shall be approved by the Directorate General of Civil Aviation and displayed on the website of respective Airport Operators. This activity shall be completed within two years from the date of issuance of the final notification.

(2)

State / Union Territory Development Authorities should take into consideration of Airport Operations requirements in the airport noise zone area for the land use planning around the airport.

5. Airport Noise Mapping. - Noise mapping in for all airports should be carried out as per the requirements specified in the Director General Civil Aviation's requirements by the airport operators considering future aircraft movement and traffic projections of the airport as per the Master Plan of the Airport. This information to be displayed at a prominent places at Airports as well as in the website of respective Airport Operator and State / Union Territory Development

Authority.

6. Protocol and Measurements Procedure. - Monitoring protocol and measurements procedure for airport noise zone displayed on the website of the Ministry of Environment, Forest and the Climate Change and the Central Pollution Control Board shall be followed.

7. Development Authorities / Regional Planning Department shall specify provisions for inclusion of sound resistance in new buildings, facilities and projects of residential, institutional, hospital and commercial facilities in the design, construction and materials selections for improving indoor environment under existing building codes and bye laws for any building constructions coming under airport noise zones.

8. All the Airport, Airline and Authority shall comply with the requirements specified in the notification within two years from the date of notification.]

[SCHEDULE III]

[Inserted by G.S.R. 1063(E), dated 26.12.1989 (w.e.f. 26.12.1989).]

(See rule 3)

AMBIENT AIR QUALITY STANDARDS IN RESPECT OF NOISE

Area Code

Category of Area

Limits in dB(A)

Leg.

Day Time

Night Time

(A)

Industrial area

75

70

(B)

Commercial area

65

55

(C)

Residential area

55

45

(D)

Silence Zone

50

40

Note (1). - Day time is reckoned in between 6 a.m. and 9 p.m.

Note (2). - Night time is reckoned in between 9 p.m. and 6 a.m.

Note (3). - Silence zone is defined as areas upto 100 metres around such premises as hospitals, educational institutions and Courts. The Silence zones are to be declared by the Competent Authority.

Use of vehicular horns, loudspeakers and bursting of crackers shall be banned in these zones.

Note (4). - Mixed categories of areas should be declared as one of the four above-mentioned categories by the Competent Authority and the corresponding standards shall apply.

[SCHEDULE IV]

[Inserted by G.S.R. 54(E), dated 5.2.1990 (w.e.f. 5.2.1990).]

(See rule 3)

STANDARDS FOR EMISSION OF SMOKE, VAPOUR, ETC., FROM MOTOR VEHICLES

(1)

Every motor vehicle shall be manufactured and maintained in such condition and shall be so driven that smoke, visible vapour, grit, sparks, ashes, cinders or oily substance do not emit therefrom.

(2)

On and from the 1st day of March, 1990, every motor vehicle in use shall comply with the following standards:-

(a)
Idling CO (Carbon monoxide) emission limit for all four wheeled petrol driven vehicles shall not exceed 3 per cent by volume;

(b)
Idling CO emission limit for all two and three wheeled petrol driven vehicles shall not exceed 4.5 per cent by volume;

(c)
Smoke density for all diesel driven vehicles shall be as follows :-

Method of Test

Maximum smoke density

Light absorption co-efficient m-1

Bosch Units

Hatridge Units

(a) Full load at a speed of 60% to 70% of maximum engine rated speed declared by the manufacturer

3.1

5.2

75

(b) Free acceleration

2.3

-

65

(3)

On and from the 1st day of April, 1991, all petrol driven vehicles shall be so manufactured that they comply with the mass emission standards as specified at Annexure "I". The breakdown of the operating cycle used for the test shall be as specified at Annexure "II" and the reference fuel for all such tests shall be as specified in Annexure "III" to this Schedule.

(4)

On and from the 1st day of April, 1991, all diesel driven vehicles shall be so manufactured that they comply with the mass emission standards based on exhaust gas opacity as specified at Annexure "IV" to this Schedule.

(5)

On and from the 1st day of April, 1992, all diesel driven vehicles shall be so manufactured that they comply with the following levels of emissions under the Indian driving cycle :-

Mass of Carbon Monoxide (CO) Maxm.

Grams per KWH

Mass of Hydro Carbons (HC) Maxm.

Grams per KWH

Mass of Nitrogen Oxides (NO) Maxm.

Grains per KWH

14

3.5

18

(6)

Each motor vehicle manufactured on and after the dates specified in paragraphs (2), (3), (4) and (5) shall be certified by the manufacturers to be conforming to the standards specified in the said paragraphs and the manufactures shall further certify that the components liable to effect the emission of gaseous pollutants are so designed, constructed and assembled as to enable the vehicle, in nominal use, despite the vibration to which it may be subjected, to comply with the provisions of the said paragraphs.

(7)

Test for smoke emission level and Carbon Monoxide level for motor vehicles.-(a) Any officer not below the rank of a Sub-Inspector of police or an Inspector of motor vehicles, who has reason to believe that a motor vehicle is by virtue of smoke emitted from it or other pollutants like Carbon Monoxide emitted from it, is likely to cause environmental pollution,

endangering the health or safety of any other user of the road or the public, may direct the driver or any person incharge of the vehicle to submit the vehicle for undergoing a test to measure the standard of black smoke or the standard of any other pollutants.

(b)

The driver or any person incharge of the vehicle shall upon demand by any officer referred to in sub-paragraph (a), submit the vehicle for testing for the purpose of measuring the standard of smoke or the levels of other pollutants or both.

(c)

The measurement of standard of smoke shall be done with a smoke meter of a type approved by the State Government and the measurement of other pollutants like Carbon Monoxide shall be done with instruments of a type approved by the State Government.

ANNEXURE I

(See paragraph 3)

MASS EMISSION STANDARDS FOR PETROL DRIVEN VEHICLES

1. Type Approval Tests:

Two and Three Wheeler Vehicles

Reference Mass, R(kg)

CO(g/km)

HC(G/km)

1

2

3

R? 150

12

8

150R? 350

12+

18(R-150)

200

8+

4(R-150)

200

R350

30

12

Light Duty Vehicles :

Reference Mass, rw(kg)

CO(g/km)

HC(g/km)

1

2

3

rw ? 1020

14.3

2.0

1020 rw? 1250

16.5

2.1

1250 rw? 1470

18.8

2.1

1470 rw? 1700

20.7

2.3

1700 rw? 1930

22.9

2.5

1930 rw? 2150

24.9

2.7

rw? 2150

27.1

2.9

2. Conformity of Production Tests :

Two and Three Wheeler Vehicles

Reference Mass, rw(kg)

CO(g/km)

HC(G/km)

1

2

3

R150

15

8

150R? 350

15+

25(R-150)

200

10+

5(R-150)

200

R350

40

15

Light Duty Vehicles :

Reference Mass, rw(kg)

CO(g/km)

HC(g/km)

1

2

3

rw ? 1020

17.3

2.7

1020rw? 1250

19.7

2.7

1250rw? 1470

22.5

2.8

1470rw? 1700

24.9

3.0
 1700rw? 1930
 27.6
 3.3
 1930rw? 2150
 29.9
 3.5
 rw? 2150
 32.6
 3.7

For any of the pollutants referred to above of the three results obtained may exceed the limit specified for the vehicle by not more than 10 per cent.

Explanation. - Mass emission standards refers to the gm. of pollutants emitted per km. run of the vehicle, as determined by a chassis dynamometer test using the Indian Driving Cycle.

ANNEXURE II

(See paragraph 3)

BREAKDOWN OF THE OPERATING CYCLE USED FOR THE TESTS

No. of Operation
 Acceleration (m/acc2)
 Speed (km/h)
 Duration of each operation(s)
 Cumulative Time(s)

1
 2
 3
 4
 5
 1.

Idling

-
 -

16
 16
 2.

Acceleration

0.65
 0-14

6
 22
 3.

Acceleration

0.56
 14-22

4
 26
 4.

Deceleration

-0.63
 22-13

4
 30

5.
Steady speed
-
13
2
32
6.
Acceleration
0.56
13-23
5
37
7.
Acceleration
0.44
23-31
5
42
8.
Deceleration
-0.56
31-25
3
45
9.
Steady speed
-
25
4
49
10.
Deceleration
-0.56
25-21
2
51
11.
Acceleration
0.45
21-34
8
59
12.
Acceleration
0.32
34-42
7
66
13.
Deceleration
-0.46

42-37
3
69
14.
Steady speed

-
37
7
76
15.
Deceleration
-0.42

34-34
2
78
16.
Acceleration
0.32
34-42

7
85
17.
Deceleration
-0.46
42-47

9
94
18.
Deceleration
-0.52
27-14

7
101
19.
Deceleration
-0.56
14-00

7
108
ANNEXURE III

(See paragraph 3)

REFERENCE FUEL FOR TYPE AND PRODUCTION CONFORMITY TESTS

Sl. No.

Characteristic

Requirements

Method of Test ref. of P: or IS: 1448*

87 Octane

93 Octane

1

2

3

4

5

1.

Colour, visual

Orange

Red

2.

Copper-strip corrosion for 3 hours at 50°C

Not worse than No.1

P:15(1968)

3.

Density at 151C

Not limited but to be reported

P:16(1967)

4.

Distillation:

P:18(1967)

(a) Initial boiling point

Not limited but to be reported

(b) Recovery up to 20°C per cent by volume, Min.

10

10

(c) Recovery upto 125° C 50 per cent by volume

50

50

(d) Recovery upto 130°C per cent by volume, Min.

90

90

(e) Final boiling point, Max.

215°C

215°C

(f) Residue per cent by volume, Max.

2

2

5.

Octane number (Research method) Max.

87

94

P:27 (1960)

6.

Oxidation stability in minutes, Min.

360

360

P:28 (1966)

7.

Residue on evaporation mg/100 ml, Max.

4.0

4.0

P:29 (1960)

(Air-jet solvent washed)

8.

Sulphur, total, per cent by weight Max.

0.25

0.20

P:34 (1966)

9.

Lead content (as Pb), g/1 Max.

0.56

0.80

P:37 (1967) or

P:38 (1967)

10.

Reid vapour pressure at 38 degree C, kgf/cm³, Max.

0.70

P:39 (1967)

*Methods of test for petroleum and its products.

ANNEXURE IV

(See paragraph 4)

LIMIT VALUES OF EXHAUST GAS OPACITY APPLICABLE FOR DIESEL DRIVEN VEHICLES THE ENGINE TESTS
AT STEADY SPEED

Nominal Flow G(1 /s)

Absorption Co-efficient K(m-1)

Nominal Flow G(1 /s)

Absorption Co-efficient K(m-1)

42

2.00

120

1.20

45

1.91

125

1.17

50

1.82

130

1.15

55

1.75

135

1.31

60

1.68

140

1.11

65

1.61

145

1.09

70

1.56

150

1.07

75
1.50
155
1.05
80
1.46
160
1.04
85
1.41
165
1.02
90
1.38
170
1.01
95
1.34
175
1.00
100
1.31
180
0.99
105
1.27
185
0.97
110
1.25
190
0.96
115
1.22
195
0.95
20(1
0.93

[SCHEDULE]

[Inserted by G.S.R. 82(E), dated 16.2.1987 (w.e.f. 16.2.1987).]

[V]

[Renumbered by G.S.R. 422(E), dated 19.5.1993 (w.e.f. 19.5.1993).]

(See rule 12)

Sl. No.

Place at which the discharge of any environment pollutant in excess of prescribed standards occurs or is apprehended to occur

Authorities or agencies to be intimated

Appointed under

1
2
3

4

1.

Factories as defined under the Factories Act, 1948 -

(a) Owned by the Central Government and engaged in carrying out the purposes of the Atomic Energy Act, 1962

(i) Atomic Energy Regulatory

The Atomic Energy Act, 1962

(ii) The Ministry of Environment and Forest

-

(b)

Factories other than those mentioned in para. (a)

(i) The Chief Inspector of Factories

The Factories Act, 1948

(ii) The Inspector of Factories having local jurisdiction

-do-

(iii) The Ministry of Environment and Forests

-

2.

Mine as defined under the Mines and Minerals (Regulation and Development) Act, 1957

(i) [The Controller-General of Mines]

[Substituted by S.O. 64(E), dated 18.1.1988 (w.e.f. 18.1.1988).]

The Mines and Minerals (Regulation and Development) Act, 1957

(ii) Regional Controller of Mines having local jurisdiction]

-

(iii) The Ministry of Environment and Forests

-

3.

Port as defined under the Indian Ports Act, 1908

(i)

Conservator of Ports

The Indian Ports Act, 1908

(ii) The Ministry of Environment and Forests

-

4.

Plantations as defined under the Plantations Labour Act, 1951

(i) The Chief Inspector of Plantations

The Plantations Labour Act, 1951

(ii) The Inspector of Plantations having local jurisdiction

-do-

(iii) The Ministry of Environment and Forests

-

5.

Motor Vehicle as defined under the Motor Vehicles Act, 1939

(i) State Transport Authority

The Motor Vehicles Act, 1939

(ii) Regional Transport Authority having regional jurisdictions

-do-

(iii) The Ministry of Environment and Forests

-

6.

Ship as defined under the Merchant Shipping Act, 1958

(i)

Director-General of Shipping

The Merchant Shipping Act, 1958

(ii) Surveyor having jurisdiction

-do-

(iii) The Ministry of Environment and Forests

-

[SCHEDULE VI]

[Inserted by G.S.R. 422(E), dated 19.5.1993 (w.e.f. 19.5.1993).]

[See rule 3(3-A)]

GENERAL STANDARDS FOR DISCHARGE OF ENVIRONMENTAL POLLUTANTS

Part A

Effluents

Sl. No.

Parameter

Standards

Inland surface water

Public Sewers

Land for irrigation

Marine coastal areas

1

2

3(a)

3(h)

3(c)

3(d)

1.

Colour and odour

See 6 of Annexure I

-

See 6 of Annexure I

See 6 of Annexure 1

2.

Suspended solids mg/l, Max.

100

600

200

(a) For process waste water-100

(b) For cooling water effluent 10 per cent. above total suspended matter of influent

3.

Particle size of suspended solids

Shall pass 850 micron IS Sieve

-

(a) Floatable solids, Max. 3 mm.

(b) Settleable solids, Max. 850 microns

4.

[***]

[Omitted by G.S.R. 801(E), dated 31-12-1993 (w.e.f. 31-12-1993).]

5.

pH value

5.5-9.0

5.5-9.0

5.5-9.0

5.5-9.0

6.

Temperature

Shall not exceed 5°C above the receiving water temperature

-

-

Shall not exceed 5°C above the receiving water temperature

7.

Oil and grease mg/1, Max.

10

20

10

[10]

[Substituted for the words "20" by Notification No. G.S.R. 739 (E) dated 9.9.2010 (w.e.f. 19.11.1986)]

8.

Total residual chlorine mg/1, Max.

1.0

-

-

1.0

9.

Ammoniacal Nitrogen (as N), mg/1, Max.

50

50

-

50

10.

Total Kjeldahl nitrogen¹[N]; mg/1, Max.

100

-

-

100

11.

Free Ammonia¹[NH₃]

mg/1, Max.

5.0

-

-

5.0

12.

Biochemical Oxygen demand (5 days at 20°C)¹[mg/1, Max]

30

350

100

100

13.

Chemical Oxygen demand, mg/1, Max.

250

-

-

250

14.

Arsenic (as As)1[mg/1], Max.

0.2

0.2

0.2

0.2

15.

Mercury (as Hg), mg/1, Max.

0.01

0.01

-

0.01

16.

Lead (as Pb)

mg/1, Max.

0.1

1.0

-

2.0

17.

Cadmium (as Cd)

mg/1, Max.

2.0

1.0

-

2.0

18.

Hexavalentchromium (as Cr+6) mg/l, Max.

0.1

2.0

-

1.0

19.

Total Chromium (as Cr) mg/1, Max.

2.0

2.0

-

2.0

20.

Copper (as Cu) mg/1, Max.

3.0

3.0

-

3.0

21.

Zinc (as Zn) mg/1, Max.

5.0

15

-

15

22.

Selenium (as Se) mg/1, Max.

0.05

0.05

-

0.05

23.

Nickel (as Ni) mg/1, Max.

3.0

3.0

5.0

[***]

[Omitted by G.S.R. 801(E), dated 31-12-1993 (w.e.f. 31-12-1993).]

[***]

[Omitted by G.S.R. 801(E), dated 31-12-1993 (w.e.f. 31-12-1993).]

[***]

[Omitted by G.S.R. 801(E), dated 31-12-1993 (w.e.f. 31-12-1993).]

27.

Cyanide (as CN) mg/1, Max.

0.2

2.0

0.2

0.2

[***]

[Omitted by G.S.R. 801(E), dated 31-12-1993 (w.e.f. 31-12-1993).]

29.

[Fluoride]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

(as F) mg/1, Max.

2.0

15

-

15

30.

Dissolved Phosphates (as P) mg/1, Max

5.0

-

-

-

[***]

[Omitted by G.S.R. 801(E), dated 31-12-1993 (w.e.f. 31-12-1993).]

32.

Sulphide(as S) mg/1, Max.

2.0

-

-

5.0

33.

Phenolic compounds[as C₆H₅OH]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

mg/1, Max.

1.0

5.0

-

5.0

34.

Radioactive materials:

(a) Alpha emitters¹[Micro curie/ml] Max.

10⁻⁷

10⁻⁷

1[10⁻⁸]

10⁻⁷

(b) Beta emitters[Micro curie/ml]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

Max.

10⁻⁶

10⁻⁶

10⁻⁷

1[10⁻⁶]

35.

Bio-assay test

90% survival of fish of after 96 hours in 100% effluent

90% survival of fish of after 96 hours in 100% effluent

90% survival of fish of after 96 hours in 100% effluent

90% survival of fish of after 96 hours in 100% effluent

36.

Manganese (as Mn)

2mg/1

2mg/1

2mg/1

37.

Iron (as Fe)

3mg/1

3mg/1

3mg/1

38.

Vanadium (as V)

0.2 mg/1

0.2 mg/1

0.2 mg/1

39.

Nitrate Nitrogen

10mg/1

-

-

20 mg/1

[***]

[Entry 40 omitted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

Part B

Waste-Water Generation Standards

Sl.

No

Industry

Quantum

1.

Integrated Iron

Steel

16 [m³/tonne]

of finished steel

2.

Sugar

0.4 [m³/tonne]

of cane crushed

3.

Pulp Paper

Industries

(a) Larger pulp

paper

(i) Pulp

paper

175 [m³/tonne]

of paper produced

(ii) [Viscose

Staple Fibre

150 m³/tonne

of product

(iii) Viscose

Filament Yarn

500 m³/tonne

of product]

(b) Small pulp

paper:

(i) Agro-residue

based

150 [m³/tonne]

of paper produced

(ii) Waste paper

based

50 [m³/tonne]

of paper produced

4.

Fermentation

Industries

(a) Maltry

3.5 [m³/tonne]

of grain produced

(b) Brewery

0.25 [m³/KL]

of beer produced

(c) Distillery

12 [m³/KL]

of alcohol produced

5.

Caustic Soda

(a) Membrane cell

process

1 [m3/tonne]

of caustic soda produced excluding cooling tower blow down

(b) Mercury cell

process

4 [m3/tonne]

of caustic soda produced (mercury bearing).

10% blow down

permitted for cooling tower

6.

Textile Industries

:

Man-made fibre

(i) Nylon

Polyester

120 [m3/tonne]

of fibre produced

(ii) Viscose rayon

150 [m3/tonne]

of product

7.

Tanneries

28 [m3/tonne]

of raw hide

8.

Starch, Glucose

and related products

8 [m3/tonne]

of maize crushed

9.

Dairy

3 [m3/KL]

of milk

10.

Natural rubber

processing industry

4 [m3/tonne]

of rubber

11. [

[Substituted by Notification No. G.S.R. 1607(E), dated 29.12.2017 (w.e.f. 19.11.1986).]

Fertiliser Industry

Naphtha, Natural Gas Mixed Feedstock

(Naphtha + Natural Gas) Based (Straight Nitrogenous Fertiliser)

3.0 m3/tonne of Urea or equivalent produced

Straight Phosphatic Fertilizer (Single Super

Phosphate (SSP) Triple Super Phosphate (TSP) excluding

manufacturing of any acid

0.4 m3/tonne of SSP or TSP

Complex Fertilizer

Standards of nitrogenous and Phosphatic
fertiliser are applicable depending on the primary product.]

12.

Natural Rubber

Processing Industry : Centrifuging and Creaming Units and Craps
and Crumb Units- 5 m³/tonne of product in Centrifuging
units;- 8 m³/tonne of product in Creaming units; and- 40
m³/tonne of product in Craps and Crumb units.

Part C

Load Based Standards

1. [Petroleum Oil Refinery :]

[Substituted by G.S.R. 186(E), dated 18.3.2008 (w.e.f. 18.3.2008).]

Parameter

Standard

1

2

Quantum limit in kg/1,000 tonne of crude processed

1.

Oil Grease

2.0

2.

BOD days, 27°C

6.0

3.

COD₃

50

4.

Suspended Solids

8.0

5.

Phenols .

0.14

6.

Sulphides

0.2

7.

CN

0.08

8.

Ammonia as N

6.0

9.

TKN

16

10.

P

1.2

11.

Cr (Hexavalent)

0.04

12.

Cr (Total)

0.8

13.

Pb

0.04

14.

Hg

0.004

15.

Zn

2.0

16.

Ni

0.4

17.

Cu

0.4

18.

V

0.8

19.

Benzene

0.04

20.

Benno (?) - Pyrene

0.08

Notes. - (i) Quantum limits shall be applicable for discharge of total effluent (process effluent, cooling water blow down including sea cooling water blow down, washings, etc.) to receiving environment (excluding direct application on land for irrigation/horticulture purposes within the premises of refinery).

(ii)

In order to measure the quantity of effluent (separately for discharge to receiving environment, application for irrigation/horticulture purposes within the premises of refinery blow-down of cooling systems), appropriate flow measuring devices (e.g. V-notch, flow meters) shall be provided with.

(iii)

Quantum of pollutants shall be calculated on the basis of daily average of concentration values (one 24-hourly composite sample or average of three grab samples, as the case may be), average flow of effluent during the day and crude throughput capacity of the refinery.

(iv)

Limit for quantity of effluent discharged (excluding blow-down from seawater cooling) shall be 400 m³/1000 tonne of crude processed. However, for refineries located in high rain fall area, limit of quantity of effluent only during rainy days shall be 700 m³/1000 tonne of crude processed.

2.

Large Pulp Paper, News Print/Rayon grade plants of capacity above 24000[tonne]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

/Annum

Parameter

Quantum

Total Organic Chloride (TOCI)

2[kg/tonne]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

of product.

3. [Natural Rubber Processing and Rubber Product Industry: Centrifuging and Creaming Units, Craps and Crumb Units.]

[Inserted by Notification No. G.S.R. 221 (E) dated 18.3.2011 (w.e.f. 19.11.1986)]

Parameter

Standards

: Quantum limit in kg/100 tonne of finished Products

Oil and

Grease

1.5

BOD, 3

days at 27°C

200

Suspended

Solids

200

Total

Chromium

0.10

Lead

0.15

Part D

General Emission Standards

1. Concentration Based Standards:

Sl. No.

Parameter Concentration not to exceed (in mg/Nm³)

Standard

1.

[Particulate Matter (PM)

150

2.

[Total Fluoride]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

[25]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

3.

Asbestos

[4 Fibres/cc and dust should not be more than 2 mg/Nm³]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

4.

Mercury

0.2

5.

Chlorine

15

[6]

[Substituted by Notification No. G.S.R. 595 (E), dated 21.8.2009 (w.e.f. 19.11.1986).]

Petroleum Oil refinery (Sulphur Recovery)

Installed Capacity of SRU* (tonne/day)

Kg/tonne of sulphur in the feed to SRU

Existing SRU

New SRU

Sulphur Dioxide

Above 20

26

10

5 to 20

80

40

Up to 5

120

80

7

[***]

[Omitted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

8.

Sulphuric acid mist

50

9.

Carbon monoxide

[1% max (v/v)]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

[***]

[Omitted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

11.

Lead

[10 mg/Nm³]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

[***]

[Omitted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

*SRU - Sulphur Recovery Unit

II. Equipment Based Standards :

[For dispersal of sulphur dioxide, a minimum stack height limit is accordingly prescribed as below :]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

Sl. No.

Parameter

Standard

1.

Sulphurdioxide

Stack-height limit in[metre]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

(i) Power generation capacity:

-500 MW and more

275

-200/210 MW and above to less than 500 MW

220

-less than 200/210 MW

H=14 (Q)^{0.3}

(ii) Steam generation capacity

Coal consumption per day

-Less than 2[tonne/hr]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

[***]

[Omitted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

2 to 5 [tonne/hr]

-5 to 10 [tonne/hr]

-10 to 15 [tonne/hr]

-15 to 20 [tonne/hr]

-20 to 25 [tonne/hr]

-25 to 30 [tonne/hr]

-More than 30[tonne/hr]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

Note. - H-Physical height of the stack in [metre]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

Q-Emission rate of SO₂ in kg/hr. [***]

[Omitted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

III. Load/Mass-based Standards :

Sl.

No.

Industry

Parameter

Standard

1.

[Fertiliser]

(Urea)

Commissioned prior

to 1-1-1982

[Particulate

Matter (PM)]

2 [kg/tonne] of

product

Commissioned after

1-1-1982

[Particulate

Matter (PM)]

0.5 [kg/tonne] of

product

Quantum Limit in

kg/tonne Plant capacity for 100% concentration of Sulphuric Acid

(tonne/day)

Existing Unit

New Unit

2. []

[Substituted by Notification No. G.S.R. 354 (E) dated 2.5.2011 (w.e.f. 19.11.1986)]

Copper, Lead or

Zinc Smelting Plant

Sulphur-Dioxide

(SO₂)

Upto 300Above

300

2.5

2.0

2.0

1.5

3.
Nitric Acid
Oxides of Nitrogen
3 [kg/tonne] of
weak acid (before concentration) produced
Quantum Limit in
kg/tonne
Plant capacity
100%
for Existing Unit
New Unit
Concentration of
Sulphuric Acid tonne/day)
4. []
[Substituted by Notification No. G.S.R. 344(E), dated 7.5.2008 (w.e.f. 19.11.1986).]
Sulphuric Acid
Plant
Sulphur dioxide
(SO₂)
Up to 300
2.5
2.0
Above 300
2.0
1.5
5.
Integrated Iron
and Steel Plant
Carbon monoxide
Particulate matter
during coke pushing in coke oven
Particulate matter
for quenching operation in Coke Oven
3kg/tonne of coke
produced
5 gramme/tonne of
coke produced
50 gramme/tonne of
coke produced.?
6.
Petroleum Oil
Refinery (Sulphur Recovery)
Installed Capacity
of SRU*
Kg/tonne of
sulphur in the feed to SRU
(tonne/day)
Existing SRU
New SRU
above 20
26

10

Sulphur Dioxide

5 to 20

80

40

up to 5

120

80]

[***]

7.

Aluminium Plants:

(i) Anode Bake

Oven

Total Fluoride

0.3 Kg/MT of

Aluminium

(ii) Pot room

(a) VSS

-do-

4.7 Kg/MT of

Aluminium

(b) HSS

-do-

6 Kg/MT of

Aluminium

(c) PBSW

-do-

2.5 Kg/MT of

Aluminium

(d) PBCW

-do-

1.0 Kg/MT of

Aluminium

Note:

VSS

= Vertical Stud

Soderberg

HSS

= Horizontal Stud

Soderberg

[PBSW]

= Pre Backed Side

Work

PBCW

= Pre Backed

Centre Work

8.

Glass Industry:

(a)Furnace

Capacity

(i) Upto the

product draw capacity of 60 MT/Day

Particulate matter

2kg/hr

(ii) Product draw

capacity more than 60 MT/Day

-do-

0.8 kg/MT of

product drawn

9.

Petrochemicals

(Basic and Intermediates)

Source

Quantum

limit in gm/hour for New/Expansion Plants (gm/hr)

Organic

Particulate

Phthalic anhydride

(PA), Maleic anhydride (MA), Toluene Di-isocyanate (TDI)

plants-process emission

100

VOC-HAPs (TDI+MDI)

(Toluene

Di-isocyanate) TDI, Methylenediphenyl Di-isocyanate (MDI)

Plants-Process emission

0.5

VOC-HAPs (Benzene

Butadiene)

Benzene, Butadiene

Plants-Process emission

25.0

VOC-HAPs (EO, VCM,

EDC, ACN+PO)

EO, VCM, EDC, ACN,

PO Plants-Process emission

50.0

Part E

Noise Standards

A. Noise Limits for Automobiles [Free Field Distance at 7.5 Metre]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

in dB(A) at the manufacturing stage

(a) Motorcycles, Scooters Three-wheelers

80

(b) Passenger Cars

82

(c) Passenger or Commercial vehicles upto 4 MT

85

(d) Passenger or Commercial vehicles above 4 MT and upto 12 MT

89

(e) Passenger or Commercial vehicles exceeding 12 MT

91

[AA. Noise limits for vehicles at manufacturing stage

The test method to be followed shall be IS: 3028-1998.]

[Substituted by G.S.R. 849(E), dated 30.12.2002 (w.e.f. 30.12.2002). Earlier it was inserted by G.S.R. 742(E), dated 25.9.2000 (w.e.f. 25-9-2000).]

(1)

Noise limits for vehicles applicable at manufacturing stage from the year 2003.

Sl. No.

Type of vehicles

Noise limits dB(A)

Date of implementation

1.

Two wheeler

Displacement upto 80 cm³

75

Displacement more than 80 cm³ but upto 175 cm³

77

1st January, 2003

Displacement more than 175 cm³

80

2.

Three wheeler

Displacement upto 175 cm³

77

Displacement more than 175 cm³

80

1st January, 2003

3.

Passenger car

75

1st January, 2003

4.

Passenger or commercial vehicles

Gross vehicle weight upto 4 tonnes

80

Gross vehicle weight more than 4 tonnes but upto 12 tonnes

83

1st July, 2003

Gross vehicle weight more than 12 tonnes

85

(2)

Noise limits for vehicles at manufacturing stage applicable on and from 1st April, 2005.

Sl. No.

Type of vehicles

Noise limits dB(A)

1.0

Two wheelers

1.1

Displacement upto 80 cc

75

1.2

Displacement more than 80 cc but upto 175 cc

77

1.3

Displacement more than 175 cc

80

2.0

Three wheelers

2.1

Displacement upto 175 cc

77

2.2

Displacement more than 175 cc

80

3.0

Vehicles used for the carriage of passengers and capable of having not more than nine seats, including the driver's seat

74

4.0

Vehicles used for the carriage of passengers having more than nine seats, including the driver's seat, and a maximum Gross Vehicle Weight (GVW) of more than 3.5 tonnes

4.1

With an engine power less than 150 KW

78

4.2

With an engine power of 150 KW or above

80

5.0

Vehicles used for the carriage of passengers having more than nine seats, including the driver's seat: vehicles used for the carriage of goods

5.1

With a maximum GVW not exceeding 2 tonnes

76

5.2

With a maximum GVW greater than 3 tonnes but not exceeding 3.5 tonnes

77

6.0

Vehicles used for the transport of goods with a maximum GVW exceeding 3.5 tonnes

6.1

With an engine power less than 75 KW

77

6.2

With an engine power of 75 KW or above but less than 150 KW

78

6.3

With an engine power of 150 KW or above:

80

[Provided that for vehicles mentioned at serial numbers 3.0 to 6.3, the noise limits for the following States shall be applicable on and from the date specified against that State,-

(i)

Himachal Pradesh with effect from 1st October, 2005

(ii)

Jammu and Kashmir with effect from 1st October, 2005

(iii)

Madhya Pradesh with effect from 1st September, 2005

(iv)

Punjab with effect from 1st October, 2005

(v)

Rajasthan with effect from 1st June, 2005

(vi)

Uttar Pradesh (Mathura, Kannauj, Muzaffarnagar, Aligarh, Farukhabad, Saharanpur, Badaun, Barreilly, Moradabad, Hathras, Rampur, Bijnor, Agra, Pilibhit, J.P. Nagar, Mainpuri, Lalitpur, Hardoi, Ferozabad, Jhansi, Shahjahanpur, Etawah, Jalon, Lakhimpur Kheri, Etah, Mahoba and Sitapur) with effect from 1st June, 2005.

(vii)

Uttanchal with effect from 1st July, 2005.]

B. Domestic appliances and construction equipments at the manufacturing stage to be achieved by 31st December, 1993.

(a)

Window Air Conditioners of 1 tonnes to 1.5 tonnes

68

(b)

Air[coolers]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

60

(c)

Refrigerators

46

[***]

[Entry (d) omitted by G.S.R. 371(E), dated 17.5.2002 (w.e.f. 17.5.2002).]

(e)

compactors(rollers), Front loaders, Concrete mixers, Cranes (movable), Vibrators and Saws.

75

ANNEXURE I

(For the purposes of Parts A, B and C)

The State Boards shall follow the following guidelines in enforcing the standards specified under Schedule VI:

1. The waste-waters and gases are to be treated with the best available technology [(BAT)]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

in order to achieve the prescribed standards.

2. The industries need to be encouraged for recycling and reuse of waste materials as far as practicable in order to minimise the discharge of wastes into the environment.

3. The industries are to be encouraged for recovery of biogas, energy and reusable materials.

4. While permitting the discharge of effluents and omissions into the environment, State Boards have to take into account the assimilative capacities of the receiving bodies, especially water bodies so that quality of the intended use of the receiving water is not affected. Where such quality is likely to be affected, discharges should not be allowed into water bodies.

5. The Central and State Boards shall put emphasis on the implementation of clean technologies by the industries in order to increase fuel efficiency and reduce the generation of environmental pollutants.

6. All efforts should be made to remove colour and unpleasant odour as far as practicable.

7. The standards mentioned in this Schedule [shall also apply to all other effluents]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

discharged such as mining, and mineral processing activities and sewage.

8. The limit given for the total concentration of mercury in the final effluent of caustic soda industry, is for the combined effluent from (a) Cell house, (b) Brine plant, (c) Chlorine handling, (d) Hydrogen handling, and (e) Hydro chloric acid plant.

[***]

[Clause (9) omitted by G.S.R. 176(E), dated 2.4.1996 (w.e.f. 3.4.1996).]

10. All effluents discharged including from the industries such as cotton textile, composite woollen mills, synthetic rubber, small pulp and paper, natural rubber, petro-chemicals, tanneries, paint, dyes, slaughter houses, food fruit processing and dairy [industries] into surface waters shall conform to the BOD limits specified above, namely, 30 mg/ 1. For discharge of an effluent having a BOD more than 30 mg/1, the standards shall conform to those given above for other receiving bodies, namely, sewers, coastal waters and land for irrigation.

[***]

[Clause (11) omitted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

12. In case of fertilizer industry the limits in respect of chromium and [fluoride]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

shall be complied with at the outlet of chromium and [fluoride]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

removal units respectively.

13. In case of pesticides :

(a)

The limits should be complied with at the end of treatment plant before dilution.

(b)

Bio-assay test should be carried out with the available species of fish in the receiving water, the COD limits to be specified in the consent conditions should be correlated with the BOD limits.

(c)

In case notabilities and isomers of the pesticides in the given list are found in significant concentrations, standards should be prescribed for these also in the same concentration as the individual pesticides.

(d)

Industries are required to analyse pesticides in waste-water by advanced analytical methods such as GLC/HPLC.

14. [The chemical oxygen demand (COD) concentration in a treated effluent, if observed to be persistently greater than 250 mg/1 before disposal to any receiving body (public sewer, land for irrigation, inland surface water and marine coastal areas), such industrial units are required to identify chemicals causing the same. In case these are found to be toxic as defined in the Schedule I of the Hazardous Wastes (Management and Handling) Rules, 1989 the State Boards in such cases shall direct the industries to install tertiary treatment stipulating time limit.

15. Standards specified in Part A of Schedule VI for discharge of effluents into the public sewer shall be applicable only if such sewer lead to a secondary treatment including biological treatment system, otherwise these discharge into sewers shall be treated as discharge into inland surface waters.]

[Inserted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

ANNEXURE II

(For the purpose of Part D)

1. The State Boards shall follow the following guidelines in enforcing the standards specified under Schedule VI:

(a)

In case of cement plants, the total dust (from all sections) shall be within 400 mg/[Nm³]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

and 250 mg/[Nm³]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

for the plants upto 200 t/d and more than 200 t/d capacities respectively.

(b)

In respect of calcination process (e.g., aluminium plants), Kilns and Step Grate Bagasse-fired-Boilers, the Particulate Matter (PM) emissions shall be within 250 mg/[Nm³]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

.

(c)

In case of thermal power plants commissioned prior to 1-1-1982 and having generation capacity less than 62.5 MW, the PM emission shall be within 350 mg/[Nm³]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

.

(d)

In case of Lime Kilns of capacity more than 5t/day and up to 40t/day, the PM emission shall be within 500 mg/[Nm³]
[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

.

(e)

In case of horse shoe/Pulsating Grate and Spreader Stoker Bagasse-fired-Boilers, the PM emission shall be within 500 (12%CD₂) and 800 (12% CO₂) mg/[Nm³] respectively. In respect of these boilers, if more than attached to a single stack, the emission standard shall be fixed, based on added capacity of all the boilers connected with the stack.

(f)

In case of asbestos dust, the same shall not exceed 2 mg/[Nm³]
[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

.

(g)

In case of the urea plants commissioned after 1-1-1982, coke ovens and lead glass units, the PM emission shall be within 50 mg/[Nm³].

(h)

In case of small boilers of capacity less than 2 tons/hr and between 2 to 5 tons/hr the PM emissions shall be within 1600 and 1200 mg/[Nm³]
[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

.

(i)

In case of integrated Iron and Steel Plants, PM emission up to 400 mg/[Nm³]
[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]
shall be allowed during oxygen lancing.

(j)

In case of stone crushing units, the suspended PM contribution value at a distance of 40 metres from a controlled, isolated as well as from a unit located in the cluster should be less than 600 micrograms/[Nm³]
[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

.

[***]

[Omitted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]
These units must also adopt the following pollution control measures:-

(i)

Dust containment-cum-suppression system for the equipment;

(ii)

Construction of wind breaking walls;

(iii)

Construction of the metalled roads within the premises;

(iv)

Regular cleaning and wetting of the ground within the premises;

(v)

Growing of a green belt along the periphery.

(k)

In case of ceramic industry, from the other sources of pollution, such as basic raw material and processing operations, heat recovery dryers, mechanical finishing operation, all possible preventive measures should be taken to control PM emissions as far as practicable.

2. The total fluoride emissions in respect of Glass and Phosphatic Fertilizers shall not exceed 5 mg/NM³ and 25 mg/NM³ respectively.

3. [In case of copper, lead and zinc smelting, the off-gases may, as far as possible, be utilised for manufacturing sulphuric acid.]

[Substituted by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

4. [In case of cupolas (Foundries) having capacity (melting rate) less than 3 tonne/hour, the particulate matter emission shall be within 450 mg/Nm³. In these cases it is essential that stack is constructed over the cupola beyond the charging door and the emissions are directed through the stack, which should be at least six times the diameter of cupola. In respect of Arc Furnaces and Induction Furnaces, provision has to be made for collecting the fumes before discharging the emissions through the stack.]

[Added by G.S.R. 801(E), dated 31.12.1993 (w.e.f. 31.12.1993).]

[SCHEDULE VII

[Substituted by Notification No. G.S.R. 826 (E) dated 16.11.2009 (w.e.f. 19.11.1986)]

[See rule 3(3-B)]

NATIONAL AMBIENT AIR QUALITY STANDARDS

S.No.

Pollutant

Time Weighted Average

Concentration in Ambient Air

Industrial, Residential, Rural and Other Area

Ecologically Sensitive Area (notified by
Central Government)

Methods of Measurement

(1)

(2)

(3)

(4)

(5)

(6)

1

Sulphur Dioxide (SO₂), µg/m³

Annual*24 hours**

5080

2080

- Improved West and Gaeke- Ultraviolet fluorescence

2

Nitrogen Dioxide (NO₂), µg/m³

Annual*

24 hours**

40

80

30

80

- Modified Jacob Hochheiser (Na-Arsenite)-

Ultraviolet fluorescence

3

Particulate Matter (size less than 10µm) or PM₁₀µg/m³

Annual*

24 hours**

60

100

60

100

- Gravimetric- TOEM- Beta attenuation

4

Particulate Matter (size less than 2.5µm) or PM_{2.5}µg/m³

Annual*

24 hours**

40

60

40

60

- Gravimetric- TOEM- Beta attenuation

5

Ozone(O₃) µg/m³

8 hours**

1 hour**

100

180

100

180

- UV photometric- Chemi Iminescence- Chemical Method

6

Lead(Pb) µg/m³

Annual*

24 hour**

0.50

1.0

0.50

1.0

- AAS/ICP method after sampling on EPM 2000 or equivalent
filter paper- ED-XRF using Teflon filter

7

Carbon Monoxide (CO) mg/m³

8 hours

1 hour

0204

0204

- Non Dispersive Infra Red (NDIR) spectroscopy

8

Ammonia(NH₃)µg/m³

Annual*24 hours**

100400

100400

- Chemiluminescence- Indophenol blue method

9

Benzene(C₆H₆) µg/m³

Annual*

05

05

- Gas chromatography based continuous analyzer- Adsorption
and Desorption followed by GC analysis

10

Benzo(?) Pyrene (BaP) - particulate phase only, ng/m³

Annual*

01

01

- Solvent extraction followed by HPLC/GC analysis

11

Arsenic(As), ng/m³

Annual*

06

06

- AAS/ICP method after sampling on EPM 2000 or equivalent

filter paper

12

Nickel (Ni), ng/m³

Annual*

20

20

- AAS/ICP method after sampling on EPM 2000 or equivalent

filter paper

* Annual arithmetic mean of minimum 104 measurements in a year at a particular site taken twice a week 24 hourly at uniform intervals.

** 24 hourly or 08 hourly or 01 hourly monitored values, as applicable, shall be complied with 98% of the time in a year. 2% of the time, they may exceed the limits but not on two consecutive days of monitoring.

Note - Whenever and wherever monitoring results on two consecutive days of monitoring exceed the limits specified above for the respective category, it shall be considered adequate reason to institute regular or continuous monitoring and further investigation.]

Substituted by Notification No. G.S.R. 826 (E) dated 16.11.2009 (w.e.f. 19.11.1986)

[Inserted by G.S.R. 272(E), dated 5.5.2005 (w.e.f. 5.5.2005).]

[SCHEDULE VII]

[Inserted by G.S.R. 176(E), dated 2.4.1996 (w.e.f. 3.4.1996).]

[See rule 3(3-B)]

NATIONAL AMBIENT AIR QUALITY STANDARDS (NAAQS)

{|

Pollutant

Time Weighted Average

Concentration in Ambient Air

Industrial Area

Residential, Rural and Other Area

Sensitive Area

Method of Measurement

(1)

(2)

(3)

(4)

(5)

(6)

SulphurDioxide

Annual Average*

80µg/m³

60

µg/m³

15

µg/m³

- Improved West and Greke method

(SO)₂

24hours**

120µg/m³

80µg/m³

30µg/m³

- Ultraviolet Fluorescence

Oxides of nitrogen as No₂

Annual Average

80µg/m³

60µg/m³

15µg/m³

- Jacab and Hochheiser modified (Na-Arsenite) Method

24hours**

120

µg/m³

80

µg/m³

30

µg/m³

- Gas Phase Chemiluminescence

Suspended Particulate Matter (SPM)

Annual Average*

360

µg/m³

140

µg/m³

70

µg/m³

- High Volume Sampling

24

hours**

500

µg/m³

200

µg/m³

100

µg/m³

- [Average flow rate not less than 1.1

m³/minute]

Respirable Particulate Matter (size less than 10 µm) (RPM)

Annual Average*

120µg/m³

60µg/m³

50µg/m³

- Respirable Particulate matter sampler.

24

hours**

150

µg/m³

100

µg/m³

75µg/m³

Lead (Pb)

Annual Average*

1.0µg/m³

0.75

µg/m³

0.50

gg/m³

- AAS method after sampling using

24

hours**

1.5µg

/m³

1.00g/m³

0.75

µg /m³

EMP 2000 or equivalent filter paper.

Carbon

8

hours**

5.0

mg/m³

2.0

mg/m³

1.0

mg/m³

- Non-dispersive, infrared spectroscopy.

Monoxide

1

hour

10.0

mg/m³

4.0

mg/m³

2.0

mg/m³

*Annual Arithmetic mean of minimum 104 measurements in a year taken twice a week 24 hourly at uniform interval.

**24 hourly/8 hourly values shall be met 98% of the time in a year. 2% of the time, it may exceed but not on two consecutive days.

Notes.- 1. National Ambient Air Quality Standard: The levels of a air quality necessary with an adequate margin of safety, to protect the public health, vegetation and property.

2. Whenever and wherever two consecutive values exceeds the limit specified above for the respective category, it shall be considered adequate reason to institute regular/continuous monitoring and further investigations.

}}

APPENDIX A

FORM I

(See rule 7)

NOTICE OF INTENTION TO HAVE SAMPLE ANALYSED

To

.....

.....

Take notice that it is intended to have analysed the sample of*which has been taken today, the

The Environment (Protection) Rules, 1986

.....day of..... 20..... from*

(Name and designation of the person who takes the sample).

(Seal)

Date

*Specify the place from where the sample is taken.

FORM II

(See rule 8)

MEMORANDUM TO GOVERNMENT ANALYST

From

.....

.....

To

The Government Analyst

.....

.....

.....

The portion of sample described below is sent herewith for analysis under rule 6 of the Environment (Protection) Rules, 1986.

The portion of the sample has been marked by me with the following mark:

Details of the portion of samples taken.

Name and designation of person who sends the sample

(SEAL)

Date

FORM III

(See rule 8)

REPORT BY GOVERNMENT ANALYST

Report No.

Date

I hereby certify that IGovernment Analyst duly appointed under section 13 of the Environment (Protection) Act, 1986 received on theday of 20..... from

*.....a sample offor analysis.

The sample was in a condition fit for analysis as reported below.

I further certify that I have analysed the aforementioned sample onand declare that the result of the analysis to be as follows:

**

.....

The condition of seals, fastening of samples on receipt was as follows:

Signed thisday of.....20.....

Address

.....

Signature

(Government Analyst)

To

.....

.....

.....

*Here write the names of the officer/authority from whom sample was obtained.

**Here write full details of analysis and method of analysis.

FORM IV

(See rule 11)

FORM OF NOTICE

By registered post-acknowledgment due

From (1)

Shri.....

.....

.....

To

.....

.....

Notice under section 19(b) of the Environment (Protection) Act, 1986

Whereas an offence under the Environment (Protection) Act, 1986 (

29 of 1986

) has been committed/is being committed by(2) I/We hereby give notice of 60 days under section 19(b) of Environment (Protection) Act, 1986 (

29 of 1986

) of my/our intention to file a complaint in the Court against(3) for violation of section.....of the Environment (Protection) Act, 1986 (

29 of 1986

).

In support of my/our notice, I am/we are enclosing the following documents (3) as evidence of proof of violation of the Environment (Protection) Act, 1986 (

29 of 1986

).

Place

Date

.....

Signature(s)

Explanation. - (1) In case the notice is given in the name of a company, documentary evidence authorising the person to sign the notice on behalf of the company shall be enclosed to this notice.

Company for this purpose means a company defined in the Explanation to sub-rule (6) of rule 4.

(2)

Here give the name and address of the alleged offender. In case of a manufacturing/processing/operating unit, indicate the name/location/nature of activity, etc.

(3)

Documentary evidence shall include photographs/technical reports/health reports of the area, etc., for enabling enquiry into the alleged violation/offence.

[FORM V]

[Substituted by G.S.R. 386(E), dated 22-4-1993 (w.e.f. 22-4-1993). Earlier it was inserted by G.S.R. 329(E), dated 13-3-1992 (w.e.f. 13-3-1992)]

(See rule 14)

Environmental Statement for the financial year ending the 31st March,.....

Part A

(i)

Name and address of the owner/occupier of the industry operation or process.

(ii)

Industry category Primary-(STC Code) Secondary-(SIC Code).

(iii)

Production capacity-Units-

(iv)

Year of establishment.

(v)

Date of the last environmental statement submitted.

Part B

Water and Raw Material Consumption

(1)

Water consumption m³/d

Process

Cooling

Domestic

Name of products

Process water consumption per unit of product output

During the previous financial year

During the current financial year

(1)

(2)

(3)

(2)

Raw material consumption

*Name of raw materials

Name of products

Consumption of raw material per unit of output

During the previous financial year

During the current financial year

*Industry may use Codes if disclosing details of raw material would violate contractual obligations, otherwise all industries have to name the raw materials used.

Part C

Pollution discharged to environment/unit of output

(Parameter as specified in the consent issued)

Pollutants

Quantity of pollutants discharged (mass/day)

Concentrations of pollutants in discharges (mass/volume)

Percentage of variation from prescribed standards with reasons

(a) Water

(b) Air

Part D

Hazardous Wastes

(As specified under Hazardous Wastes (Management and Handling) Rules, 1989)

Hazardous Wastes

Total Quantity (Kg)

During the previous financial year

During the current financial year

(a) From process

(b) From pollution control facilities.

Part E

Solid Wastes

Total Quantity (Kg)

During the previous financial year

During the current financial year

(a) From process

(b) From pollution control facility

(c) (1) Quantity recycled or re-utilised within the unit

(2) Sold

(3) Disposed

Part F

Please specify the characterisations (in terms of composition and quantum) of hazardous as well as solid wastes and indicate disposal practice adopted for both these categories of wastes.

Part G

Impact of the pollution abatement measures taken on conservation of natural resources and on the cost of production.

Part H

Additional measures/investment proposal for environmental protection including abatement of pollution, prevention of pollution.

Part I

Any other particulars for improving the quality of the environment.

Related AI tags, queries and research notes

Translation